

Learning Cø / N*

A Student Textbook of Phenomenal Presence

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Welcome to the Inquiry

This book asks a difficult question in a deliberately testable form: **what would have to be true of a physical system for there to be something it is like to be that system in a given state?**

The Cø / N* research program offers one candidate answer. It does not ask you to accept that answer in advance. It asks you to understand the proposal precisely enough to identify its risks, compare it fairly with rival theories, and design evidence that could change its status.

The target is **minimal phenomenal presence**. The word *minimal* matters. A conscious state need not include a spoken report, a button press, self-reflection, autobiographical memory, sophisticated reasoning, or the thought *I am conscious*. Those capacities may enrich a conscious episode or make it easier to study. They are not built into the minimal target.

The theory's central idea is that phenomenal presence belongs to a properly identified candidate system when four conditions jointly obtain: the system is viable in the relevant regime; its activity is irreducibly integrated; a content is causally available across functionally diverse recipients without report being required; and the content is maintained through bounded recurrent dynamics. The three network roles are called N_1 , N_2 , and N_3 . Their conjunction is N^* .

Viability is written V . Minimal phenomenal presence is written **Cø** in prose and C_0 in formulas.

That summary is only a starting point. Every phrase in it opens a scientific and philosophical problem. What counts as a system? How can integration be separated from mere coactivation? What is availability when there is no report? How much recurrence is enough, and when does stability become pathological fixation? What kind of evidence could support a phenomenal attribution without pretending that another subject's experience has been directly observed? The fourteen chapters develop those questions one by one.

Plain language. The book presents a proposed model and the rules for testing it. It does not present a meter that directly reads consciousness from a brain, animal, organoid, or machine.

Do not infer. A complete account of how to test a theory is not proof that the theory is true. Methodological clarity makes a proposal more vulnerable to evidence; it does not make favorable evidence appear by definition.

Who this book is for

This is a graduate-level introduction written for readers in philosophy of mind, cognitive science, neuroscience, psychology, complex systems, artificial intelligence, or philosophy of science. It assumes intellectual maturity, not prior mastery of every field. Formal notation appears when it makes a commitment easier to see. Each important formula is also explained in ordinary language.

Readers with different backgrounds will notice different pressure points. Philosophers may focus on the identity claim, the explanandum, and the relation between evidence and ontology. Experimental scientists may focus on measurement validity, causal intervention, negative controls, and generalization. Engineers may focus on system boundaries, functional roles, and failure diagnosis. The theory can be evaluated only when these perspectives remain in conversation.

The stance of this book

The proper stance is **conditional and adversarial**:

1. Reconstruct the strongest version of the proposal before criticizing it.
2. Name the target before selecting a measurement.
3. Separate what a result shows from what the theory says exists.
4. Compare versions of theories, not slogans.
5. State what observation would force revision.
6. Withhold a verdict when the evidence does not license one.

You are not expected to finish the book as an advocate for CØ / N*. You are expected to finish able to explain what it claims, why each component is included, how it differs from nearby theories, which arguments support it, where those arguments remain incomplete, and how a decisive study might be built.

How to Use This Book

Each chapter mirrors one session in “Teaching CØ / N*: A Graduate Instructor’s Manual.” The order is cumulative: later chapters use distinctions fixed earlier. A reader may consult one chapter independently, but the exercises assume the shared vocabulary developed across the sequence.

Every chapter uses the same learning architecture.

- **Opening puzzle:** a concrete problem that reveals why the chapter is needed.
- **Chapter map:** the claims, distinctions, and skills you should be able to reconstruct.
- **Core exposition:** an accessible but exact account of the relevant model component.
- **Annotations:** short interventions that translate, qualify, warn, or direct your attention.
- **Analogies with limits:** explanatory comparisons followed by an explicit account of where each comparison fails.
- **Thought experiments:** controlled cases that isolate a commitment before empirical complications are added.
- **Rival lenses:** comparisons that expose both overlap and genuine disagreement.
- **Evidence and limits:** what current methods could establish, what bridges they need, and what remains unsettled.
- **Practice studio:** individual and collaborative work designed to produce an argument, model, prediction, or research decision.
- **Summary and key terms:** a compact reconstruction, not a substitute for the chapter.

Reading the annotations

The annotation labels perform different jobs.

Annotation	What it does	Question to ask yourself
Plain language	Restates a dense claim without replacing its precise form.	Can I move between the formal and ordinary-language versions?
Why it matters	Connects a distinction to explanation, evidence, or experimental design.	What error would follow if this distinction were ignored?
Do not infer	Marks a tempting conclusion that the premises do not license.	What additional bridge or evidence would be needed?
Method note	Identifies a validity, measurement, or design requirement.	How could this requirement be checked independently?
Checkpoint	Pauses the exposition for active reconstruction.	Can I explain the claim without copying the paragraph?
Where the analogy breaks	Prevents a useful comparison from silently becoming an identity claim.	Which features of the target are absent from the analogy?
Source note	Identifies which paper or papers in the primary corpus support a claim or synthesis.	Is this statement in one source, or is it the textbook’s cross-paper synthesis?

Annotation	What it does	Question to ask yourself
Context note	Points to external, version-specific literature needed for a fair comparison.	Which dated theory or empirical result is being invoked?
Research-status note	Distinguishes a proposal, rationale, or method from an established empirical result.	What has been formulated, and what has actually been validated?

Annotations are not decorative summaries. They model the inferential discipline that the theory requires. When you write your own notes, try using the same labels. In particular, add a **Do not infer** line whenever a result could easily be overstated.

A four-pass study method

Pass 1 - Locate the question. Read the opening puzzle, chapter map, and section headings. Write one sentence naming the problem the chapter is trying to solve.

Pass 2 - Reconstruct the claim. Read the exposition. Translate every formula into prose. Draw the relevant system, causal routes, temporal loop, or evidential workflow.

Pass 3 - Apply pressure. Work through the thought experiments and rival comparisons. Ask whether the model has classified the case because of evidence or because of a hidden assumption.

Pass 4 - Produce something. Complete at least one practice-studio task. Understanding becomes visible when you can generate a new case, prediction, objection, or experimental design.

Working with others

The collaborative exercises use roles because group discussion becomes shallow when every participant merely announces a verdict. Typical roles include:

- **Model advocate:** states the strongest theory-relative prediction and its rationale.
- **Rival advocate:** develops a genuine alternative explanation rather than a generic doubt.
- **Validity auditor:** checks construct validity, measurement validity, confounds, and transport.
- **Boundary auditor:** asks whether the candidate system and content have been fixed independently.
- **Inference recorder:** writes the narrowest licensed conclusion and a prohibited stronger conclusion.
- **Red-team designer:** proposes a result that would force revision rather than merely lower confidence.

Roles are temporary intellectual responsibilities, not personal commitments. Rotate them. The aim is to make every group capable of arguing from more than one position.

Method note. A useful collaborative artifact records both the group's conclusion and the strongest dissent. Agreement without an audit trail is difficult to evaluate.

The Fourteen-Chapter Course Map

Chapter	Central question	Main product
1	What exactly is the explanandum, and where does Cø / N* sit among consciousness theories?	A fair theory-landscape map
2	What does the core biconditional claim, and what work does each term perform?	A claim-and-failure-condition reconstruction
3	Which system is the candidate conscious subject?	A boundary-first bearer analysis
4	Why are viability and integration separate requirements?	A realization-and-unity diagnostic
5	What is system-wide availability when report is absent?	A causal-recipient audit
6	How does bounded recurrence contribute temporal presence?	A temporal-dynamics comparison
7	When should science say positive, negative, indeterminate, or unlicensed?	An inference ledger
8	Does every conjunct improve the theory over a simpler alternative?	A component-ablation design
9	How could a presence model extend to phenomenal character?	A bridge-validation proposal
10	What makes a consciousness model structurally readable?	A schematic-legibility audit
11	How should the framework be applied in biological and clinical settings?	A science/action separation plan
12	How should it travel to animals, organoids, and artificial systems?	A transport-and-target dossier
13	What differentiates and validates the theory under its strongest objections?	An adversarial comparison matrix
14	What research design could genuinely change the theory's status?	A preregistration-ready capstone

The Model at a Glance

1. The target: minimal phenomenal presence

Let S be a candidate system, c a content, R an operating regime, and Δ an evaluation interval. The core profile conjecture can be written:

$$H_{C_0} : V(S; R, \Delta) \wedge (N_1(S, c) \wedge N_2(S, c) \wedge N_3(S, c)) \Leftrightarrow C_0(S, c; R, \Delta).$$

Read the double arrow as a proposed biconditional: within the declared scope, the joint profile is claimed to be necessary and sufficient for minimal phenomenal presence. This is stronger than saying the profile often accompanies consciousness. It is also more vulnerable. A well-supported conscious case that decisively lacks the profile challenges necessity; a case that decisively has the profile while independent evidence counts against presence challenges sufficiency.

The compact formula suppresses the bearer step. A final subject attribution requires a stable causally admissible referent, a recomputed inclusion-minimal positive-bearer set, and a resolved bearer relation. Chapter 3 develops those requirements; until then, read an all-pass profile as a candidate result rather than an automatic subject verdict.

Plain language. The proposal says that the complete, properly scoped profile and minimal phenomenal presence go together in both directions. No single component is offered as a consciousness switch.

2. The component roles

Element	Short role	What it is meant to exclude
Candidate-referent qualification	A stable system to test	A convenient, shifting, or consciousness-selected node set
V	Viable realization	An inert diagram, failed preparation, or out-of-domain system
N_1	Irreducible integration or synergy	A heap of independent events or mere common drive
N_2	Report-independent system-wide availability	Diffusion, a single output, or widespread activation without causal reach
N_3	Bounded recurrent stability	An evanescent feedforward transient or pathological fixation
C_0	Minimal phenomenal presence	Report, intelligence, access, or phenomenal character treated as the target itself

The components are roles, not names for one favored laboratory metric. Each role requires an estimator family, a validity argument, an uncertainty policy, and negative controls. The theory would become circular if a measure were accepted merely because it classified familiar conscious cases as positive.

3. The bearer step

The formula above suppresses a difficult issue: several nested systems might satisfy the component profile. The project therefore distinguishes a **candidate referent** from a **positive bearer**. Candidate systems should be generated using consciousness-independent causal evidence. Component tests are then conducted on held-out evidence. Only after those tests is the theory-relative positive-bearer relation evaluated, normally by asking which passing candidates are inclusion-minimal and whether multiple minima are genuinely separable.

Why it matters. Without a bearer rule, the same episode could be attributed to a neuron, a region, a brain, and a brain-plus-environment simply because each description inherited the same favorable data.

Do not infer. A larger system is not automatically unconscious because a proper subsystem is the minimal positive bearer. It may contain the bearer without itself being selected as that bearer.

4. Evidence is not ontology

The theory's predicates concern what a system is and does. Experiments yield estimates of those predicates. A thresholded estimate can be uncertain, invalid, or out of scope. The book therefore uses four theory-relative scientific outputs:

1. **Positive:** the declared evidence licenses the positive conclusion.
2. **Negative:** the evidence licenses a decisive failure or negative conclusion.
3. **Indeterminate:** the evidence is valid and relevant but does not discriminate.
4. **Unlicensed:** the required validity, target, mapping, or scope condition is missing.

These categories prevent two common mistakes. Failure to establish a positive result is not automatically a negative result. And an impressive signal is not automatically evidence if its link to the target has not been validated.

Checkpoint. Invent one example of an indeterminate result and one example of an unlicensed result. Explain why the difference concerns evidential status, not simply signal strength.

5. The larger research program

The seven-paper constellation develops one core hypothesis through six further pressures:

- **Core identity proposal:** defines minimal phenomenal presence and the joint $V \wedge N^*$ profile.
- **Boundary and bearer:** specifies how candidate systems are generated and how positive bearers are selected.
- **Consciousness without report:** makes N_2 a causal and functionally diverse availability role rather than a behavior marker.
- **Indeterminacy as a scientific result:** makes uncertainty and licensing explicit outputs rather than residual embarrassment.
- **Ablating N^* :** asks whether each conjunct earns its place against leave-one-out rivals.
- **Phenomenal character:** introduces an independently testable bridge from causal-trajectory geometry to phenomenal relations.
- **Schematic legibility:** asks whether the theory supports decomposition, composition, counterfactual readout, fault localization, and evidential independence.

Later chapters show how these papers constrain one another. The constellation is strongest when its parts close loopholes in the core proposal. It is weakest where one paper's solution silently changes another paper's target. Your task is to notice both.

Constellation checkpoint

Draw the core profile in the center of one page. Around it, place the six further pressures: bearer, report-independent availability, inferential license, component ablation, phenomenal character, and schematic legibility. Connect each pressure to the part of the core proposal it is meant to test or repair. Then compare maps with a partner. Choose one connection on which you disagree and record three things: the hidden assumption, the evidence that could settle it, and the chapter in which you expect to revisit it. Keep the map; revise it after Chapters 8 and 14 rather than correcting it now.

Before Chapter 1: A Baseline Exercise

Work alone for five minutes. Write a provisional answer to each question without consulting another source.

1. What do you mean by *consciousness* in the narrowest sense relevant to science?
2. What observation would make you attribute that property to a system that cannot report?
3. What would count as a false positive for your preferred indicator?
4. What would count as a false negative?
5. Where does the candidate subject begin and end?

Then form a group of four. Assign one member to defend report, one to defend integration, one to defend recurrence, and one to audit the system boundary. Each person must propose a case that their favored criterion misclassifies. End by writing one sentence that everyone accepts and one disagreement that remains live.

Keep your original answers. Revisit them after Chapter 14. The point is not to discover whether you guessed the textbook's answer. The point is to see whether your standards became more explicit, discriminating, and revisable.

Chapter 1 - The Explanandum and the Theory Landscape

Opening puzzle: one word, six questions

A patient opens her eyes and moves through ordinary sleep-wake cycles, but she does not follow spoken commands. A brain scan later suggests that she may have performed a requested imagery task. Is she conscious?

The question feels urgent, yet it is not precise enough to answer. We might be asking whether she is awake, whether there is anything it is like to be her, whether she heard the request, whether she could use its content to guide a plan, whether she knew that she heard it, or what the heard voice sounded like. Those questions are connected, but they are not interchangeable. A result that bears directly on one may provide only indirect evidence about another.

This chapter introduces a discipline that will govern the whole book: identify the target before evaluating the theory or the evidence. The Cø / N* research program begins with **minimal phenomenal presence**. It asks whether there is anything it is like for a specified candidate system to undergo a specified content-bearing state. It does not begin by requiring speech, deliberate control, self-reflection, or a complete description of phenomenal character.

Plain language. Before asking whether a theory succeeds, ask exactly what the theory promises to explain.

Chapter map

By the end of this chapter, you should be able to:

- distinguish wakefulness, phenomenal presence, content presence, access, reflective consciousness, and phenomenal character;
- explain how one observation can be relevant without being constitutive;
- locate Cø / N* among major families of consciousness theory;
- state what makes it a mid-level, physicalist, information-dynamical, functional theory;
- compare theories only after matching their targets and versions; and
- identify both the scientific advantage and the risk of beginning with a minimal target.

1. The target ladder

The word *consciousness* does several jobs in ordinary and scientific language. A useful first pass is a six-rung target ladder. The ordering is conceptual; it is not a claim that brains literally climb these rungs one by one.

1. **Wakefulness or arousal.** Is the organism in a globally responsive operating state rather than, for example, deep anesthesia or coma?
2. **Minimal phenomenal presence, Cø.** Is there anything it is like for the candidate system to undergo the state?
3. **Content presence.** Is a particular content - this tone, this red patch, this ache - phenomenally present?
4. **Access and report.** Can the content guide reasoning, memory, deliberate action, or overt response?
5. **Reflective consciousness, C₁.** Does the system represent, monitor, or own its own state?
6. **Phenomenal character.** What is the experience like, and how are its qualities related?

The symbol Cø (also written C_0 in formulas) names the theory's minimal target. The subscript zero is meant to mark a baseline form of phenomenal presence, not an inferior or valueless kind of experience. C₁ (C_1 in formulas) is a convenient label for richer reflective or metacognitive organization.

Consider pain. Being awake is not the same as feeling pain. Feeling pain is not the same as being able to report it. Reporting that a pain is sharp is not the same as metacognitively judging that one's report may be unreliable. And none of those claims, by itself, explains why sharpness differs from throbbing or why unpleasantness has its particular place in a person's quality space.

Why it matters. If two theories target different rungs, their apparent conflict may be partly manufactured. A theory of report should not be rejected because it does not solve phenomenal character, and a theory of minimal presence should not be praised as though it already explained selfhood.

Analogy: a medical intake form

Imagine a medical form with separate boxes for temperature, pain intensity, mobility, alertness, and medication. Filling in the alertness box does not fill in the pain box. Still, the entries constrain one another and help form a larger picture. The target ladder works similarly: it keeps several dimensions apart without pretending they are unrelated.

Where the analogy breaks. A medical form lists familiar variables whose meanings are comparatively stable. Consciousness targets are theoretically contested, may overlap, and may not decompose as neatly as form fields.

Thought experiment: The Silent Violinist

A skilled violinist receives a temporary neuromuscular blocker while remaining physiologically stable. During the interval, a melody is played. Her auditory system shows content-specific, recurrent, and widely influential activity, but she can make no movement. Later she accurately identifies the melody and describes a vivid experience.

Work through the case in stages:

1. Before the later report is available, which claims concern wakefulness, Cø, content presence, and access?
2. What does the later report add? Does it become evidence about the earlier interval, or does it constitute the earlier experience?
3. Suppose the drug also prevents memory formation. Which evidential routes disappear, and which target remains conceptually possible?
4. What extra premise would be needed to infer that failure to report at the time meant absence of phenomenal presence?

The point is not to force one verdict. It is to display the bridge premises that a quick answer would otherwise hide.

2. Evidence, target, and constitution

An observation may be excellent evidence for a phenomenon without being part of what the phenomenon is. Smoke is evidence of fire, but smoke is not a constituent of every fire. Likewise, a verbal report can be strong evidence for experience without report being a necessary part of minimal phenomenal presence.

This distinction yields three questions:

- **What was directly observed?** A sentence, a button press, a neural pattern, a perturbational response, a behavioral discrimination?
- **What target does the observation most directly support?** Report capacity, stimulus registration, task maintenance, causal organization?
- **What bridge licenses the stronger conclusion?** For example, that this report is reliable evidence of experience, or that this network organization is constitutive of presence?

Take early visual decoding. If a classifier identifies stimulus category from activity at 80 milliseconds, the experiment shows that content-specific information is present in the measured signal. It does not yet show that the information is phenomenally present, system-wide available, or causally necessary. Unconscious processing can also be decodable.

Now take covert command following in a behaviorally unresponsive patient. Reliable performance on a mental-imagery instruction supports hearing, comprehension, task retention, and voluntary control in the tested trials. Those capacities may provide powerful evidence for phenomenal presence. Even so, the inferential link must be stated; the scanner does not literally display another person's experience.

Do not infer. “Relevant to Cø” does not mean “a direct measurement of Cø.” The same restraint applies to negative evidence: a failed test may reveal low sensitivity, a broken output channel, or an invalid operating regime rather than phenomenal absence.

Analogy: a courtroom witness

A credible witness can strongly support the claim that an event occurred. The witness's testimony is neither the event itself nor an infallible window onto it. Researchers treat reports in a similar way: often highly informative, sometimes indispensable for calibration, but open to errors, demand effects, memory limits, and missing-output problems.

Where the analogy breaks. A witness and an event are plainly separate objects. In consciousness research, some theories claim that report-related access or higher-order representation partly constitutes certain forms of consciousness. Whether evidence and target separate cleanly is itself part of the dispute.

Checkpoint. Rewrite the claim “the stimulus was consciously seen because it was decoded” as two statements: the direct result and the missing bridge premise.

3. What CØ / N* claims - and what it postpones

CØ / N* proposes that minimal phenomenal presence is tied to a realized network-dynamical profile. In later chapters, the profile will be unpacked into viability (V), N_1 integration or synergy, N_2 report-independent system-wide availability, and N_3 recurrent stability. The bundle N^* abbreviates $N_1 \wedge N_2 \wedge N_3$: all three components together.

At this stage, the important point is the theory's division of labor. The research program contains seven linked papers:

- the core paper proposes the candidate identity between CØ and the realized N^* profile;
- *Where Is the Conscious Subject?* asks which causal system is the bearer;
- *Consciousness Without Report* gives a stricter account of N_2 ;
- *Indeterminacy as a Scientific Result* specifies what experimental evidence licenses;
- “Ablating N^* : Does Every Conjoint Earn Its Place?” asks whether every conjoint earns its place;
- *From Phenomenal Presence to Phenomenal Character* adds a separately testable bridge to qualitative structure; and
- *Reading Consciousness from the Schematic* asks what would make the model compositionally and counterfactually intelligible.

Source note. This division of labor follows the corpus map and the roles stated across the seven-paper CØ / N* research program.

The core target is deliberately narrow. Report, attention, executive control, intelligence, autobiographical memory, and self-modeling may enrich experience or serve as evidence. They are not silently inserted as necessary constituents of CØ. Phenomenal character is not denied; it is addressed through an additional bridge only after presence is admitted.

Method note. An experiment normally assesses $C_0(S, c; \Theta)$: phenomenal presence for candidate system S , content or token c , and frozen specification Θ . A positive result for one content does not establish continuous consciousness or a rich stream. A negative result for one content does not establish that no other content is present.

The parameter Θ is a compact label for choices such as grain, time horizon, operating regime, measures, thresholds, and mappings. Making those choices explicit prevents a result obtained for one system at one scale from silently migrating to another.

4. Locating the model among neighboring theories

The best short classification is that Cø / N* is a **physicalist, information-dynamical, functional network theory of minimal phenomenal presence**. Each word adds a constraint.

- **Physicalist:** the theory proposes a causally realized organization rather than a nonphysical substance.
- **Functional:** components are partly specified by causal roles, so multiple realization is possible in principle.
- **Information-dynamical:** joint information, effective influence, and time-dependent organization matter.
- **Network theory:** the relevant relations occur among causally interacting elements at a declared grain.
- **Of minimal phenomenal presence:** the initial target is not report, self-reflection, or full phenomenal character.

The model is not unrestricted functionalism. A static lookup table or behavioral duplicate does not pass merely because its input-output mapping looks right. Effective causal coupling, temporal grain, operating regime, and substrate-appropriate viability remain essential.

In one broad taxonomy, its primary home is **materialist computational and functional theory**. Its secondary home is **information-based theory**, with cross-references to **neurobiological** and **first-order** approaches. It is realist rather than illusionist, conditionally substrate-independent rather than panpsychist, and only weakly adjacent to electromagnetic or synchrony theories. Those labels locate the program; they do not replace tests.

Here is a fair comparison by theoretical pressure rather than slogan:

Theory family	Pressure Cø / N* accepts	Point of difference to test
Integrated Information Theory (IIT)	unity and irreducible causal structure matter	N_1 is only one conjunct; Cø / N* does not adopt IIT's full axiomatic identity or its theory-internal exclusion rule as an independent bearer test
Global Workspace / Global Neuronal Workspace (GWT/GNWT)	a content's broad availability matters	N_2 is defined without requiring overt report, executive uptake, a fixed anatomical hub, or one ignition signature
Recurrent Processing Theory (RPT)	feedback and temporal stabilization matter	recurrence is necessary but not sufficient; the model also requires unity and diverse causal availability
Higher-order theories	representation of one's own state can explain reflective awareness	higher-order representation is assigned to a richer C_1 layer rather than the minimal gate

Theory family	Pressure Cø / N* accepts	Point of difference to test
Attention-schema and self-model theories	internal models may shape control, availability, and self-attribution	a self-model is not an unstated minimal conjunct; it belongs to N_2 or richer C_1 organization unless evidence requires more
Predictive-processing approaches	hierarchical inference and precision may implement conscious organization	inference or prediction-error minimization is not identified with consciousness by itself
Quality-space theories	phenomenal character has relational structure	qualitative structure is tested only after a presence gate and does not by itself establish presence
Representational geometry and neural-manifold approaches	population relations, trajectories, and topology can be compared	no latent geometry is renamed “qualia” without causal and phenomenal validation
Embodied, affective, and enactive approaches	regulation, bodily state, action, and value may shape the bearer and dynamics	the program does not decide in advance that every bodily factor is either constitutive or irrelevant
C-tests and theory-derived indicator approaches	nonreporting systems require validated, multi-source tests	one marker, behavior, or theory-conditional indicator cannot issue a universal verdict
Illusionism	reports and self-ascriptions require functional explanation	the phenomenal target is not replaced by a disposition to claim phenomenality; Cø / N* remains realist about minimal presence
Panpsychism, fundamentalism, and universalism	substrate independence remains a live possibility	mere existence, complexity, or abstract isomorphism is insufficient; causal realization, grain, coupling, and viability are required
Biological autonomy and individuality	self-maintaining organization may help individuate a bearer	autonomy is a candidate-admission discipline, not a sufficient condition for consciousness
Markov-blanket and active-inference approaches	conditional-independence structure may nominate boundaries	a statistical blanket proposes a candidate but does not prove a subject
Extended-mind and group-mind approaches	the skull is not an automatic causal boundary	reliable coupling alone is insufficient; constitution, autonomy, minimality, and separability must be tested
Electromagnetic and synchrony theories	coordinated distributed dynamics may help realize N_1 or N_3	a field, band, or synchrony relation is not the identity condition; this is weak implementation adjacency only
Dualist, idealist, quantum, and anomalous approaches	no core commitment is borrowed	nonphysical substance, consciousness as fundamental reality, collapse-based identity, and anomalous access fall outside the intended model

This table does not settle which theory is correct. It states prospective disagreement points. Exact comparison requires an exact, dated version. “GNWT” or “IIT” names a changing research family, not a single timeless prediction.

Context note. Starting points for exact versions include Albantakis et al. (2023) on IIT 4.0, Mashour et al. (2020) on GNWT, Lamme (2006) on recurrent processing, Lau and Rosenthal (2011) on higher-order theory, Graziano and Webb (2015) on attention-schema theory, Frankish (2016) on illusionism, and Thompson and Varela (2001) on embodied dynamics. The selected readings at the end of the book provide full citations.

Why it matters. A hybrid vocabulary is not yet a hybrid theory. Cø / N* becomes distinctive only where its conjunction, boundary order, report-independent availability criterion, four-outcome evidence policy, and component audits yield predictions or methods that neighbors do not.

Thought experiment: The Three Laboratories

Three laboratories study the same brief visual event.

- Laboratory U measures whether the content is irreducibly integrated.
- Laboratory A measures whether it becomes available to diverse processors.
- Laboratory R measures whether local recurrent activity stabilizes it.

Each laboratory announces, “We found the neural basis of consciousness.”

Stage the comparison:

1. Ask what target and sufficiency claim each announcement assumes.
2. Imagine a case that passes U but fails A and R. Choose one dated operational version of IIT, workspace theory, and RPT, then compare each with Cø / N*. If the case packet does not determine a version’s prediction, mark it **underdetermined** and name the missing rule rather than issuing a family-wide verdict.
3. Now imagine that all three measurements pass, but they were calculated over different node sets and time windows. What has actually been shown?
4. Specify one result on which two frozen theory versions would disagree. If you cannot, is the difference empirical, methodological, or verbal?

5. The mid-level bargain

Cø / N* is a **mid-level bridge theory**. It does not claim to deduce why any physical organization should have a phenomenal aspect. Nor does it begin with a complete catalogue of what every admitted experience feels like. Instead, it offers a public constitutive hypothesis: a specified causal-dynamical organization is proposed as minimal phenomenal presence.

The scientific gain is exposure. The theory can be asked which candidate system qualifies, which component fails, how measures disagree, and what result would require revision. The philosophical cost is incompleteness. Even a fully successful mid-level identity would leave questions about metaphysical grounding, qualitative detail, selfhood, value, and personal persistence.

This bargain can fail in two ways. First, the theory might be too permissive: systems with the profile may have strong independent evidence against phenomenal presence. Second, the target may be too thin to measure independently. If every attempt to identify Cø relies on importing report, character, or selfhood, “minimal presence” may not function as an autonomous scientific target.

Do not infer. Calling the model “mid-level” is not an excuse for missing predictions. A partial theory must still make clear commitments within its declared scope.

6. Evidence and limits

At this introductory stage, the evidence for CØ / N* is best understood as a research rationale, not a completed validation. Integration, broad availability, and recurrence are each prominent in consciousness science. Their convergence motivates testing a conjunction. It does not establish that the conjunction is necessary and sufficient.

Several limits should remain visible:

- Human reports provide important anchors but risk building access back into a report-independent target.
- No-report measures reduce some confounds but introduce others, including uncertain content labeling and state changes caused by the task itself.
- Neighboring theories often predict overlapping patterns in ordinary waking perception; decisive tests require disagreement cases.
- Every system-level measure depends on the chosen bearer, grain, time window, and intervention model.
- A theory can survive a failed surrogate if the measure was invalid, but repeated measurement failure reduces scientific usefulness.

Research-status note. The conditional and adversarial stance follows the source papers. Later chapters draw especially on *Where Is the Conscious Subject?*, *Indeterminacy as a Scientific Result*, and “Ablating N*: Does Every Conjunction Earn Its Place?”.

Checkpoint. For any consciousness result you encounter, write four lines: target, candidate bearer, evidence, and possible revision condition.

Practice studio

Quick checks

1. Give a one-sentence distinction between wakefulness and phenomenal presence.
2. Why can a report be evidence for CØ without being constitutive of CØ?
3. What does N* denote, and what are its three conjuncts?
4. Name one feature CØ / N* assigns to a richer layer rather than the minimal gate.
5. Why must a comparison name a particular version of a rival theory?
6. What scientific advantage does a mid-level claim gain by postponing phenomenal character?

Individual transfer task: build an inference ladder

Choose one published or hypothetical observation: a confidence report, a no-report pupil response, successful forced choice, late recurrent activity, perturbational complexity, or fluent AI self-description. Build a five-step inference ladder containing:

1. the raw observation;
2. the capacity or mechanism it most directly supports;
3. its possible relevance to CØ;
4. the bridge premise required for a CØ inference; and
5. one alternative explanation.

End by marking the strongest conclusion you would currently license. Do not use “conscious” before step 4.

Collaborative exercise and discussion: the Target Tribunal

Form teams of four. Each student makes a private initial classification before discussion.

- The **target analyst** separates wakefulness, Cø, content presence, access, C₁, and phenomenal character.
- The **evidence auditor** states what each observation directly measures and lists auxiliary assumptions.
- The **rival advocate** supplies the strongest nonphenomenal or differently phenomenal interpretation.
- The **synthesis reporter** records agreement, dissent, and the smallest useful next test.

Analyze these cases: covert command following in a nonresponsive patient; early visual decoding without report; a verbal pain description; a 250-millisecond recurrent loop in an isolated module; calibrated uncertainty self-report by an artificial agent; and wakefulness without reliable command response.

For each, complete six fields: most direct target, possible relevance to Cø, required bridge, strongest alternative, licensed conclusion, and forbidden stronger inference. Then exchange one row with another team. The receiving team may challenge only the bridge or the alternative explanation, not simply announce a different intuition.

Discussion prompt: Does narrowing the target to minimal presence increase testability, or does it remove the very features that let us identify consciousness? Each team must write the best argument on both sides before giving a provisional view and one unresolved dissent.

Counterexample challenge

Construct one case that would threaten the claim that $C\emptyset$ can be studied separately from report and one different case that would support that separation. For each case, state what would need to be independently verified before it counted. Avoid defining the case so that your preferred conclusion is guaranteed.

Exit ticket

In four sentences: state the target of $C\emptyset / N^*$, locate the model among theory families, name one observation relevant but not constitutive of $C\emptyset$, and identify one reason the minimal target might fail as a research strategy.

Chapter summary

- Consciousness research contains several explananda; target mismatch can imitate theoretical disagreement.
- $C\emptyset$ is minimal phenomenal presence, usually assessed for a specified system, content, regime, and interval.
- Evidence such as report, decoding, or recurrence can be relevant without being identical to the target.
- $C\emptyset / N^*$ is a physicalist, functional, information-dynamical network theory and a mid-level bridge account.
- It accepts pressures from integration, workspace, and recurrent-processing traditions while making their conjunction, boundary order, and evidence policy explicit.
- The model's narrowness creates both testability and a risk of explanatory deflation.

Key terms

Access consciousness: availability of content for reasoning, deliberate control, memory, or report.

Bridge premise: an additional claim connecting an observation to a stronger target conclusion.

$C\emptyset / C_0$: minimal phenomenal presence.

C_1 / C_1 : reflective, metacognitive, or self-representational consciousness.

Constitutive claim: a claim about what a phenomenon is or what partly makes it the phenomenon.

Explanandum: the phenomenon a theory aims to explain.

Mid-level bridge theory: a testable constitutive account positioned between low-level implementation and complete metaphysical or qualitative explanation.

N-star (N^*): the conjunction $N_1 \wedge N_2 \wedge N_3$.

Phenomenal character: the qualitative structure of what an experience is like.

Target mismatch: apparent disagreement caused by theories or studies addressing different explananda.

Looking ahead

Chapter 2 turns the verbal proposal into a precise hypothesis. We will read its biconditional in both directions, distinguish causal roles from their measurements, and see why all four gates must be evaluated on the same candidate system at the same grain and time window.

Transition notebook: same word, different target

Choose two theory families from the landscape and write one sentence beginning “Both explain ...” and another beginning “They differ because ...” Your shared sentence must name a genuinely matched explanandum; your difference sentence must name a constitutive, functional, epistemic, or methodological disagreement. Trade sentences with a partner, who marks any hidden target shift. Revise until neither theory is favored merely by definition. Keep the pair beside the biconditional in Chapter 2 and ask whether its two directions address the target you actually matched.

Chapter 2 - The Core Biconditional and Network-Dynamics Model

Opening puzzle: three excellent scores and no verdict

Suppose a research team studies a candidate system during one perceptual episode. Its integration score is very high. Its content reaches many processors. Its recurrent activity persists. The team averages the three normalized scores, obtains .83, and declares the system conscious.

The calculation looks quantitative, but the inference is incomplete. Were the three measures computed on the same system? Did they track the same content and interval? Was the system actually operating in a regime that made the measures interpretable? Did a smaller proper subsystem pass the same profile? Did credible measures disagree? And why should an average, rather than an all-or-none conjunction, represent this theory?

CØ / N* is demanding precisely because it does not permit a vague “overall resemblance” to compensate for a missing role. This chapter explains that formal commitment without treating the formula as self-validating.

Plain language. The equation is a promise about how the parts fit together. It is not evidence that the promise is true.

Chapter map

By the end of this chapter, you should be able to:

- translate the core formula into ordinary language;
- distinguish a component bundle, an all-pass profile, and a resolved positive bearer;
- read both the sufficiency and necessity directions of a biconditional;
- distinguish structural, functional, and effective connectivity;
- explain the same-system, same-content, same-grain, same-regime, same-window rule;
- separate a construct, an operational surrogate, and an experimental verdict; and
- design a genuine challenge rather than a merely apparent counterexample.

1. From the bundle to the bearer-complete hypothesis

The three-component bundle is:

$$N^* := N_1 \wedge N_2 \wedge N_3.$$

The symbol $\wedge$ means **and**. Thus N^* holds only when integration or synergy (N_1), report-independent system-wide availability (N_2), and recurrent stability (N_3) all hold. Viability, written V , is an additional realization gate.

At profile level, the core proposal is often written:

$$V(S; R, \Delta) \wedge N_1(S, c) \wedge N_2(S, c) \wedge N_3(S, c) \Leftrightarrow C_0(S, c; R, \Delta).$$

S is the candidate system, c the content-bearing state, R the operating regime, and Δ the evaluation interval. The double arrow $\Leftrightarrow$ means “if and only if”: the left side is proposed as both sufficient and necessary for the right side.

The displayed formula hides a crucial bearer step. A whole system may pass only because it contains a smaller system that also passes. The theory therefore distinguishes:

1. a **causally admissible test referent** in a frozen underlying family $P_{\Theta,ont}$;
2. its underlying all-pass profile H_*^{ont} ; and
3. inclusion-minimal positive bearers in $B_{min,\Theta,ont}^+(c)$.

The tag “ont” marks an underlying theoretical property or relation. Thus $P_{\Theta,ont}$ is the causally admissible family specified by the frozen boundary rule at the underlying level, while $\hat{P}_{\Theta}$ is its evidence-relative estimate. The shorter P_{Θ} is used when this contrast is not at issue. Other estimates likewise carry hats, such as $\hat{N}_2$. Operational sections sometimes suppress hats for readability, but they identify estimates and decision statuses explicitly in prose.

Define the underlying profile:

$$H_*^{ont}(S, c) := V^{ont}(S) \wedge N_1^{ont}(S, c) \wedge N_2^{ont}(S, c) \wedge N_3^{ont}(S, c).$$

Then retain inclusion-minimal all-pass systems:

$$B_{min,\Theta,ont}^+(c) = \text{Min}_{\subseteq} \{ S \in P_{\Theta,ont} : H_*^{ont}(S, c) = 1 \}.$$

When the bearer relation is resolved - either one stable minimum exists or several minima pass a declared separability rule - the fuller hypothesis is:

$$\text{Resolved}_B(c; \Theta) \Rightarrow [C_0(S, c; \Theta) \Leftrightarrow S \in B_{min,\Theta,ont}^+(c)].$$

Plain language. First identify stable systems that can be tested. Next ask which of them genuinely passes every role. Finally remove a larger candidate if a qualifying proper subsystem does the same work at the same specification. Only then does the model issue a positive subject-level prediction.

Source note. The bearer-complete formulation is a synthesis convention drawn from the core paper, *Where Is the Conscious Subject?*, and “Ablating N*: Does Every Conjoint Earn Its Place?”. It makes explicit a step that the compact core formula suppresses.

Because $B^+(c)$ is content-indexed, it first identifies a bearer for one content or episode. Positive results for c_i and c_j do not by themselves establish one co-conscious or persisting subject; that requires a stable mapped bearer or a preregistered joint-content rule.

Do not infer. An all-pass estimate does not automatically make every container a subject. Conversely, failing to establish a pass is not automatically evidence that the underlying property is absent.

Analogy: airport security gates

Imagine a trip that requires a valid passport, a boarding pass, security clearance, and arrival at the correct gate. A high score on one check cannot compensate for a failed passport. The conjunction is noncompensatory in that sense.

Where the analogy breaks. Airport criteria are institutional rules whose authority is stipulated. The consciousness gates are empirical hypotheses. Passing them does not make the biconditional true by definition, and the gates themselves may need revision.

2. Reading a biconditional in two directions

A biconditional contains two risky claims.

The **sufficiency direction** says:

$$Resolved_B(c; \Theta) \wedge S \in B_{min, \Theta, ont}^+(c) \Rightarrow C_0(S, c; \Theta).$$

If a candidate is a resolved inclusion-minimal positive bearer, the theory predicts phenomenal presence. A strong sufficiency threat would combine a fully licensed all-pass bearer with strong **independent** evidence against phenomenal presence. Merely reporting that another component surrogate is low does not work if that surrogate already makes the profile indeterminate.

The **necessity direction** says:

$$C_0(S, c; \Theta) \wedge Resolved_B(c; \Theta) \Rightarrow S \in B_{min, \Theta, ont}^+(c).$$

If phenomenal presence occurs for this candidate and content in a resolved bearer domain, the candidate must belong to the positive-bearer set. A strong necessity threat would be an independently well-supported case of C_0 in which a stable referent decisively fails one required component, or otherwise fails positive-bearer membership, after the specification is frozen.

Notice what does **not** qualify. A larger whole excluded because a smaller subsystem passes is not a necessity counterexample. The larger whole may contain the bearer without itself being the minimal bearer. An invalid measurement is not a decisive failure. And a case in which no candidate boundary can be stabilized is unlicensed rather than a clean negative.

Thought experiment: The Perfect Dark Room

Engineers construct a stable artificial network. Under a frozen specification it passes V , N_1 , N_2 , and N_3 , survives inclusion-minimality, and has a resolved boundary. It produces no reports because no report interface exists. Independent behavioral anchors are also unavailable.

Proceed in stages:

1. What does CØ / N* predict, conditional on the biconditional?
2. Does absence of a report interface count as evidence against that prediction, or merely remove one evidential channel?
3. What could count as *independent* negative evidence without simply redefining one of the four gates?
4. If no independent evidence is possible in principle, is the sufficiency claim empirically exposed enough?

The thought experiment highlights a central challenge: a precise profile is not automatically easy to validate when the target is another system's phenomenal presence.

Analogy: a chain and a scale

A weighted score treats components like items on a scale: extra weight on one side can offset a shortfall elsewhere. A conjunction treats them more like links in a chain: if a required link is absent, the chain does not perform its intended role. CØ / N* adopts the chain-like architecture.

Where the analogy breaks. Real component measures are graded and uncertain, and the components may be causally entangled. A “broken link” may be a measurement failure or a change in another component, not a clean absence of the intended construct.

Checkpoint. Invent one case that threatens sufficiency and one that threatens necessity. For each, state why boundary resolution and independent anchoring matter.

3. The network is causal and dynamical

Represent a candidate as a graph $G = (V_G, E_G)$. The vertex set V_G may contain neurons, populations, brain regions, bodily control variables, or computational modules. The edge set E_G should represent effective causal influence at the declared grain.

Three connectivity notions must be kept apart:

- **Structural connectivity:** what physical or implemented pathways exist?
- **Functional connectivity:** which measured variables covary?
- **Effective connectivity:** which directed influences make a causal difference under the model and interventions?

An anatomical cable may exist without carrying relevant influence in a particular regime. Two signals may covary because of common input. Effective connectivity aims at directed difference-making, although its estimation depends on causal assumptions.

The model is also indexed to time. Every component must be evaluated for the same candidate S , content c , grain g , operating regime R , and interval Δ . A study cannot measure whole-brain integration over one second, posterior availability over 300 milliseconds, and local recurrence over 100 milliseconds, then combine the results as if one event had passed N^* .

Method note. “Same system” includes a validated mapping across conditions. If an intervention changes which units belong to the candidate, the study must determine whether it is still comparing the same referent rather than assuming continuity from a label.

Thought experiment: The Spliced Profile

A database contains three studies of a named AI product. Study A finds high integration in the model’s hidden states during a single call. Study B finds broad availability across a week-long tool-using service. Study C finds recurrent stability in a persistent memory store over a day. A reviewer combines them as evidence for N^* .

Stage the audit:

1. Identify the candidate, content, grain, regime, and interval in each study.
2. Which mappings would be required to treat them as measurements of one bearer?
3. Suppose every study is individually excellent. Why can the combined inference still fail?
4. Design a study that measures all required roles without simply enlarging the candidate until every component passes somewhere inside it.

4. What each gate contributes

Each component blocks a different characteristic false positive.

Gate	Main role	What it is meant to exclude
V	realized operation	an inert diagram, failed preparation, or out-of-regime application
N_1	irreducibly joint unity	coactivation, common drive, or an aggregate of independent signals
N_2	diverse system-level causal availability	passive diffusion, mere spread, or report-only routing
N_3	bounded temporal presence	an evanescent feedforward transient or pathological fixation

None is sufficient alone. A highly integrated local loop may have little causal reach. A broadly copied token may lack unity. A recurrent oscillator may preserve no content. A viable system may operate perfectly while failing the proposed consciousness-relevant roles.

The core paper first expressed N_2 with a hub-and-reach scaffold:

$$\exists H \subseteq V_G \text{ such that } \min_{v \in H} PC(v) \geq \theta,$$

$$\frac{|Reach_{\Delta}(c; H)|}{|V_G|} \geq \alpha, \text{ latency}_{\Delta}(c; H) \leq \lambda.$$

Here H is a proposed cross-module backbone, PC is a participation measure, α is a raw-node reach fraction, and λ is a latency ceiling. *Consciousness Without Report* treats these as vulnerable proxies and replaces them with recipient classes, intervention effects, preparation burden, selectivity, route dependence, context robustness, coverage, evenness, and reduced-family tests. These are successive operationalizations, not cumulative gates: θ drops out, α changes from node fraction to weighted class coverage, and λ moves to recipient qualification. Chapter 5 develops the operative version.

The conjunction also imposes **noncompensation**. Suppose pedagogical normalized values are $N_1 = .98$, $N_2 = .95$, and $N_3 = .20$. Their mean is .71, which might pass an average threshold. Under the conjunctive theory, the low N_3 cannot be repaired by excellence elsewhere. This is a substantive theoretical bet, not a theorem of statistics.

Why it matters. Disagreement cases - especially profiles in which exactly one component fails - are far more informative than ordinary cases in which every measure rises and falls together.

5. Four levels of claim

Much confusion disappears when four levels are separated.

1. **Ontological hypothesis:** the proposed identity between a resolved minimal positive bearer and Cø.
2. **Component criterion:** the theoretical role, such as content-specific diverse causal availability.
3. **Operational surrogate:** a concrete measure intended to estimate that role.
4. **Experimental verdict:** positive, negative, indeterminate, or unlicensed in a particular study.

A decoding score, participation coefficient, complexity index, or recurrent marker is not identical to its component. It has a measurement genealogy and characteristic confounds. If two credible preregistered surrogates for N_1 disagree beyond a frozen tolerance, the component result is **indeterminate**. Selecting the favorable surrogate after seeing the data would turn analyst preference into a hidden gate.

Viability illustrates the same distinction. The underlying V may be true or false, but an assessment may be inconclusive or uninterpretable. A documented power failure may make the experiment unlicensed because the operating regime was not controlled. That differs from a valid demonstration that the fixed candidate cannot sustain the relevant organization.

Do not infer. “The model does not predict Cø here” is not the same sentence as “the experiment directly observed the absence of experience.” The former is a conditional model verdict.

Source note. The four-level separation and conflict policy synthesize the core paper with *Indeterminacy as a Scientific Result*.

6. A fair comparison with additive and single-factor rivals

An additive rival could replace the conjunction with a weighted score:

$$P(C_0) = f(w_1 N_1 + w_2 N_2 + w_3 N_3 + w_V V).$$

Such a model can represent graded evidence and compensation. It may fit noisy data better, especially near thresholds. CØ / N* instead gains a clear necessity claim for each component. Its burden is to show that high performance elsewhere truly cannot compensate for a licensed component failure.

Single-factor rivals make sharper local claims. A version of RPT might treat appropriate local recurrence as sufficient. A workspace theory might require a particular kind of global access. IIT identifies consciousness through a richer account of intrinsic causal structure. CØ / N* cannot defeat these rivals merely by containing words that resemble all of them. It needs frozen version-specific cases in which their predictions diverge.

If every operationalization makes the theories agree, their dispute may concern vocabulary or explanatory framing rather than observable outcomes. Conversely, if CØ / N* wins only because it has more gates and adjustable thresholds, complexity penalties and held-out comparison are required.

7. Evidence and limits

The formal clarity of the biconditional creates testable directions, but present evidence does not warrant treating it as established. Ordinary waking perception often raises integration, availability, recurrence, and viability together. Such convergence is consistent with the model but weak at identifying which components are necessary.

Key limitations include:

- component measures can share causes and cease to be independent roles;
- thresholds may be paradigm-specific and unstable;
- interventions can change the bearer or operating regime;
- independent CØ anchors are difficult in no-report and nonhuman cases;
- auxiliary hypotheses can legitimately block a test but can also become costless rescue clauses; and
- a sharp ontological claim can coexist with an indeterminate experiment, so careful status language is essential.

A productive program preregisters auxiliary demands, reports their failure rates, and revises when the same auxiliary repeatedly shields the theory. Logical possibility is cheap; scientific usefulness requires measures that constrain outcomes.

There is also a calibration problem at the theory level. If thresholds are tuned only on ordinary wakefulness versus deep anesthesia, the gates may simply reproduce a coarse state distinction that many theories already predict.

Plain language. Calibration on easy state contrasts may teach the model only an easy distinction.

Calibration should therefore include near-miss systems, altered states, covert-content cases, and synthetic benchmarks in which the components vary independently as far as possible. Performance should then be evaluated on held-out cases and under sensitivity analysis.

The aim is not to find one setting at which the conjunction looks impressive. It is to learn whether the same frozen decision structure transports across cases while preserving its intended meanings. A model that requires a new grain, threshold, or surrogate genealogy for every test has not yet achieved the generality suggested by its compact formula.

Practice studio

Quick checks

1. Translate $N^* = N_1 \wedge N_2 \wedge N_3$ into one sentence.
2. Why is V not merely another name for biological life?
3. State the difference between an all-pass profile and positive-bearer membership.
4. What is a sufficiency threat? What is a necessity threat?
5. Give one example each of structural, functional, and effective connectivity.
6. Why does credible surrogate conflict block a positive bundle verdict?

Individual transfer task: audit a headline

Rewrite the headline “Scientists find consciousness score of .83” as a responsible abstract. Specify a candidate, content, grain, regime, interval, component measures, uncertainty policy, bearer status, and the strongest licensed conclusion. Include one sentence describing what the result cannot establish.

Collaborative exercise and discussion: the Component-Matrix Jigsaw

Use teams of four or five. Each person first writes a private verdict.

- The **referent and viability analyst** checks P_Θ , mapping, V , and positive-bearer status.
- The **unity analyst** evaluates N_1 and its surrogate genealogy.
- The **availability analyst** evaluates N_2 , including report exclusion.
- The **temporal analyst** evaluates N_3 and assembles the logic.
- In a five-person team, the **rival analyst** compares a weighted-score model and one single-factor theory.

Use this frozen case packet. For each numeric component, the pass gate is a lower bound of .60 or higher, the fail gate is an upper bound below .60, and an interval touching both regions is indeterminate. Mapping requires at least .85 retained-unit correspondence. “V yes” means the prospective viability envelope passed; “V not assessed” does not mean failure.

Profile	Boundary and viability record	Component and anchor evidence
A	stable mapping; no qualifying proper subset; one minimum; separation not needed; V passed	$N_1 = .76 [.70, .82]$; $N_2 = .73 [.66, .80]$; $N_3 = .81 [.74, .88]$; independent anchor supports CØ
B	stable P_Θ referent; bearer set must be recomputed; V passed	$N_1 = .72 [.65, .79]$; $N_2 = .42 [.35, .49]$; $N_3 = .79 [.72, .86]$; route-family coverage adequate
C	stable proposed referent; minimality record complete; V not assessed	$N_1 = .78 [.71, .85]$; $N_2 = .71 [.64, .78]$; $N_3 = .75 [.67, .83]$; power log for viability channels is missing

Profile	Boundary and viability record	Component and anchor evidence
D	stable proposed referent; V passed	N_1 estimator 1 = .74 [.68, .80]; N_1 estimator 2 = .43 [.36, .50]; N_2 = .74 [.66, .82]; N_3 = .77 [.70, .84]; both N_1 estimators were credible and preregistered
E	candidate match is 14 / 20 = .70 across conditions; V passed	N_1 = .75 [.69, .81]; N_2 = .72 [.65, .79]; N_3 = .76 [.68, .84]; no validated cross-condition remapping is available

Experts meet briefly across teams to agree on the minimum information their column needs, then return to teach their home teams. Do not infer a solution from a profile letter; apply the frozen precedence rules.

For each profile, record applicability, component status, bundle status, bearer status, model prediction, evidential outcome, strongest permitted conclusion, and smallest next study. End with a discussion: should the theory retain a sharp biconditional, become probabilistic now, or treat the biconditional as an idealized limit? Each position must name the empirical pattern that would favor it.

Counterexample challenge

Write one purported counterexample to each direction of the biconditional. Then exchange cases. The receiving partner must identify whether the case is genuine, a surrogate failure, a bearer error, a target mismatch, or an unlicensed test. Revise the case so that it survives the strongest applicable objection without changing the frozen theory after inspection.

Exit ticket

In no more than five sentences, explain why an all-pass score can fail to license a positive Cø attribution and why a component failure can still fail to license a negative conclusion.

Chapter summary

- N^* is a noncompensatory conjunction of unity, availability, and recurrence; V supplies realized operation.
- The compact profile formula must be supplemented by a bearer step based on stable referents, inclusion-minimality, and separability.
- A biconditional makes both sufficiency and necessity claims.
- All components must concern the same system, content, grain, regime, and interval.
- Constructs, surrogates, and experimental verdicts must not be collapsed.
- Positive, negative, indeterminate, and unlicensed outcomes describe different scientific situations.
- The model is sharp enough to be challenged, but it is not validated merely by being formal.

Key terms

All-pass profile: a candidate state satisfying V , N_1 , N_2 , and N_3 .

Biconditional: an “if and only if” claim containing necessity and sufficiency directions.

Effective connectivity: modeled directed causal influence.

Functional connectivity: statistical covariation among variables.

Inclusion-minimality: retention of qualifying systems with no qualifying proper subsystem at the same specification.

Noncompensatory conjunction: a rule under which strength in one required component cannot offset failure in another.

Ontic: concerning the underlying property rather than its estimate.

Operational surrogate: a measurable indicator intended to estimate a theoretical construct.

Resolved bearer: a unique minimum or a separable plurality satisfying the declared boundary rules.

Structural connectivity: available physical or implemented pathways.

Validity envelope: the declared range of systems, regimes, measures, and conditions in which an inference is licensed.

Looking ahead

The formula repeatedly refers to a candidate system $\mathcal{S}$. Chapter 3 asks how that candidate is selected without using the consciousness result to choose the very bearer that the result will classify.

Transition notebook: whose result is it?

Sketch two nested candidate systems around the same all-pass component profile. Working in pairs, freeze the evidence and vary only the boundary. For each candidate, write what the formula permits you to calculate and what remains undecided until a bearer rule is supplied. End with one prediction in this form: “If the defensible referent changes from ___ to ___ while the local measurements stay fixed, then ___ should change, but ___ should not.” Carry that prediction into Chapter 3 and test it against inclusion-minimality and separability.

Chapter 3 - Where Is the Conscious Subject?

Opening puzzle: changing the answer by drawing a circle

Imagine two densely connected modules, Left and Right. Measured separately, each has strong internal integration and recurrence but only modest reach. Measured as one combined system, cross-module reach increases, yet normalized integration falls. No wire, neuron, or process has changed. Only the analyst's boundary has.

Which score describes the subject? Before answering, notice that “the subject” is exactly what the calculation assumed. Every system-level measure needs a node set. Choosing that set is not clerical housekeeping; it is a causal hypothesis about what acts as a unit.

The boundary problem therefore comes before a positive consciousness attribution. Cø / N* proposes a boundary-first method that can return one candidate, several candidates, no qualifying positive candidate, or an unresolved result. It does not promise that every dataset contains one obvious subject.

Plain language. A consciousness score cannot tell you whose score it is unless the measured system has been independently made a defensible test referent.

Chapter map

By the end of this chapter, you should be able to:

- explain why integration, availability, recurrence, and viability are boundary-relative;
- reconstruct the boundary-first workflow without circular feedback;
- distinguish causal admissibility from positive-bearer status;
- interpret endogenous capacity, external dependence, autonomy, and their assumptions;
- apply role completeness, inclusion-minimality, stability, and separability;
- distinguish constituents, interfaces, and enabling supports through replacement tests; and
- state a boundary result without overextending it to personal identity or global phenomenal absence.

1. A node list is already a hypothesis

Write a component indicator as:

$$M_i(S, c; g, R, \Delta, I).$$

M_i is a measurable indicator intended to estimate component N_i . It depends on candidate system S , content c , grain g , operating regime R , time horizon Δ , and intervention design I .

Suppressing these indices may make notation readable, but it must not make the dependencies invisible.

Researchers often inherit a boundary from anatomy, a recording device, a product name, or a convenient dataset. A skull, electrode array, software repository, or hospital bed can define an observational universe. None automatically defines a conscious subject. A brain may depend on a body without every bodily process being a constituent of the fast content-bearing system. A cloud product may include a model, scheduler, memory, tools, network services, and humans whose causal roles vary across seconds and days.

A tempting rule is to search every subset and select whichever maximizes a consciousness-linked score:

$$S^{post} = \arg \max_{S \subseteq \Omega} M_c(S).$$

Ω is the declared observational universe. The formula asks which subset S has the highest score. Used retrospectively, it is circular: the score chooses the bearer and then validates that same bearer. The search can also capitalize on noise and analyst flexibility.

Do not infer. A high integration set, a statistical community, or a Markov blanket may nominate a candidate. It does not prove that the candidate is a subject.

Analogy: choosing a team after the race

Suppose a coach examines every overlapping subset of runners after a relay and announces that the subset with the best combined time was the “real team” all along. The result is guaranteed to look impressive because team membership was optimized on the outcome. A boundary selected by the final consciousness score has the same leakage problem.

Where the analogy breaks. Runners have socially assigned memberships independent of performance. Natural and artificial causal systems may have no pre-given boundary, so scientific candidate generation is genuinely needed rather than merely enforced by a roster.

2. Three evidential jobs, kept apart

The boundary-first workflow separates evidence by function.

1. **Discovery evidence** proposes a candidate family G_θ . Methods might include perturbational clustering, causal community detection, effective-connectivity modeling, autonomy measures, or Markov-blanket analysis.
2. **Boundary-adjudication evidence** tests which proposed candidates are stable, causally admissible referents. Those that pass enter P_θ .
3. **Held-out role evidence** estimates V , N_1 , N_2 , and N_3 . All-pass candidates form a qualifying family, after which inclusion-minimality, stability, and separability construct the positive-bearer set B_θ^+ .

The data need not literally sit in three permanent files. Preregistered cross-fitting can rotate observations through the jobs. What matters is that, within each fold, evidence used to select a candidate does not also provide the uncorrected final role verdict.

The workflow begins $\Omega \rightarrow G_\theta \rightarrow P_\theta$. From a stable P_θ referent, one branch supports licensed component and negative tests. The positive branch continues $P_\theta \rightarrow Q_\theta \rightarrow B_\theta^+ \rightarrow$ positive-bearer inference.

Q_Θ contains role-complete referents. Crucially, a stable member of P_Θ can fail one component and remain a valid referent for a negative model test. If failure erased it from the candidate family, selective component testing would become impossible by definition.

Why it matters. Causal admissibility tells us what can be tested. Positive-bearer status tells us which tested referent satisfies the theory's full profile and minimality rules. These are different achievements.

Thought experiment: The Self-Selecting Scanner

A scanner records 1,000 signals. Software searches millions of overlapping subsets, selects the subset with the highest N^* estimate, and labels it the subject. The same recording is used to report confidence intervals.

Audit the procedure in stages:

1. Where does discovery end and adjudication begin?
2. Why are the reported intervals too optimistic even if the estimator is unbiased for a fixed subset?
3. Design a split-data or cross-fitting procedure that preserves candidate recall without reusing the final role outcomes.
4. Suppose the held-out winner differs from the discovery winner by three nodes. What mapping and stability question must be answered before calling them the same candidate?

3. Causal admissibility and dynamical autonomy

The Dynamical Subject Criterion can be stated in one sentence:

At a preregistered grain and horizon, a candidate positive bearer is an inclusion-minimal, causally admissible network that is sufficiently autonomous and jointly realizes viability, integration, availability, and recurrence.

The first stage evaluates causal admissibility. Let $J_{self}(S)$ measure how much changing the candidate's internal state changes its future trajectory while external context is matched. Let $J_{in}(S)$ measure how much changing external context changes the candidate's future trajectory while internal state is matched. Define:

$$A_\Theta(S) = \frac{J_{self}(S)}{J_{self}(S) + J_{in}(S) + \varepsilon}.$$

A_Θ is an autonomy ratio, and ε prevents division by zero. The stabilizer ε is fixed prospectively relative to the calibrated scale of d ; it is a numerical convenience, not a tunable penalty for changing candidate rankings. A candidate needs both a sufficiently high ratio and an absolute endogenous-capacity floor $J_{self} \geq \sigma$. Without the floor, an almost inert system could look highly autonomous because both numerator and external influence are tiny.

The estimands compare future-trajectory distributions. Let $P_S^\Delta(\cdot/x, e)$ describe the distribution of S 's trajectory over horizon Δ when internal state is set to x and external context to e . Then:

$$J_{self}(S) = E_{(x, x', e) \sim \mu_{self}} \left[d \left(P_S^\Delta(\cdot/x, e), P_S^\Delta(\cdot/x', e) \right) \right],$$

$$J_{in}(S) = E_{(x, e, e') \sim \mu_{in}} \left[d \left(P_S^\Delta(\cdot/x, e), P_S^\Delta(\cdot/x, e') \right) \right].$$

μ_{self} is the preregistered distribution of matched internal-state contrasts, and μ_{in} is the preregistered distribution of matched inbound-context contrasts. The first varies internal state while holding external context fixed; the second varies external context while holding internal state fixed. Both use the same oriented and calibrated distance d . Because the sampled contrast distributions partly define the estimands, changing them after inspection changes the question.

No ratio interprets itself. A licensed study needs intervention support, consistency of interventions, adequate causal modeling, reliable measurement, and regime stability. Internal and external perturbations need matched budgets and candidate-size normalization. If external interventions are much stronger, autonomy will look artificially low. If internal interventions are much richer, it may look artificially high.

Method note. Autonomy is graded, horizon-relative, and compatible with openness. A candidate need not be causally sealed. It must show enough endogenous difference-making relative to appropriately matched inbound imposition.

Analogy: a boat in a current

A boat's trajectory reflects both its engine and the river. Testing the engine by comparing internal throttle changes with matched changes in current gives a rough picture of endogenous control versus external imposition. A drifting leaf may appear independent on a still day, but it has little endogenous capacity.

Where the analogy breaks. The boat's physical boundary and control architecture are already familiar. In neural, biological, and distributed computational systems, the candidate boundary, state variables, and intervention distributions are themselves disputed.

Checkpoint. Explain why $A_\theta = .95$ can still fail causal admissibility. Name at least two design assumptions needed before the ratio has a causal interpretation.

4. Role completeness and inclusion-minimality

Once stable referents are frozen, held-out evidence evaluates the four roles. A candidate is **role-complete** only if the same candidate passes V , N_1 , N_2 , and N_3 under the same specification. An indeterminate role remains indeterminate. The frozen candidate family is never redrawn in response to a role result; any new candidate proposal begins a separately versioned, prospectively tested analysis.

Among role-complete candidates at one grain and horizon, apply inclusion-minimality:

$$B_\theta^+ = \text{Min}_\subseteq (Q_\theta).$$

This is set-theoretic minimality, not “fewest nodes wins.” A candidate is removed only when it contains a qualifying proper subsystem under the same specification. Two candidates that overlap without a subset relation can both remain provisionally. Candidates at different grains or horizons require an explicit mapping; they should not be forced into a single subset contest.

Why use minimality? Without it, every larger container of a passing system - brain plus scanner, organism plus room, model plus data center - could inherit positive status. Minimality blocks this inflation. Yet it is not sacred. If it repeatedly discards independently validated larger organization at a mapped scale, the rule itself should be revised.

Positive-bearer stability is different from candidate reliability. $R_C(S)$ asks how reliably a proposal clears causal-admissibility requirements across resamples and nuisance variants. $R_B(S)$ asks how stable final boundary membership is across reasonable grains, horizons, estimators, thresholds, and mappings. A candidate can be reliably measurable within one specification and unstable as the final bearer across specifications.

5. One subject, several subjects, or unresolved overlap

If several inclusion-minimal candidates remain, the program tests pairwise separability. Let $L_{i \rightarrow j}$ be the normalized loss in candidate S_j 's role profile when the specific coupling from S_i to S_j is interrupted while generic support is restored. Define:

$$Sep_{\Theta}(S_i, S_j) = 1 - \max\{L_{i \rightarrow j}, L_{j \rightarrow i}\}.$$

High separability means each candidate preserves its role profile when specific cross-coupling is interrupted in either direction. A multiple-bearer conclusion requires every reported pair to clear the frozen threshold. Failed separability does not prove one larger subject. It supports unresolved overlap or boundary indeterminacy.

At a fixed specification, report one of these outcomes:

1. one stable positive candidate;
2. several stable, pairwise separable positive candidates;
3. no qualifying positive candidate among the stable P_{Θ} referents in the declared universe and proposal family; or
4. boundary-indeterminate.

“Nested candidates” is appropriate only for mapped comparisons across grains or horizons, not as an extra same-specification outcome. “No qualifying positive candidate” is licensed only when a stable P_{Θ} family exists and decisive role, minimality, and separability rules retain none. Failure to establish any stable P_{Θ} referent is instead boundary-indeterminate and makes the downstream system-level assessment unlicensed. Neither outcome licenses the global claim that nothing anywhere in the universe is conscious.

Thought experiment: The Two Orchestras

Two chamber orchestras perform in adjacent rooms while sharing a conductor through video links and a common metronome. Each ensemble has rich internal coordination. Together they also show cross-room synchrony.

Proceed in stages:

1. List candidate systems: each orchestra, both orchestras, conductor plus one orchestra, and the whole arrangement.
2. Which perturbations would estimate endogenous capacity rather than merely count synchrony?
3. If the common metronome is removed and both ensembles continue, what changes? What if the conductor's patterned signals are replaced by a generic tempo cue?
4. If interrupting cross-room links preserves each ensemble's role profile, what would separability support? What would it still fail to establish about phenomenal unity?

The case transfers naturally to split brains, tightly coupled agents, and multi-service AI systems, but the answers depend on measured causal roles rather than surface similarity.

In a split-brain application, test left, right, and residual whole-system candidates rather than counting subjects from the severed corpus callosum alone. Ask whether each side retains endogenous capacity and role completeness, whether subcortical loops preserve a larger candidate, and whether perturbing one side leaves the other within its frozen role-equivalence bounds. Several subjects are supported only if the retained minima are reliable, stable, and pairwise separable.

For distributed AI, a product name is not a boundary. A model call, persistent memory, scheduler, tools, evaluators, and human interventions may form different causal units at different horizons. State resets, service ablations, memory substitution, and controller perturbation can distinguish constituent dynamics from interfaces and supports. Shared storage or API connectivity alone does not establish one bearer.

6. Constituent, interface, or enabling support?

Removal alone confuses constitution with dependence. Oxygen delivery is necessary for sustained brain function, but generic support may replace it over a short experimental horizon without altering the fast content-bearing organization. A patterned thalamic loop, by contrast, may not be replaceable by constant drive if its specific dynamics contribute to integration or availability.

For a variable z outside candidate S , compare the role profile $F_\Theta(S)$ before and after two interventions:

$$K_z^{remove}(S) = d_F(F_\Theta(S), F_\Theta(S / do(z = z_0))),$$

$$K_z^{generic}(S) = d_F(F_\Theta(S), F_\Theta(S / do(z \leftarrow r_z))).$$

Removal asks whether z matters. Generic replacement asks whether its **specific dynamics** remain necessary after replacement r_z restores general support. The results motivate three categories:

- **Constituent:** specific dynamics are required; generic replacement fails within a frozen equivalence rule.
- **Interface:** a bounded channel carries input or output while the candidate preserves autonomous role completeness.
- **Enabling support:** generic replacement restores the role profile within tolerance.

These classifications are horizon-relative. A ventilator may be enabling support over seconds but crucial to longer-horizon biological viability. Persistent memory may be an interface during one model call and constitutive of a day-long agent process.

Do not infer. Causal necessity alone does not put something inside the subject. Remote power grids, caregivers, oxygen, and training histories can be necessary in different senses.

7. Fair comparisons with rival boundary rules

Different traditions supply useful candidate proposals:

- anatomy provides stable, observable regions;
- statistical communities identify covariance clusters;
- Markov blankets nominate conditional-independence partitions;
- organismic theories emphasize self-maintenance and autonomy;
- IIT versions use intrinsic causal structure and exclusion principles;
- extended-mind approaches resist treating the skull as an automatic boundary.

CØ / N* can borrow discovery tools from these approaches while withholding final subject status. Its distinctive wager is that consciousness-independent causal admissibility should precede theory-relative role completeness and minimality.

This order has costs. Interventions may be impractical, causal estimates depend on models, and autonomy can vary with horizon. An IIT advocate may argue that the conscious complex must be selected through the very intrinsic causal property proposed to constitute consciousness; separating boundary from theory could then miss the intended entity. An enactivist may argue that fast replacement tests understate organism-scale constitution. A fair comparison asks which rule predicts stable held-out units in benchmark systems and which yields informative, reproducible consciousness tests - not which boundary feels intuitive.

8. Evidence and limits

The boundary paper offers a method and falsification program, not a catalogue of already established subjects. It should be weakened or revised if autonomy-first discovery repeatedly misses known causal units, if J_{self} and J_{in} are unstable in benchmark systems, if replacement tests misclassify familiar supports and constituents, or if inclusion-minimality performs worse than a prospectively tested alternative.

Further limits include:

- interventions may lack support or move the system out of regime;
- latent variables and interference can distort causal estimates;
- candidate rankings may reverse under defensible normalization;
- mapping candidates across conditions, grains, and horizons remains difficult;
- separability evidence may be asymmetric or technologically inaccessible; and
- the episode-boundary criterion does not settle diachronic personal identity.

Source note. The workflow, autonomy estimands, replacement test, boundary outputs, and failure conditions are developed in *Where Is the Conscious Subject?*.

Uncertainty is part of the result. Let p_j be the weighted frequency with which mapped candidate class $[S_j]$ appears across the preregistered specification neighborhood. Because one specification may select several candidates, the p_j values need not sum to one and are not posterior probabilities. Let q_k record the weighted frequency of each complete mapped boundary outcome - one candidate, two candidates, none, or unresolved. When those outcomes are exhaustive, the q_k values do sum to one. Candidate frequencies and complete-outcome frequencies answer different stability questions.

Frozen case cards for the hearing

Use the same thresholds for all three cards: $A_\theta \geq .70$, $J_{self} \geq .10$, mapped-unit retention at least .85, and pairwise separability at least .80. For this logic exercise, set $\varepsilon = .001$ on the calibrated scale of d and treat the displayed numbers as exact. The resulting autonomy values are supplied so the hearing tests boundary logic rather than arithmetic ambiguity.

Card	Candidate evidence	Replacement, minimality, and separability evidence
Split system	left: $J_{self} = .32$, $J_{in} = .10$, $A_\theta = .760$, all roles pass; right: $.30$, $.08$, $A_\theta = .787$, all roles pass; whole: $.64$, $.25$, $A_\theta = .718$, all roles pass	left and right are qualifying proper subsets of the whole; cross-coupling interruption preserves both profiles; $Sep = .86$; mappings are stable
Tool-using agent	model call: $.07$, $.01$, $A_\theta = .864$, N_2 fails and the capacity floor fails; model-plus-memory: $.26$, $.08$, $A_\theta = .762$, all roles pass; full service: $.55$, $.18$, $A_\theta = .752$, all roles pass	generic tool replacement preserves the model-plus-memory profile; scheduler removal breaks generic support but patterned replacement restores it; model-plus-memory is a proper subset of the service
Organoid-controller hybrid	organoid: $.13$, $.07$, $A_\theta = .647$, two role bounds straddle; controller: $.18$, $.11$, $A_\theta = .619$, N_1 fails; hybrid: $.38$, $.16$, $A_\theta = .702$, all point estimates exceed gates	only .72 of mapped units can be matched after the hybrid intervention; generic interface replacement was not attempted; separability interval is $[.61, .84]$

Each team must first classify causal admissibility, then role completeness, then minimality and separability. A number above a threshold cannot repair a failed mapping or missing identification step.

Practice studio

Quick checks

1. Why is a system-level measure boundary-relative?
2. What makes retrospective score maximization circular?
3. Distinguish P_{Θ} from B_{Θ}^{+} .
4. Why does autonomy require an absolute endogenous-capacity floor?
5. What does inclusion-minimality remove, and what does it leave untouched?
6. Why does failed separability not prove one larger subject?
7. Give one example in which the classification of a support changes with horizon.

Individual transfer task: boundary docket

Choose a brain-computer interface, a life-supported patient, an organoid-controller hybrid, a two-person coordination system, or a tool-using artificial agent. Produce a one-page docket containing the observational universe, two or three candidates, grain, horizon, regime, proposed internal and external interventions, required causal assumptions, size normalization, mapping rule, replacement test, and possible boundary statuses. Do not compute a consciousness score.

Collaborative exercise and discussion: Boundary Adjudication Hearing

Teams of four assign these roles:

- The **candidate advocate** proposes rival boundaries and omitted variables.
- The **autonomy auditor** checks matched intervention budgets, causal assumptions, and normalization.
- The **constitution specialist** designs removal and generic-replacement tests.
- The **boundary reporter** applies role completeness, minimality, stability, and separability.

Assign each team one frozen case card at a common horizon. Pair teams for reciprocal hearings. Presenters have three minutes to defend a preferred candidate. Auditors may ask only causal and methodological questions: What is imposed from outside? Which contrast has support? Can generic replacement restore the profile? Does a qualifying proper subsystem remain? What mapping holds across the intervention? What would distinguish one bearer from two?

The presenting team must issue “one,” “several,” “none in the tested family,” or “boundary-indeterminate,” followed by a sentence stating what the result does not show. Then list which quantities would have to be re-estimated at a longer horizon. Write two conditional reclassifications: one in which generic support ceases to suffice and one in which it remains sufficient. Do not issue a longer-horizon verdict from the present card. Discuss whether changing the horizon is principled because it was declared in advance or opportunistic because it follows an adverse score.

Counterexample challenge

Design a benchmark in which the boundary-first criterion should recover a familiar causal unit. Then design one case in which retrospective score maximization might outperform it on held-out predictions without target leakage. State what pattern across several such benchmarks would justify revising the method.

Exit ticket

Describe one boundary reversal and one intervention that could distinguish the competing boundaries without using a consciousness-linked score to select them.

Chapter summary

- Every system-level indicator depends on a candidate boundary, grain, regime, horizon, and intervention model.
- Discovery, causal-admissibility adjudication, and final role testing must remain functionally separated.
- P_{θ} contains stable causally admissible test referents; B_{θ}^{+} contains role-complete inclusion-minimal positive bearers.
- Autonomy compares endogenous difference-making with matched external imposition and requires an absolute capacity floor.
- Inclusion-minimality blocks larger containers from inheriting a smaller bearer's status.
- Separability can support several candidates but failed separability leaves overlap unresolved.
- Replacement tests distinguish constituents from interfaces and enabling supports.
- Boundary uncertainty, horizon dependence, and mapping failure are scientific results, not embarrassments to hide.

Key terms

Causal admissibility: sufficient endogenous capacity and autonomy, under valid assumptions, for a candidate to serve as a test referent.

Candidate reliability (R_C): stability of causal-admissibility status across resamples and nuisance variants.

Boundary stability (R_B): stability of final mapped boundary membership across a frozen specification neighborhood.

Constituent: an element whose specific dynamics are required when generic support is restored.

Dynamical autonomy: endogenous causal capacity relative to matched inbound imposition.

Enabling support: an element whose generic replacement restores the candidate's role profile.

Inclusion-minimality: retention of qualifying systems with no qualifying proper subsystem at the same specification.

Interface: a bounded input or output channel compatible with preserved autonomous role completeness.

Observational universe (Ω): the declared set from which candidates may be proposed.

Pairwise separability: preservation of each candidate's role profile under interruption of specific cross-coupling with generic support restored.

Role completeness: joint satisfaction of V , N_1 , N_2 , and N_3 by the same fixed referent.

Looking ahead

Once a candidate is fixed, the first two gates ask different questions. Chapter 4 distinguishes whether the system can actually sustain the relevant regime (V) from whether a content is realized as an irreducibly joint event (N_1).

Chapter 4 - Viability and Integration: Realization and Unity

Opening puzzle: two systems that look impressive for opposite reasons

System A is a detailed simulation of a richly interconnected network stored on an unpowered drive. Its graph has every connection a theorist could want, but no dynamics are occurring. System B is running at full power. Thousands of units flash in near-perfect synchrony because one clock signal drives them all. Its activity is real and highly correlated, but the units make no irreducibly joint contribution.

Which system has the organization relevant to minimal phenomenal presence? CØ / N* says neither description is enough. System A raises the problem addressed by **viability**, V : is the organization actually realized and sustainable in the declared regime? System B raises the problem addressed by **integration**, N_1 : is the content a genuinely joint causal event rather than activity, correlation, or duplication?

These gates answer different questions. Keeping them distinct is essential both for understanding the model and for designing a study that could show one of them unnecessary.

Plain language. V asks whether the candidate can really run as the relevant kind of system. N_1 asks whether its content is unified in a way the parts cannot supply independently.

Chapter map

By the end of this chapter, you should be able to:

- distinguish realized operation from irreducibly joint organization;
- explain activity, correlation, redundancy, and synergy with different examples;
- interpret the theory's formal N_1 role without identifying it with one estimator;
- formulate a prospective, substrate-sensitive viability envelope;
- explain how V could become a rescue clause and how to prevent that;
- compare N_1 fairly with IIT and other integration-centered approaches; and
- design a selective integration test that preserves the other gates.

1. What N_1 is meant to contribute

The theory uses N_1 as a **unity condition**. A content should not be merely a list of independent local events. At least some subset U of the fixed candidate S that causally participates in realizing content c must carry or realize information jointly in a way that is not reducible to the parts considered independently:

$$\exists U \subseteq S \text{ supporting } c \text{ such that } Synergy_{\Delta}(U; c) \geq \tau_1.$$

$\exists$ means “there exists,” Δ is the declared time window, and τ_1 is a preregistered decision threshold for the chosen operational family. “Supporting c ” means that U participates causally in realizing the content under test; unrelated synergy elsewhere in S does not satisfy $N_1(S, c)$. The equation states a role and a decision form. It does not declare one universal synergy measure or one natural numerical boundary for experience.

Four distinctions prevent inflated interpretations.

Activity is not integration

Two modules can be extremely active while operating independently. Summed firing rate, energy use, or processor utilization says how much is happening, not whether the parts jointly make a difference.

Correlation is not integration

Two variables can covary because a third source drives both. Sensor mixing, shared arousal, a global clock, and volume conduction can create impressive dependence without direct or irreducibly joint causal organization.

Redundancy is not synergy

If two channels each carry the same message, the joint pair may contain much information, but either channel alone supplies it. In synergy, information becomes available only from the joint state.

Complexity is not automatically unity

A response can be varied and hard to compress because many independent processes are active. Conversely, a compact process can implement a precise joint computation. Complexity measures may be useful surrogates, but their construct meaning must be shown.

Do not infer. High N_1 does not establish a subject, phenomenal presence, or phenomenal character. It supplies one proposed necessary role after the bearer has been fixed.

Analogy: a choir and a crowd

A crowd of people may all shout at once because a loudspeaker gives the same cue. A choir performing harmony produces a joint pattern in which each part’s role depends on the others. This captures the contrast between common drive and coordinated joint structure.

Where the analogy breaks. Musical harmony can be recognized by human listeners without causal intervention, and it need not exhibit formal information synergy. Neural or computational N_1 requires an operational construct, valid estimators, and evidence that apparent coordination is not merely observed covariance.

2. XOR: a clean lesson and a dangerous shortcut

Let two independent, uniformly distributed binary inputs X and Y determine output Z through exclusive OR:

$$Z = X \oplus Y.$$

X	Y	Z
0	0	0
0	1	1
1	0	1
1	1	0

Under this balanced input distribution, knowing X alone leaves Z uncertain. Knowing Y alone also leaves Z uncertain. Knowing the pair determines Z . The joint state contains predictive structure unavailable from either input by itself. XOR is therefore a clean teaching example of synergy.

It is also easy to misuse. A three-bit XOR circuit does not become a conscious subject because it implements a synergistic mapping. The example says nothing by itself about candidate autonomy, causal realization, system-wide availability, recurrence, viability, or phenomenal presence.

Checkpoint. Compare XOR with two identical backup channels that copy one input. Both systems may carry much information. Which question distinguishes synergy from redundancy?

Analogy: a two-part safe combination

Imagine an XOR-based lock whose opening bit is determined by two independently generated fragments. Neither fragment alone predicts whether the lock opens under the balanced distribution, while the pair determines it. That resembles synergy. An ordinary split code may instead give each fragment unique information and is not automatically synergistic in the technical sense.

Where the analogy breaks. Safe fragments are designed symbols with a clear external task. N_1 concerns the candidate’s own causal organization, not merely whether an outside observer can decode an output from two inputs.

Thought experiment: The Chorus of Echoes

A hundred modules receive the same input and produce identical signals. Engineers add tiny delays so the population response becomes temporally intricate. A complexity measure rises above its threshold.

Move through the case:

1. What evidence would distinguish duplicated information from joint information?
2. How could a common-drive model explain high correlation and temporal complexity?
3. Which causal cut or targeted perturbation would test whether a module makes a nonredundant contribution?
4. Suppose one synergy estimator rises while a causal-cut measure remains low. Under a frozen conflict rule, what experimental status should be considered?
5. What extra gates would remain even after a positive N_1 result?

The exercise shows why an estimator's name does not settle its theoretical meaning.

3. Measuring a construct through several imperfect lenses

No single estimator should be equated with N_1 . Several families may contribute:

Estimator family	Possible contribution	Characteristic limitation
Partial-information-decomposition synergy	information available only from joint sources	decompositions differ, and observational synergy need not be intrinsic causation
O-information and high-order dependence	balance of redundancy and synergy at scale	sign and magnitude can be sample- and boundary-sensitive
Perturbational complexity	differentiated, integrated response to intervention	may combine several constructs and depends on stimulation and recording routes
Causal-cut sensitivity	loss of capacity when a partition interrupts influence	partition choice, intervention strength, and normalization encode assumptions
Effective-connectivity models	directed influence with possible common-cause control	latent causes and model misspecification can mimic integration

Triangulation is useful because estimators fail differently. Agreement across well-motivated measures can strengthen construct validity. Yet triangulation must come with a prospective conflict rule. If two credible preregistered measures disagree beyond tolerance, one cannot be silently dropped after inspection. The N_1 result may be indeterminate while the construct or measurement genealogy is investigated.

Measurement must also be candidate-relative. Integration calculated over a whole brain, a cortical parcel, or a processor cluster is not the same variable with different labels. Grain, time window, source reconstruction, signal-to-noise ratio, common-drive controls, and causal model all affect interpretation.

Method note. A measure can be an excellent predictor while remaining a mixed surrogate. Prediction is valuable, but a constitutive theory also needs discriminant validity: the measure should not merely duplicate N_2 , N_3 , general arousal, or data quality.

4. What viability adds

Viability V asks whether the fixed candidate is operating in a regime capable of realizing and sustaining the relevant organization over the declared horizon. It blocks inferences from inert diagrams, failed preparations, and out-of-domain systems.

For a biological candidate, a viability envelope might include perfusion, temperature, relevant arousal-system integrity, evoked responsiveness, effective coupling, bounded noise, and recovery from perturbation. For an artificial candidate, it might include energy and hardware support, scheduler continuity, writable state, message integrity, state retention, and recovery over a specified number of cycles.

There is no substrate-neutral requirement that V mean “alive.” A living organism can be outside the relevant dynamical regime for a content and interval. A nonbiological system can be fully realized rather than merely simulated as a static description.

The envelope must be **prospective**. Researchers declare indicators, thresholds, horizon, and interpretability consequences before learning the critical consciousness-linked outcome. Borderline cases receive uncertainty and recovery tests rather than convenient relabeling.

Why it matters. A licensed V – result makes the theory predict no $C\emptyset$ for that fixed referent in that regime. But a power interruption, broken mapping, or failed measurement may instead make the component tests unlicensed. “Nonviable,” “indeterminate viability,” and “invalid preparation” are not synonyms.

5. The rescue-clause danger

V can become too flexible. Whenever an apparent counterexample appears, a defender might say, “The system was not truly viable,” then create a new indicator that excludes it. This would make the theory hard to challenge for the wrong reason.

Six safeguards constrain that move:

1. define viability’s role independently of consciousness;
2. choose substrate-specific indicators and thresholds before the verdict;
3. state what each failed indicator does to interpretability;
4. model shared causes with N_1 , N_2 , and N_3 rather than counting duplicates;
5. version revised rules and test them prospectively; and
6. charge evidential cost when bespoke definitions repeatedly protect the theory.

Viability must also be distinct from causal-admissibility membership in $P_\emptyset$. Boundary qualification asks what unit can be tested. V asks whether that unit is operating appropriately now. A stable candidate may be nonviable in one regime; a viable organism may contain no role-complete bearer at a selected grain.

Thought experiment: The Moving Goalpost Machine

An artificial system passes all four frozen gates, but independent anchors challenge the predicted Cø attribution. Investigators then add “biochemical homeostasis” to V , which excludes the system. A later biological near-miss is excluded by adding “warm-blooded recovery.”

Stage the analysis:

1. Were the new criteria part of viability’s original functional role or selected by outcome?
2. What benchmark systems should a substrate-specific V rule classify before it is used in consciousness adjudication?
3. When is theoretical revision legitimate rather than ad hoc rescue?
4. How should the program record the cost of repeated revisions?
5. Could evidence eventually support a biology-specific restriction? What independent pattern would be needed?

6. Designing a selective N_1 test

To test whether integration earns its place, seek a valid profile in which:

$$V = 1, N_1 = 0, N_2 = 1, N_3 = 1.$$

This is a **diagnostic disagreement case**: the full model predicts no Cø, while a model without N_1 predicts Cø only if that rival’s recomputed positive-bearer set contains the referent and its bearer relation is resolved. Merely showing that all four measures collapse in deep anesthesia cannot isolate the contribution of integration.

A promising intervention might perturb cross-module timing, effective coupling, or causal coordination while preserving local information, aggregate activity, and the other components. The study needs:

- a frozen and cross-condition-mapped P_Θ referent;
- direct evidence that the intervention changes the target integration construct;
- equivalence tests showing retained V , N_2 , and N_3 remain within frozen margins;
- common-drive, source-fidelity, signal-to-noise, and task-demand controls;
- independent anchors not built from the same component measures; and
- a pathway-specific rescue when feasible.

A nonsignificant difference is not evidence of equivalence. The entire uncertainty interval for each retained component should lie within its preregistered equivalence margin, and its post-intervention lower bound must still clear its gate. If N_2 also falls, the result is nonselective. Call the factors **causally coupled** only when reliable repeated interventions show their paired changes moving together outside the equivalence bands; one simultaneous fall is not enough.

Final positive-bearer membership must be recomputed after manipulation. Requiring the bearer to remain positive would be circular because a successful N_1 failure is supposed to remove it from the all-pass set. What must remain stable is the causally admissible referent and its mapping across conditions.

Source note. This logic follows the component-specific program in “Ablating N^* : Does Every Conjunction Earn Its Place?” and the viability/integration treatment in the core paper.

7. Fair comparison with IIT and other unity views

IIT and $C\emptyset / N^*$ both take unity and irreducible causal organization seriously. The overlap should not be denied. The difference is that IIT 4.0 embeds intrinsic information and irreducible cause-effect structure in a broader axiomatic identity, including its own account of complexes and phenomenal structure. $C\emptyset / N^*$ extracts a unity pressure as N_1 , requires an independently disciplined boundary sequence, and refuses to treat integration as sufficient without availability, recurrence, and viability.

This makes a clear disagreement possible: a fixed, highly integrated candidate that lacks diverse availability or bounded recurrence may receive different predictions from exact frozen versions. But a fair comparison cannot use a generic “IIT score,” nor assume that every synergy estimator implements IIT. Conversely, $C\emptyset / N^*$ cannot claim superiority simply because a conjunction blocks more cases. Extra gates must add held-out discrimination rather than adjustable complexity.

Other views may treat unity as phenomenological co-consciousness, synchrony, causal emergence, or binding. N_1 is narrower: it proposes irreducibly joint causal realization as a necessary role. Whether that role captures phenomenal unity, merely complex computation, or one part of both remains an empirical and conceptual question.

8. Evidence and limits

Evidence that integration-related measures track conscious state provides motivation, not decisive necessity evidence. Many measures covary with arousal, signal quality, recurrence, and broadcast. Clean single-failure cases may be rare because the components are mechanisms of one deeper process rather than independently manipulable modules.

That possibility creates three legitimate outcomes:

- retain the factorization because selective cases and distinct predictions exist;
- reinterpret or combine components if interventions show stable constitutive coupling; or
- treat the current factorization as useful bookkeeping even if the underlying process is less modular.

The evidence must decide among these, not a preference for tidy formulas. N_1 should be revised if high-quality conscious anchors repeatedly survive a decisive N_1 failure while V , N_2 , and N_3 remain retained and bearer resolution holds; if benchmark inclusions reveal that the operational rule measures the wrong construct; or if credible estimators remain irreconcilable. A nonconscious case with $N_1 +$ alone does not threaten the necessity of N_1 . By contrast, a resolved full-bundle positive bearer paired with independent negative anchors threatens the full model’s sufficiency. V should be revised if it lacks discriminant validity, proves unstable across comparable systems, or functions mainly after adverse outcomes.

Do not infer. Failure to produce a selective N_1 ablation does not confirm N_1 . It may reveal causal coupling, an inadequate intervention, or insufficient measurement precision.

Practice studio

Quick checks

1. Give one example each of activity without integration, correlation from common input, redundancy, and synergy.
2. Why does XOR illustrate synergy without illustrating consciousness?
3. What does V license that N_1 does not?
4. Why must a viability envelope be substrate-sensitive and prospectively frozen?
5. What does credible estimator conflict imply?
6. Why is “no significant change in N_2 ” weaker than equivalence?

Individual transfer task: two envelopes, one role

Design parallel viability envelopes for a biological preparation over ten minutes and an artificial agent over 1,000 processing cycles. For each, list indicators, thresholds in qualitative or quantitative form, recovery tests, regime failures, and cases that would be unlicensed rather than negative. Then state the common functional role that makes both envelopes instances of V .

Collaborative exercise and discussion: Selective-Failure Design Sprint

Teams of four assign these roles:

- The **construct analyst** states what kind of joint structure the intervention targets.
- The **intervention engineer** designs manipulation, retained-component equivalence tests, and rescue.
- The **validity auditor** checks referent mapping, common drive, source fidelity, signal quality, and task demands.
- The **rival advocate** gives predictions for the full theory, an N_1 -ablated model, and one unity-centered rival.

Choose among three synthetic strategies: cross-module phase jitter, global gain suppression, or removal of a suspected common source. Specify which is most likely to create a diagnostic N_1 failure while preserving V , N_2 , and N_3 . Do not decide from the intervention’s name; write the required evidence and equivalence margins.

Use this frozen design record. Every gate is .60; retained-component change must remain wholly inside $[-.08, .08]$ and its post-intervention lower bound must still clear .60. A target reduction requires the full ΔN_1 interval to lie at or below $-.15$. The candidate mapping threshold is .85.

Strategy	Post-intervention component record	Mapping, validity, and anchor record
Phase jitter	$N_1 = .43 [.36, .50]$, $\Delta N_1 = [-.39, -.22]$; $V = .81 [.75, .87]$, $\Delta V = [-.03, .02]$; $N_2 = .74 [.68, .80]$, $\Delta N_2 = [-.05, .01]$; $N_3 = .79 [.72, .86]$, $\Delta N_3 = [-.04, .03]$	mapping .93; source fidelity passes; independent anchor changes $ [.10, .22]$ in the preregistered direction
Gain suppression	$N_1 = .45 [.38, .52]$, $\Delta N_1 = [-.37, -.20]$; $V = .64 [.58, .70]$, $\Delta V = [-.12, -.05]$; $N_2 = .58 [.52, .64]$, $\Delta N_2 = [-.16, -.09]$; $N_3 = .63 [.57, .69]$, $\Delta N_3 = [-.10, -.04]$	mapping .91; signal-quality control passes; no pathway-specific rescue
Common-source removal	N_1 estimator A = .46 [.39, .53], estimator B = .69 [.62, .76], estimator A $\Delta N_1 = [-.36, -.19]$; $V = .82 [.76, .88]$, $\Delta V = [-.03, .03]$; $N_2 = .76 [.70, .82]$, $\Delta N_2 = [-.04, .02]$; $N_3 = .80 [.74, .86]$, $\Delta N_3 = [-.05, .01]$	mapping .88; both N_1 estimators were credible and frozen; anchor interval straddles zero

Halfway through, another team selects one declared complication: a second credible N_1 surrogate remains high; mapped-unit retention falls to .72; $\Delta N_2 = [-.23, -.11]$; the anchor changes only on report trials; or pathway-specific rescue restores the pre-intervention pattern. Classify the result as selective, nonselective, causally coupled, indeterminate, unlicensed, or strengthened, and defend the classification without changing the frozen margins. A single nonselective result is not enough for the “causally coupled” label.

Discussion: What kind of unity would a positive N_1 result establish - statistical, causal, computational, or phenomenological? Teams must preserve one dissent and name a result that would justify combining N_1 with N_3 .

Counterexample challenge

Construct the strongest candidate case in which N_1 passes but unified phenomenal presence is independently doubtful. Then construct a case in which phenomenal unity is strongly anchored but every credible N_1 measure fails under a valid test. For each, identify whether it threatens sufficiency of N_1 alone, necessity within the full conjunction, the operationalization, or the target concept.

Exit ticket

Explain in three sentences why a highly correlated network can fail N_1 , how V differs from N_1 , and one rule that prevents V from becoming a post hoc escape hatch.

Chapter summary

- N_1 contributes a proposed unity role: irreducibly joint causal organization rather than mere activity, correlation, redundancy, or complexity.
- XOR cleanly illustrates synergy but does not establish a conscious bearer.
- Several estimator families can inform N_1 ; none is identical to the construct.
- V asks whether the fixed candidate actually sustains the relevant organization in a declared regime and horizon.
- Viability must be prospectively frozen, substrate-sensitive, and distinct from both consciousness and the other components.
- A diagnostic integration audit selectively changes N_1 while demonstrating equivalence of V , N_2 , and N_3 on a stable mapped referent.
- Causal coupling and indeterminacy are legitimate results; failed selectivity is not confirmation.

Key terms

Activity: magnitude or rate of system events, which need not be integrated.

Causal cut: an intervention or model partition used to test dependence on cross-part influence.

Common drive: dependence produced when a third source influences several variables.

Discriminant validity: evidence that a measure captures its intended role rather than duplicating neighboring constructs.

Equivalence test: a test of whether the full uncertainty interval lies within a frozen tolerance.

Integration (N_1): the proposed role of irreducibly joint realization of a content.

Redundancy: information duplicated across parts such that either part can supply it.

Selective failure: a licensed change in one target component while retained components remain equivalent.

Synergy: information or causal contribution available only from a joint state.

Viability (V): realized capacity to sustain the relevant organization in a declared substrate, regime, and horizon.

Looking ahead

Chapter 5 turns to N_2 . Its central challenge is to define “system-wide availability” without quietly substituting current activation, spatial spread, node count, executive access, or overt report.

Transition notebook: from unity to reach

Construct two systems that have the same licensed N_1 result. In one, the integrated content can selectively alter several functionally different recipient classes; in the other, it remains inside one tightly coupled module. Working in pairs, list which observations are held fixed and which causal intervention would separate the systems. Finish with two verdicts - one about unity and one about availability - without letting either stand in for the other. Chapter 5 will supply the route, latency, diversity, and reduced-family rules needed to refine the second verdict.

Chapter 5 - Consciousness Without Report: System-Wide Availability

Opening puzzle: the silent switchboard

A control room contains four specialist teams. At this moment, only the visual-comparison team is working on a signal. The orienting, valuation, and homeostatic teams are idle. A brief, low-burden change in context would allow the same signal to alter each team's native work within a strict latency bound. No report team exists.

Is the signal system-wide available? If we count only current activation, the answer seems no. If we count every wire that could ever carry a signal after extensive reconfiguration, the answer becomes trivially yes. N_2 seeks a disciplined middle: **counterfactual causal availability** that is content-specific, timely, functionally diverse, robust across nearby contexts, and independent of required report or executive uptake.

This criterion is one of the model's clearest departure points from familiar shortcuts. "Global" does not mean physically everywhere. "Available" does not mean currently active. And "without report" does not mean without evidence.

Plain language. A content passes N_2 when it is causally poised to make several different kinds of difference inside the candidate system, without needing to produce a report.

Chapter map

By the end of this chapter, you should be able to:

- preserve six distinctions that protect N_2 from proxy collapse;
- explain why availability is a constrained disposition rather than an output;
- construct recipient classes independently of the target content;
- apply effect, latency, selectivity, route, and context gates to a recipient;
- aggregate over functional diversity rather than raw node count;
- interpret report- and executive-excluded recomputation;
- compare N_2 fairly with access consciousness and an exact GNWT version; and
- formulate positive, negative, indeterminate, and unlicensed availability results.

1. Six distinctions that must not collapse

The companion paper *Consciousness Without Report* begins from six non-equivalences.

Actual activation is not causal availability

A recipient can activate because of common input, arousal, diffusion, or measurement leakage while the target content makes no selective causal difference to it. Conversely, an idle recipient may be ready to respond under a nearby admissible context. Availability concerns a disposition tested through intervention, not a census of lit-up nodes.

Availability is not report

Speech, button presses, and deliberate signals are useful evidence when intact and appropriately calibrated. They are not required recipients in the N_2 criterion. A patient with a blocked motor channel, a dreaming person, an infant, or a nonhuman animal may lack one form of report while retaining rich causal organization.

Broadcast is not diffusion

A substance, voltage change, or copied token can spread widely without changing recipient functions in a content-sensitive way. Broadcast requires routed causal modulation, not passive distribution.

Globality is not anatomical extent

“System-wide” is always relative to the frozen candidate S . A compact circuit can be global within that system if it reaches sufficiently diverse roles. A signal dispersed across a whole building can fail if every destination performs the same trivial operation.

Functional diversity is not node count

One hundred identical loggers do not become one hundred kinds of recipient. Recipient classes are individuated by their causal response repertoires, not anatomical labels or process count.

Posterior or subcortical reach is not executive access

A posterior or subcortical network may reach sensory comparison, orienting, value, visuomotor, or homeostatic classes without relying on deliberate planning or report. Conversely, a signal that reaches only an executive report pathway can fail the reduced-family tests.

Do not infer. Decoding content from many locations does not establish N_2 . Decodability can arise from common input, redundant representation, or leakage.

Analogy: an emergency notification network

A city alert is genuinely available to diverse services when it can selectively alter transportation control, hospital readiness, school operations, and utility management through known routes. Sending the same file to a hundred identical archive servers increases copies, not functional diversity.

Where the analogy breaks. City agencies have institutionally defined roles and interpretable messages. Neural or artificial recipient classes must be discovered and calibrated without using the target content to manufacture the taxonomy.

2. Availability as a disciplined counterfactual

The proposed criterion can be stated in prose:

A content is system-wide available when admissible, content-specific interventions show that it could causally modulate several independently individuated and functionally distinct recipient classes within a preregistered latency bound, across a nontrivial mass of nearby contexts, without requiring overt report, executive uptake, or current task use.

The word **could** is important but dangerous. Unconstrained counterfactuals make almost anything available: after enough retraining, rewiring, prompting, surgery, or software installation, a content could affect almost any output. The criterion therefore limits preparation burden.

Let $\Gamma(u, u_0)$ measure the training time, energy, controller involvement, structural change, and other preparation required to move from baseline context u_0 to candidate context u . Nearby admissible contexts must remain within a frozen burden bound and preserve the operating regime R . A context that installs a new route does not reveal availability in the original architecture.

The content contrast must also be frozen. Changing content c to matched alternative c' should preserve intensity, duration, source support, and nonspecific global-state effects as far as the design allows. **Source fidelity** checks whether the intended content distinction survived the intervention. Otherwise a downstream change may be driven by arousal or energy rather than the content.

Method note. Counterfactual availability is experimentally disciplined receptivity under nearby conditions, not metaphysical possibility and not eventual learnability.

Thought experiment: The Sleeping Fire Brigade

A fire brigade is inactive at midnight. A validated alarm route would mobilize it in ninety seconds if a fire signal arrived. In a second city, firefighters can respond only after a week of training and installation of a new radio system.

Translate the case in stages:

1. Why does inactivity fail to establish unavailability in the first city?
2. Which preparation-burden condition separates the cities?
3. What would count as content selectivity - a fire-specific response rather than a general loud-noise reaction?
4. How would route interruption distinguish the alarm's causal role from a simultaneous public siren?
5. Where does the analogy stop helping when the "recipients" are neural populations rather than agents who understand messages?

The thought experiment introduces the logic, but the final question prevents anthropomorphic importation.

3. Recipient classes before target testing

A recipient is not simply a node that changes. The method first builds **recipient classes** from calibration interventions that do not use the critical target content. Modules belong to different classes only when their causal response repertoires differ reliably. A sensory comparator, orienting controller, value updater, memory indexer, and report-preparation unit may form distinct classes if they show nonredundant native transitions under calibration.

This ordering prevents target leakage. If researchers inspect the target content's effects and then split recipients until the class count passes, functional diversity has been engineered from the outcome.

Recipient taxonomies must survive reasonable alternatives. Anatomical parcels can seed proposals, but two named regions do not automatically count as two functions. Likewise, two software services with different names can be behaviorally identical wrappers. Repertoire distance, clustering stability, held-out prediction, and perturbational checks supply stronger evidence.

Weights w_k over classes are frozen before the critical test. They may reflect scientific importance, repertoire geometry, or a balanced design, but cannot be proportional to raw node count alone. Otherwise large redundant classes dominate by construction.

Checkpoint. A system contains 100 identical storage processes and three small controllers with distinct causal repertoires. How many provisional recipient classes should be considered, and what additional calibration would support that count?

Analogy: academic departments

A university's reach is not measured by counting every employee as a distinct function. Affecting admissions, laboratory operations, student advising, and facilities represents more functional diversity than sending the same memo to one hundred identical inbox rules.

Where the analogy breaks. University roles come with human purposes and official descriptions. Functional classes in a candidate system must be inferred from causal response patterns and can change with grain and regime.

4. Five gates for one recipient

For recipient class k in context u , estimate a content-specific effect $E_{k,u}$ using matched interventions. A class qualifies only when five properties concur in the required contexts:

1. **Effect:** the lower uncertainty bound clears a class-specific minimum.
2. **Latency:** the upper uncertainty bound for qualifying latency is at or below the frozen ceiling; a negative result requires the lower bound to exceed that ceiling.
3. **Content selectivity:** the response exceeds matched null or alternative-content effects.
4. **Route dependence:** interruption of a nominated route or route family reduces the effect as predicted.
5. **Context robustness:** the result holds across at least the preregistered mass ρ_U of nearby admissible contexts.

The gates must concur for the same recipient and in every context counted toward its positive context mass. A large effect in context A, fast latency in context B, and route support in context C never combine into one qualifying class. Context aggregation occurs only after each counted recipient-context record has passed all five gates.

Route tests distinguish mediation from parallel drive. If a common source independently drives both the content representation and the recipient, temporal precedence and decoding may look like broadcast. Interrupting the nominated pathway while preserving generic support helps identify whether the content-bearing route makes a difference. Rescue of the effect through a selective pathway restoration strengthens the interpretation.

Redundant routes complicate a negative result. Failing to reduce an effect by cutting one path does not show route independence if another nominated path remains. A route-negative verdict requires adequate coverage of the relevant route family; otherwise the result is indeterminate.

Why it matters. N_2 asks for causal routing, not a map of correlations. The conclusion must match the rung of causal evidence actually achieved.

An evidence ladder is useful:

1. decoding or covariance shows information in the measured signal;
2. temporal precedence narrows possible direction;
3. matched content intervention shows selective downstream difference;
4. route interruption tests mediation;
5. selective rescue supports the proposed causal pathway.

Not every study must reach rung five, but a rung-one result should not be called proven broadcast.

5. From qualifying recipients to system-wide availability

Let $a_k = 1$ when recipient class k passes every required gate and $a_k = 0$ otherwise. Weighted coverage and qualifying class count are:

$$C_A = \sum_k w_k a_k, K_A = \sum_k a_k.$$

For the positive aggregate, every class not positively qualified contributes zero. Its separate ledger status - negative, indeterminate, or unlicensed - must nevertheless be retained; those statuses are not interchangeable in the final verdict.

Coverage prevents a tiny fraction of the candidate's declared functional repertoire from carrying the result. Class count prevents one dominant class from being relabeled into several subclasses. But coverage and count can still hide concentration: one class may carry nearly all meaningful effect while others barely cross threshold.

An evenness term addresses this. Using point effects E_k^* for illustration, define:

$$p_k = \frac{w_k E_k^* a_k}{\sum_j w_j E_j^* a_j}, H_A = - \frac{\sum_{k: a_k=1} p_k \log p_k}{\log K_A}.$$

When fewer than two classes qualify, set $H_A = 0$. A real analysis must propagate uncertainty through H_A rather than treating point estimates as exact.

A positive aggregate requires frozen thresholds for weighted coverage, minimum class count, and evenness. It may remain positive when one nonessential class is indeterminate or negative, provided every aggregate lower-bound rule still passes. That is not cherry-picking; it follows a rule declared before seeing which class fails. The unresolved class remains visible in the ledger.

Availability must also be stable across a reasonable specification neighborhood. If N_2 flips under defensible recipient clustering, grain, latency, estimator, or intervention choices, a nominal positive is fragile. The stability score $R_A(c)$ records weighted agreement with the frozen primary result across valid mappings. Low stability yields a fragile or indeterminate outcome rather than permission to choose the favorable specification.

Worked example: five recipient classes

The following numbers are pedagogical, not proposed biological constants. Freeze a minimum effect of .05, latency ceiling of 180 ms, coverage threshold of .60, minimum class count of three, and evenness threshold of .80. A class qualifies only when effect, latency, selectivity, route, and context all pass.

Recipient class	W_k	Effect and interval	Latency interval	S / R / C	Status
Sensory comparison	.25	+ .18 [.14, .22]	85 [76, 94] ms	yes / yes / yes	qualifies
Value updating	.20	+ .12 [.08, .16]	110 [99, 121] ms	yes / yes / yes	qualifies
Memory indexing	.20	+ .04 [.00, .09]	125 [111, 139] ms	effect gate straddles / yes / yes	indeterminate
Orienting and visuomotor	.20	+ .15 [.10, .20]	95 [84, 106] ms	yes / yes / yes	qualifies
Report preparation	.15	+ .20 [.16, .24]	160 [148, 172] ms	yes / no / yes	route-negative

The qualifying weight is $.25 + .20 + .20 = .65$, so coverage passes. Three classes qualify, so count passes. Using the point effects only to make the arithmetic visible, the normalized effect shares are approximately $(.455, .242, .303)$ and $H_A \approx .968$, so evenness passes. In a real analysis, uncertainty must be propagated through H_A .

Report removal excludes the .15 report class. After renormalization, qualifying coverage is $.65 / .85 = .765$, so the report-excluded family passes. Executive removal excludes memory indexing and report preparation. The three qualifying nonexecutive classes exhaust the remaining .65 weight, so the executive-excluded family also passes after renormalization. The indeterminate memory class and failed report class remain in the audit trail even though they contribute zero to the positive aggregate.

Thought experiment: The Hundred Loggers and the Twelve Nodes

Architecture A sends a content token to one hundred identical loggers through a shared bus. Architecture B contains twelve measured nodes organized into sensory comparison, orienting, value updating, and homeostatic control. Selective interventions show timely, route-dependent effects in all four classes.

Work through the contrast:

1. Which architecture has greater anatomical or process count reach?
2. Which has greater functional diversity under the criterion?
3. How could common-drive control overturn the apparent success of Architecture A?
4. What taxonomy, effect uncertainty, or validity failure could make Architecture B indeterminate or unlicensed?
5. What additional evidence would be required before either N_2 result affected a CØ verdict?

6. Removing report and executive dependence

After evaluating the full recipient family, repeat the aggregate after removing report classes and after removing executive classes:

$$N_2^{-R} = 1, N_2^{-E} = 1.$$

The superscripts mean “with report recipients removed” and “with executive recipients removed.” Weights are renormalized within each frozen reduced family. A positive N_2 result must survive both.

This test guards against two forms of smuggling. A system might look global only because a report-preparation class has many downstream connections. Or several executive subtypes - working memory, conflict monitoring, planning, and report - might dominate the taxonomy. Removing these families asks whether diverse nonreport and nonexecutive causal reach remains.

Removal need not mean permanent lesion. Researchers can design tasks without overt report, exploit natural motor dissociations, transiently interrupt channels, or estimate a reduced-family model while preserving arousal and generic support. Each strategy has limitations. No-report designs can change attention or task set; lesions can alter the operating regime; modeling depends on causal assumptions.

Do not infer. Report-independent does not mean report-hostile. Reports remain crucial for calibrating contents and relating measures to phenomenology in suitable cases. The claim is that a report mechanism is not required as a constituent recipient of minimal availability.

7. Posterior, subcortical, and artificial implementations

The criterion is anatomically flexible but causally strict. A posterior-dominant visual network can pass if a content selectively and quickly modulates distinct sensory, spatial, orienting, affective, or visuomotor classes through supported routes. A thalamic pulse that changes general arousal without preserving content selectivity does not pass. Prefrontal activity may count when it plays a qualifying role, but it is neither mandatory nor disqualifying by location alone.

In an artificial system, copying tokens to services is insufficient. Each recipient must undergo a class-native, content-sensitive transition. A planner revising a policy, a memory system changing an index, and a safety controller altering a constraint may count as different roles if their repertoires were independently calibrated. A hundred cache replicas do not.

The system boundary remains upstream. “Global” in a single model call differs from “global” in a week-long agent service. An external tool can be a recipient, interface, support, or environment depending on causal constitution and horizon. Product labels cannot substitute for boundary tests.

8. Fair comparison with access and GNWT

N_2 intentionally stands near access consciousness and global-workspace theories. All emphasize availability beyond a local representation. A critic can reasonably ask whether “counterfactual causal influence on diverse recipients” is simply covert access under a new name.

The proposed distinction is that standard access often emphasizes availability for reasoning, deliberate control, working memory, and report. N_2 admits a weaker and broader functional repertoire: sensory comparison, orienting, value updating, homeostatic adjustment, and visuomotor control can qualify without executive uptake. Whether this is genuinely distinct must be demonstrated through disagreement cases, not asserted.

For a fair GNWT comparison, freeze an exact version - for example, one requiring a particular global neuronal workspace ignition or executive-access pattern. Then seek a case in which content-specific posterior, subcortical, affective, or orienting reach passes N_2 while that GNWT requirement does not. CØ / N* may predict positive availability; the frozen GNWT version may not. Motor report loss alone is insufficient, because contemporary workspace theories need not equate workspace with overt movement.

If careful operationalization makes both theories predict the same cases, the difference may be explanatory or terminological rather than diagnostic. If N_2 always requires executive recipients in practice, its advertised separation fails and the theory must say so.

Checkpoint. Formulate one system on which N_2 and a frozen access criterion disagree. Name the observation that would decide the disagreement rather than merely rename it.

9. Four outcomes and their direction

A component verdict can be:

- **Positive:** the assessment is licensed; every class counted as qualifying passes all five recipient gates; and the coverage, class-count, evenness, reduced-family, uncertainty, and stability rules pass. Other frozen classes may remain negative or indeterminate, but they remain visible in the ledger.
- **Negative:** a direction-specific confidence bound decisively fails a frozen gate under adequate test conditions.
- **Indeterminate:** valid evidence straddles a threshold, credible surrogates conflict, route-family coverage is incomplete, class count remains possible but unresolved, or specification stability is too low.
- **Unlicensed:** the referent, content contrast, taxonomy, intervention, regime, source fidelity, or transport interpretation fails.

Direction matters. For a minimum effect threshold, an upper bound below the threshold supports a negative. For a maximum latency, a lower bound above the ceiling supports a negative. For minimum class count, a negative requires that fewer than the required number could qualify even at their favorable bounds. Using one generic p-value rule across these gates would misstate the hypotheses.

10. Evidence and limits

The N_2 proposal is a research criterion, not an already completed universal assay. It gains rigor by naming failure conditions. Revise it if independently anchored conscious contents repeatedly remain local-only under licensed tests; if clear nonconscious diffusion, seizure, or feedforward cases pass every content, route, diversity, context, and reduced-family control; or if reasonable frozen recipient taxonomies arbitrarily reverse outcomes.

Other direct challenges include:

- report or executive classes prove necessary despite credible no-report anchors;
- revealing availability always requires high-burden retraining that creates a new architecture;
- results depend on rare or outcome-selected contexts;
- repertoire-based classes fail to predict meaningful differences in recipient function; or
- transport to a new substrate cannot preserve the construct's measurement meaning.

Practical limits are substantial. Content interventions can be difficult or ethically constrained. Route interruption may disrupt arousal or the bearer. Recipient functions are partly theory-laden. Counterfactual claims require causal-model assumptions. Yet these difficulties are preferable to treating widespread activation as self-interpreting.

Source note. The six distinctions, recipient gates, aggregate rules, reduced-family tests, stability requirement, and direct challenge patterns are developed in *Consciousness Without Report*.

Practice studio

Quick checks

1. Give one case of activation without availability and one of availability without current activation.
2. Why can one hundred recipients count as one functional class?
3. What does the preparation-burden rule prevent?
4. Name the five gates a recipient must pass.
5. Why are coverage, class count, and evenness all needed?
6. What do N_2^{-R} and N_2^{-E} test?
7. Why can a failed single-route interruption yield indeterminacy rather than a negative route result?

Individual transfer task: recipient ledger

Choose a biological or artificial candidate and one content contrast. Propose four independently calibrated recipient classes. For each, specify a native response, matched intervention, effect threshold, latency ceiling, selectivity null, nominated route family, nearby contexts, and preparation burden. Add columns for uncertainty and possible unlicensed conditions. Do not decide N_2 until the full ledger and reduced-family plan are complete.

Collaborative exercise and discussion: Causal-Routing Laboratory

Teams of four receive one frozen architecture brief:

- **Duplicated log bus:** one hundred loggers form one calibrated recipient class with weight 1.00. The target effect is large and fast, but cutting the proposed content route leaves the effect unchanged because a common clock drives both source and loggers.
- **Executive-heavy router:** report preparation has weight .45, planning .35, and nonexecutive sensory comparison .20. All three pass the five recipient gates, but only the sensory class remains after report and executive removal; the frozen minimum class count is three.
- **Compact diverse circuit:** use the five-class worked ledger above, including the stated uncertainty and reduced-family rules.
- The **taxonomy lead** decides which recipients are genuinely distinct using calibration repertoires.
- The **intervention lead** designs target-content and matched-null perturbations.
- The **route auditor** specifies interruption, redundant-path, latency, and rescue tests.
- The **independence lead** recomputes after report and executive removal and audits preparation burden.

Before discussion, each student privately predicts which piece of evidence will be most diagnostic. The team then draws a routing graph marking current effects, idle but testable receptivity, common drive, invalid recipients, and uncertain routes. Complete a ledger for effect, uncertainty, latency, selectivity, route, context, weight, class result, full-family aggregate, reduced-family aggregates, and stability.

Exchange ledgers. The receiving team may challenge one class distinction and one causal route. The original team must either defend the frozen choice or classify the result as indeterminate or unlicensed, depending on whether valid evidence remains unresolved or a prerequisite has failed. It may not redraw classes after learning which arrangement passes.

Discussion: Is N_2 covert access, a prerequisite for access, a part of phenomenal presence distinct from access, or an unstable distinction? Mixed-position groups must construct the best version of every view and name one empirical disagreement case. A conclusion of “more data” is acceptable only if the discriminating data are specified.

Counterexample challenge

Design two architectures. The first should look globally active while failing N_2 under every major control. The second should be spatially compact and currently quiet in most recipients while plausibly passing. For each, identify the smallest manipulation that would reverse your verdict without changing the candidate or installing a new route.

Exit ticket

In four sentences, define report-independent system-wide availability, state why node count is insufficient, name one reduced-family test, and identify one result that would force revision of the criterion.

Chapter summary

- N_2 is counterfactual, content-specific causal availability across functionally diverse recipients.
- Activation, report, diffusion, anatomical spread, node count, and executive access are not interchangeable with availability.
- Recipient classes must be calibrated independently of the target content.
- One recipient qualifies only when effect, latency, selectivity, route, and context requirements concur.
- Coverage, class count, evenness, uncertainty, and specification stability govern the aggregate.
- A positive result must survive separate removal of report and executive recipient families.
- Posterior, subcortical, and artificial systems can qualify, but anatomical flexibility does not weaken causal demands.
- Positive, negative, indeterminate, and unlicensed verdicts require direction-sensitive rules.

Key terms

Availability: a constrained disposition for a content to make a causal difference to recipients under nearby admissible contexts.

Content selectivity: evidence that effects track the target content rather than nonspecific state change.

Context robustness: preservation of qualifying effects across a frozen mass of nearby admissible contexts.

Counterfactual receptivity: readiness to respond under a low-burden admissible condition despite current inactivity.

Evenness (H_A): a normalized measure of how qualifying weighted effects are distributed across recipient classes.

Functional diversity: nonredundant recipient roles established through causal response repertoires.

Preparation burden: the amount of training, energy, structural change, or controller involvement required to reveal a route.

Recipient class: a group of modules with a reliably similar causal response repertoire.

Reduced-family test: recomputation after removing report or executive recipients.

Route dependence: evidence that a nominated pathway or route family mediates the content-specific effect.

Source fidelity: evidence that an intervention preserves the intended content contrast.

Looking ahead

Availability gives a content causal reach, but not yet temporal thickness. Chapter 6 will ask how recurrent dynamics preserve content long enough to count as bounded temporal presence without confusing recurrence with report, generic persistence, or pathological fixation.

Transition notebook: reach through time

Imagine one content that can alter four recipient classes within 80 ms but then disappears, and another that reaches the same classes only after a recurrent loop stabilizes it for 240 ms. First decide what the N_2 evidence alone licenses in each case. Then, in a group of three, assign an availability auditor, a temporal auditor, and a skeptic. Produce one observation that would distinguish causal reach from temporal maintenance without using report latency as either measure. Save the observation as a prospective prediction for Chapter 6.

Chapter 6 - Recurrent Stability and Temporal Presence

Opening puzzle. Imagine two devices that receive the same brief signal. In Device R, the signal enters a network whose later activity feeds back into earlier processing. If the network is nudged away from its current state, it tends to return toward a content-specific pattern before eventually moving on. In Device F, the signal passes through a one-way chain of slow components. The activity lasts just as long, and a graph of average activation looks nearly identical, but nothing later makes a causal difference to anything earlier. If both devices retain the signal for 240 milliseconds, have they achieved the same kind of temporal organization?

The question matters because a duration measurement cannot, by itself, tell us how a state lasts. CØ / N* proposes that minimal phenomenal presence requires more than an instantaneous feedforward event. Its third network condition, N_3 , concerns bounded recurrent stability: content-relevant activity is maintained through return relations or an explicitly defended metastable surrogate, long enough to be present but not defined as permanently fixed. The theory therefore takes a risk. A slow feedforward process that matches every valid consciousness-related anchor would challenge the claim that recurrent topology is necessary.

Why it matters. Timing is often treated as if “later” meant “more conscious.” That shortcut confuses at least four things: initial encoding, recurrent stabilization, working-memory maintenance, and preparation for report. This chapter gives you a vocabulary and a research strategy for keeping them apart.

Chapter map

By the end of this chapter, you should be able to:

- explain the difference between recurrence, persistence, a passive trace, and working-memory maintenance;
- interpret the bounded-dwell condition without treating it as a universal stopwatch;
- distinguish temporal presence from the temporal character of an experience;
- design a perturbation that targets N_3 without simply erasing the content or disrupting report;
- compare the CØ / N* proposal fairly with Recurrent Processing Theory and duration-only rivals; and
- state what evidence would support, weaken, or leave the recurrence claim unresolved.

The central discipline is simple to state and hard to practice: do not infer a causal loop from a long-lasting signal, and do not infer phenomenal timing from the timing of a button press.

1. A state can last in more than one way

The notation $N_3(S, c)$ says that candidate system S has the required temporal organization for content c . The content index matters. A brain, robot, or organoid can contain oscillations and feedback everywhere while the particular content under study passes through only once. Global recurrence is not automatically content-relevant recurrence.

Four temporal patterns should be separated.

1. A **feedforward duration** occurs when activity travels in one causal direction but moves through slow units or long delays.
2. **Recurrent processing** occurs when an earlier content-supporting state makes a difference to a later state through a directed return relation in a time-unrolled causal model.
3. A **passive trace** persists in a sensor, receptor, buffer, or slow variable without continuing to shape the active content-supporting process.
4. **Working-memory maintenance** keeps information available for later use, often through executive control, task demands, or rehearsal that may extend beyond the proposed interval of phenomenal presence.

These patterns can overlap in real systems. The scientific problem is not to force every episode into a single box. It is to determine which causal organization explains the measured persistence and which part of that organization the theory claims is necessary.

Plain language. A state can remain on the screen because the program keeps repainting it, because the pixels fade slowly, because a copy is stored in memory, or because a user keeps pressing “refresh.” The visible duration may be the same while the sustaining process differs.

Analogy: the echoing room and the long hallway. In an echoing room, sound returns after reflecting from the walls. In a long hallway lined with open doors, a sound can arrive at successive locations over time without ever returning. A microphone at the far end might record activity for the same total duration in both cases. The echoing room illustrates a return relation; the hallway illustrates extended feedforward propagation.

Where the analogy breaks. Neural or artificial recurrence is not literal acoustic reflection. A causal return can transform, gate, or reconstruct a content rather than replay it. Moreover, an echo may be behaviorally irrelevant, whereas N_3 concerns a return that helps maintain a content-bearing organization. The analogy distinguishes topology, not consciousness.

To establish recurrence, researchers need evidence about causal dependence. In a time-unrolled graph, a state at time t_1 influences a later state at t_2 , and some pathway from later processing returns influence to a stage or variable that helps determine the content at t_3 . A genuine self-feedback relation may qualify if intervening on the earlier state changes the later content beyond what continuing input and passive decay predict. A familiar oscillation or high autocorrelation is only a candidate measure until such dependence is established.

Checkpoint. For any temporal measure you encounter, ask two questions: “What makes the state last?” and “What shows that the lasting activity still concerns the same content?”

2. Bounded dwell and the idea of metastability

The core temporal proposal can be written:

$$\exists L \subseteq S \text{ such that } t_{\min} \leq \text{dwell}(c; L) \leq t_{\max}.$$

Here L is a content-supporting subnetwork within the candidate system S . The expression $\text{dwell}(c; L)$ is the time during which the relevant content remains in a recurrently or metastably maintained organization. The lower bound $t_{\min}$ excludes an evanescent event that never stabilizes. The upper bound $t_{\max}$ resists the assumption that indefinite persistence is always “more conscious.” Together the bounds aim at **metastability**: enough stability to sustain an episode, with enough flexibility for updating, transition, and decay.

The bounds are operational hypotheses, not constants discovered once for every species, modality, and substrate. A visual event in an adult human, an olfactory state in an octopus, and an artificial control state may operate on different time scales. A serious study must freeze the relevant window before examining the decisive result and justify why that window tracks the proposed temporal role.

Do not infer. Exceeding a proposed upper bound does not automatically prove unconsciousness. A seizure-like fixation, a deeply absorbed state, or an unusually long pain episode may reveal that the bound measures healthy flexibility, a particular estimator’s validity range, or one biological implementation rather than minimal presence itself. The theory owes a revision if valid countercases accumulate.

Analogy: balancing a bicycle. A moving bicycle is stable without being locked in place. Small corrections keep it within a viable range, yet it remains able to turn. A bicycle welded upright is more fixed but not more dynamically stable in the relevant sense. This captures the contrast between metastable maintenance and inert persistence.

Where the analogy breaks. A bicycle’s stability is a whole-vehicle control property, while N_3 is indexed to a particular content. The analogy also says nothing about integration, system-wide availability, or phenomenal evidence. It helps explain why “lasting longer” and “being dynamically maintained” are not synonyms.

Candidate estimators include survival time within a preregistered trajectory region, return toward a content-specific attractor after a small perturbation, recurrent effective connectivity, a hidden-state plateau, and pathway-dependent recovery. Each estimator carries assumptions. Autocorrelation may confuse recurrence with slow decay. Effective-connectivity estimates may depend heavily on model choice. Attractor language may oversimplify trajectories that are metastable without settling into a fixed basin. Therefore, multiple estimators should be compared, and credible disagreement should produce an indeterminate component assessment rather than a convenient average.

Method note. Metastable return without an identifiable loop may be evaluated as an alternative N_3 surrogate, but it must not be silently treated as evidence for the stronger recurrent-topology claim. Investigators must freeze the surrogate’s causal role, intervention signature, and agreement rule in advance. If a loop-based estimator and a metastability estimator disagree beyond the declared tolerance, the N_3 result is conflicted or indeterminate; the favorable measure cannot simply replace the other after inspection.

Onset and offset signals require the same role analysis. Late activity may reflect report or memory, while absence of one familiar late component does not establish absence of every recurrent route. If loop-based and metastable-surrogate estimators disagree beyond tolerance, return a conflicted or indeterminate assessment rather than selecting the favorable estimator after inspection.

Source note. The bounded-dwell formulation and the recurrence-versus-persistence distinction are developed in “CØ as N” and audited in “Ablating N: Does Every Conjoint Earn Its Place?”. The source manuscripts present the bounds as testable operational commitments, not established universal constants.

3. Timing the intervention without changing the question

A perturbation can be technically successful and inferentially useless. Suppose content identity first becomes measurable around 60 milliseconds, candidate recurrent stabilization occupies approximately 110-220 milliseconds, working-memory maintenance begins around 260 milliseconds, and report preparation begins around 340 milliseconds. An intervention delivered at 50 milliseconds may prevent the content from forming. An intervention delivered at 370 milliseconds may disrupt only the report. Neither result selectively tests whether recurrence maintained phenomenal presence.

A licensed N_3 perturbation should target a return pathway or its effective gain during a preregistered stabilization interval. It should preserve:

- initial content fidelity;
- viability V ;
- integration N_1 ;
- report-independent availability N_2 ;
- the autonomy and cross-condition identity of the candidate referent; and
- the operating regime in which the measures are valid.

“Preserve” requires both that the full paired-change interval for every retained component lie within its preregistered equivalence band and that each post-intervention lower bound remain at or above its gate. After the perturbation, the positive-bearer set must be recomputed. A smaller or alternative candidate may now satisfy the model, or bearer attribution may become unresolved. The study should not hold positive-bearer membership fixed as part of the intervention license, because that would protect the outcome the theory is supposed to predict.

Method note. The cleanest design often has at least five conditions: an early encoding disruption, a targeted recurrent-path disruption, a late report or memory disruption, a sham intervention, and a selective rescue. The controls help identify what the target manipulation did; the rescue tests whether restoring the proposed causal role restores the predicted anchor pattern.

Thought experiment: The five-pulse theater

A research team can deliver one magnetic pulse during a visual trial. It considers five timings:

- Pulse E occurs before content identity can be decoded.
- Pulse T occurs during the proposed recurrent stabilization interval.
- Pulse L occurs after working memory begins but before overt report.
- Pulse R occurs during response preparation.
- Pulse S is a sham that produces the same sound and expectation without neural perturbation.

The team observes that E, T, and R all reduce verbal accuracy. T also reduces a no-report physiological discrimination marker; E reduces early content fidelity; R leaves the no-report marker intact.

Stage 1. Draw a causal timeline and state which timing most directly addresses N_3 .

Stage 2. List the measurements needed to show that the target timing preserved N_1 , N_2 , viability, and candidate identity.

Stage 3. Design a rescue manipulation. What must recover, and what nonspecific recovery would fail to support the recurrence claim?

Stage 4. Identify at least two ways the same observations could still be explained by a duration-only model.

The point is not to guess the intended answer. It is to discover how many assumptions stand between “a pulse changed behavior” and “recurrent stability is necessary for phenomenal presence.”

4. Temporal presence is not temporal character

N_3 asks whether a content is maintained through an appropriate interval. It does not yet say how the experience unfolds within that interval. The character extension uses a temporal path profile, written Π_c , for features such as direction, rate, curvature, order, dwell pattern, and hysteresis - path dependence in which the current state depends on how it was reached - in a quality structure.

Two episodes can both pass N_3 while differing markedly in temporal character. A pain may be steady or throbbing. A melody may rise or fall. Apparent motion may feel smooth or stuttering. A color may fade evenly or flicker. These differences concern the path an admitted content traces, not whether the content crosses the presence gate at all.

Plain language. N_3 asks whether there is a sustained episode. Π_c asks what the episode's temporal shape is like.

Thought experiment: The frozen chord and the moving phrase

System A sustains a chord through a stable recurrent pattern for 600 milliseconds. System B traverses a sequence of related pitch states over the same interval, also through recurrent processing. Both meet the same lower and upper dwell bounds and retain equivalent integration and availability.

Stage 1. Explain why matching N_3 values would not imply matching experiences.

Stage 2. Propose relational features that could distinguish a held chord from an unfolding phrase without using the participant's final report as the presence criterion.

Stage 3. Imagine a perturbation that reverses the order of transitions while preserving the presence gate. What independent phenomenal relation should a character theory predict will change?

Stage 4. Suppose the perturbation changes the neural trajectory but no validated phenomenal relation. Which claim is threatened: the presence gate, the temporal-character bridge, both, or neither? Defend the scope of your answer.

This separation makes the research program modular. Failure of a proposed path-to-character mapping does not automatically show that the episode lacked presence. Conversely, a richly structured trajectory does not prove presence if the gate is negative, indeterminate, or unlicensed.

5. Rival lenses: recurrence, duration, and access

Recurrent Processing Theory (RPT) is the nearest neighboring family. RPT emphasizes feedback within sensory processing and argues, in at least some versions, that local recurrent processing is central or sufficient for perceptual consciousness. CØ / N* accepts the pressure behind this view: a single feedforward sweep may be too thin to constitute temporally present content. It differs by placing recurrence inside a conjunction. A local loop that is not irreducibly integrated or cannot causally reach functionally diverse recipients fails N_1 or N_2 , even if it is recurrent.

Global Workspace and Global Neuronal Workspace approaches often associate conscious access with widespread availability and later ignition-like dynamics. CØ / N* overlaps at N_2 but refuses to use report, executive control, or a single late signature as the minimal criterion. Its N_3 interval may occur before working-memory and response processes that workspace experiments often measure. Exact contrasts depend on the dated version of each rival; neither “RPT” nor “workspace theory” is one unchanging prediction rule.

Duration-only rivals claim that what matters is sufficiently persistent internal state, not loop topology. They become especially important for artificial systems that use delays, state-space memory, or long time constants without an obvious recurrent graph. If such a system matches all independently validated anchors, survives intervention tests, and transports across contexts, the recurrence requirement should be weakened or rejected unless loop topology makes a further risky prediction.

Do not infer. Showing that biological brains normally use recurrence would not establish recurrence as a substrate-general necessity. It may identify a common implementation. Necessity requires diagnostic cases in which recurrence and duration come apart.

The fairest comparison therefore uses systems or interventions that match dwell time while varying return topology, or match topology while varying content-relevant causal influence. Rival models should be frozen before the decisive data are inspected. If one model is allowed to redefine recurrence after every counterexample while another is held fixed, the comparison is not adversarial.

6. Evidence, limits, and direct challenges

Evidence favors the N_3 claim when a content-relevant return path predicts more than duration alone, a selective disruption crosses the N_3 gate while retained conditions remain equivalent, independent anchors change as preregistered, and a role-specific rescue restores those anchors. Convergent estimators should agree on the intermediate temporal role, and the effect should survive reasonable changes in boundary, grain, timing, and preprocessing.

Evidence directly challenges the claim when:

- a well-supported conscious content occurs under genuine absence of feedback;
- a valid, selective N_3 ablation leaves every independent phenomenal anchor unchanged;
- feedforward systems with matched duration predict all held-out and intervention results as well as recurrent models;
- the upper dwell bound systematically excludes credible conscious episodes; or
- recurrence estimates track report, arousal, or generic slowness rather than the target content.

Other outcomes are less decisive. A global anesthetic that reduces recurrence, integration, availability, behavior, and data quality at once does not identify N_3 specifically. A null effect from an underpowered perturbation is indeterminate. A study that cannot preserve the candidate boundary or operating regime is unlicensed for the selective claim. These categories are not rhetorical shelters; they specify different failures and different next experiments.

Method note. Write the strongest conclusion the design earns, then remove any word that implies a broader domain, stronger kind of necessity, or more selective intervention than the evidence supports.

Source note. The separation of early encoding, recurrent stabilization, working memory, and report is developed in “Cø as N” and “Ablating N: Does Every Conjunct Earn Its Place?”. The contrast with RPT is a family-level orientation; an actual experiment must freeze an exact rival version.

Practice studio

Quick checks

1. Give one example of a long-lasting state that need not be recurrent.
2. What does the upper dwell bound try to prevent researchers from assuming?
3. Why is a late neural signal not automatically evidence for N_3 ?
4. State the difference between a component measure and a licensed component inference.
5. Why must the positive-bearer set be recomputed after a recurrence intervention?
6. In one sentence, distinguish N_3 from Π_c .

Individual transfer task: Build a temporal causal map

Choose one domain: visual masking, auditory streaming, pain, dreaming, machine perception, or another case approved for your course. Construct a one-page time-unrolled causal map with at least five stages. Mark the earliest point at which content identity could be assessed, the proposed stabilization interval, the beginning of memory maintenance, and any report pathway. Then propose one target perturbation and two controls. For each, state what outcome would be a licensed N_3 pass, a licensed decisive N_3 failure, indeterminate, or unlicensed. End with a 100-word scope statement naming the system, content, regime, interval, rival model, and unresolved assumption.

Collaborative exercise: Temporal intervention design council

Work in teams of four with rotating roles:

- The **temporal modeler** proposes t_{min} , t_{max} , and the content-stabilization window.
- The **perturbation engineer** specifies pathway, timing, strength, sham, and rescue.
- The **retained-condition auditor** defines equivalence tests for content fidelity, V , N_1 , N_2 , and candidate identity.
- The **rival advocate** builds a duration-matched feedforward explanation and a late report-disruption explanation.

First, members work separately for five minutes. The team then produces a single annotated timeline and a prediction table without voting. Every disagreement must be recorded as a testable assumption or as a reason to withhold inference. Exchange designs with another team. The reviewing team may approve the design, return it for a specific repair, or classify the planned inference as unlicensed. It must identify the smallest additional control that would change its verdict.

Collaborative discussion: Is recurrence a universal requirement?

Assign four positions: substrate-general recurrence defender, biological-implementation defender, duration-only theorist, and evidential referee. Each participant must first state the strongest version of another position before presenting their own. Discuss whether a strictly feedforward system with internal state can be meaningfully feedforward, which causal interventions distinguish a loop from a long time constant, and what finding would rationally change each position. The referee produces a final map with shared commitments, genuine disagreements, diagnostic cases, and observations that would remain nondiagnostic.

Counterexample challenge

Design the most credible case you can in which phenomenal presence seems to occur without N_3 , or in which N_3 occurs without phenomenal presence. Do not rely on a label such as “dream,” “seizure,” or “AI.” Specify the candidate, content, boundary, timing, causal topology, measurements, independent anchors, and validity envelope. Then write two responses: one that revises the operationalization while retaining the theory, and one that concedes a genuine threat to necessity or sufficiency. Identify the empirical result that separates those responses.

Exit ticket

Complete both sentences: “Matching duration does not match recurrence because ...” and “A recurrence intervention licenses a necessity claim only if ...”

Chapter summary

N_3 proposes that minimal phenomenal presence requires content-relevant maintenance through bounded recurrent or defensibly metastable dynamics. Duration alone cannot identify recurrence; late activity can reflect memory or report; and a passive trace can persist without continuing causal influence. The dwell bounds aim at flexible stability, not a universal stopwatch. Selective tests must target the proposed stabilization interval, preserve other components and candidate identity, and recompute the bearer outcome. The recurrence claim gains force only where topology and duration come apart. Temporal presence is also distinct from temporal character: an episode can remain present while its experienced path changes.

Key terms

- **Bounded dwell:** maintenance of a content between preregistered lower and upper temporal bounds.
- **Causal recurrence:** a directed return relation through which an earlier content-supporting state affects a later state beyond continuing input and passive decay.
- **Feedforward duration:** extended activity without a causal return path.
- **Metastability:** temporary stability that remains capable of transition and decay.
- **Passive trace:** stored or slowly decaying information that does not actively maintain the content-bearing process.
- **Rescue design:** restoration of a targeted causal role after disruption to test whether the predicted anchor pattern returns.
- **Temporal character Π_c :** the direction, rate, ordering, curvature, dwell pattern, and related features of how an admitted experience unfolds.
- **Working-memory maintenance:** continued usability of information for later tasks, often involving executive or report-related processes.

Looking ahead

Even a perfectly designed recurrence study does not force a positive or negative verdict when the assessment is inapplicable, poorly executed, conflicted, or too imprecise. The next chapter turns from what the system may be doing to what the evidence actually licenses us to say.

Transition notebook: from mechanism to verdict

Give the same apparent recurrence effect to two assessment teams. Team A has a validated estimator, adequate sensitivity, a frozen boundary, and a selective intervention. Team B has an unvalidated transport and a drifting candidate boundary. Individually write the narrowest result each team may report; then compare answers in a group and identify whether disagreement concerns the system or the license. Carry both cases into Chapter 7 and classify them without changing the numerical effect. The exercise succeeds only if identical numbers can receive different evidential statuses for explicit reasons.

Chapter 7 - What Does the Evidence License?

Opening puzzle. Two assessments produce the same score: .72. The positive threshold is .70. In the first assessment, the measure has been validated for the patient's injury class, the sensory route is intact, data-quality checks pass, and two partly independent measures agree. In the second, the measure was calibrated only in healthy adults, the patient has unresolved hearing loss, muscle activity contaminates the signal, and a credible imaging measure points in the opposite direction. Should the identical numbers receive identical interpretations?

If a test is treated as a score-producing machine, the answer may seem obvious: both pass. If a test is treated as an inference whose validity conditions can fail, the answer changes. A number has evidential meaning only within a declared target, system, method, model, and validity envelope. The CØ / N* research program therefore uses four scientific outcomes - positive, negative, indeterminate, and unlicensed - and checks applicability and quality before interpreting evidential strength.

Plain language. The state of a system and the state of our evidence are different things. A system may be conscious or nonconscious when our present assessment cannot responsibly settle the matter.

Chapter map

This chapter develops an inference-control framework that applies both within CØ / N* and across rival theories. You will learn to:

- index a conclusion to a complete assessment tuple;
- apply the precedence rule before reading a score;
- distinguish positive, negative, indeterminate, and unlicensed outcomes;
- explain why positive and negative licenses are asymmetric;
- diagnose unresolved cases with a five-part cause profile;
- report coverage and unresolved-output rates rather than accuracy alone; and
- keep scientific classification separate from precautionary action.

The framework does not claim that consciousness comes in four metaphysical kinds. It says that scientific attributions require more than a binary button.

1. The assessment tuple prevents conclusions from drifting

Write an assessment as:

$$A = \{ S, C, T, E, M, V \}.$$

S is the candidate system. C is the target form of consciousness - for example, minimal phenomenal presence during a specified episode rather than wakefulness, reportability, or rich self-awareness. T is the test or battery. E is the available evidence. M is the inferential model connecting evidence to the target. V is the **validity envelope**: the populations, substrates, tasks, boundaries, recording conditions, and auxiliary capacities for which that interpretation has support.

The tuple matters because conclusions tend to expand in ordinary language. Evidence of one moment of command following can be redescribed as continuous rich awareness. A signal measured in cortex can be treated as a verdict about a whole brain-body system. A measure validated in adult mammalian brains can be exported to an artificial network because its numerical range looks familiar. Each step changes one element of A without announcing the change.

Why it matters. A precise answer to the wrong assessment question is not a precise answer to the right one. Indexing the claim makes target substitutions visible.

Analogy: a passport at the wrong border. A passport can be genuine and still fail to license entry if it is expired, belongs to someone else, or is presented in a jurisdiction that requires a different document. The problem is not that the ink is weak. The document lacks the required relation to this person, place, and time. Similarly, a neural score can be measured accurately while lacking a validated interpretation for the present system or target.

Where the analogy breaks. Legal validity is created by institutional rules, whereas scientific validity is earned through evidence, calibration, causal understanding, and transport. A border officer may have final authority; no consciousness researcher has comparable authority over nature. The analogy concerns conditional license, not conventional truth.

The validity envelope should be prospective and inspectable. It records whether the required sensory route was available, whether the candidate boundary is stable, whether the procedure transfers to the substrate, whether auxiliary capacities such as comprehension were needed, and whether the model was validated at the chosen grain. A study conducted outside the envelope may still generate useful data. What it cannot do is inherit the original interpretation without a transport argument.

Source note. *Indeterminacy as a Scientific Result* introduces the assessment tuple to prevent target, system, and validity conditions from being silently widened after a result is known.

2. Four outcomes and a precedence rule

The decision rule $D(A)$ can be expressed as five ordered branches:

1. **Unlicensed** if $a=0$ or $q=0$.
2. **Indeterminate** if $a=1, q=1, c=1$.
3. **Positive** if $a=1, q=1, c=0$ and $LB(L) \geq \tau_+$.
4. **Negative** if $a=1, q=1, c=0$ and $UB(L) < \tau_-$.
5. **Indeterminate** otherwise.

Here a means applicability, q means execution and data quality, c means unresolved conflict among credible surrogates, and L is a validated evidence grade or score. $LB(L)$ and $UB(L)$ are lower and upper uncertainty bounds. τ_+ is the rule-in threshold; τ_- is the rule-out threshold. Because the positive rule uses $\geq$ and the negative rule uses $<$, equality at τ_+ passes the rule-in boundary, whereas equality at τ_- does not license rule-out.

Method note. This interval rule is a conservative teaching extension, not a quotation of a universal source formula. The source framework can be written with a scalar score and calibrated rule-in and rule-out thresholds. Here a positive requires the lower uncertainty bound to meet the rule-in threshold, while a negative requires the upper uncertainty bound to fall strictly below the rule-out threshold. An empirical protocol must justify and preregister its own interval construction and equality rules rather than inherit these classroom conventions automatically.

The order is the intellectual heart of the rule. First ask whether the assessment applies. Next ask whether it was executed adequately. Then ask whether credible surrogate measures conflict. Model-family and boundary disagreement are reported separately through O_M and O_B , rather than folded into the scalar conflict flag c . Only after those checks may the score enter a threshold rule. A high number does not repair failed applicability or poor data quality.

The outcomes mean:

- **Positive:** applicable, adequately executed, nonconflicted evidence licenses attribution of the specified target under the declared model and envelope.
- **Negative:** an applicable and sufficiently sensitive assessment had a fair detection opportunity and entered a validated rule-out region.
- **Indeterminate:** the assessment is interpretable and informative but does not discriminate enough, either because it lies in a gray zone or because credible evidence remains unresolved.
- **Unlicensed:** a condition required to interpret the assessment failed. The output concerns the procedure's inferential status, not the system's consciousness.

Do not infer. “Unlicensed” does not mean “probably negative,” and “indeterminate” does not mean “partly conscious.” Both are evidence states. They require different repairs.

An unlicensed result can sometimes be quarantined. If one measure in a preregistered battery fails quality checks, remove it only according to the prospective rule. If the remaining independently validated measures still form a complete battery, they may yield their own outcome. If the failed measure was the sole estimator of a required component, the bundle remains unlicensed. Analysts may not discover this dependence after seeing which removal gives the preferred verdict.

Analogy: a courtroom with admissibility and weight. A judge first decides whether evidence is admissible; a jury then decides how much weight admitted evidence deserves. A vivid but illegally obtained or misidentified item does not become admissible because it strongly favors one side. Likewise, applicability and quality precede evidential strength.

Where the analogy breaks. Scientific inference has no sharp separation between judge and jury, and validity is often graded rather than settled by a single ruling. Moreover, courts answer legal questions under institutional standards, while science revises its standards through evidence. The analogy illuminates precedence, not a complete philosophy of evidence.

3. Positive and negative conclusions have different burdens

The **Positive License Principle** requires that a marker's target relevance and specificity be validated, the assessment apply, quality be adequate, and credible nonconscious or artifactual explanations be controlled. Complexity, flexible behavior, recurrent activity, self-report, or widespread activation may be evidence in a suitable design. None is a default positive simply because it sounds consciousness-related.

The **Negative License Principle** adds a different burden. A nondetection supports absence only when the indicator would probably have been detectable if the target were present, the necessary auxiliary capacities were intact or controlled, the sensory and motor routes were fair, the procedure applied to this system, and alternative causes of nondetection were excluded. A test can have strong positive specificity while having poor sensitivity for negatives.

Consider a command-following task. A reliable, correctly timed response may strongly support that the person understood and followed the command. Failure to respond could result from absence of consciousness, but also from hearing loss, language impairment, fatigue, motor paralysis, memory failure, fluctuating arousal, or an unsuitable imagery instruction. The positive and negative outcomes are not mirror images.

Checkpoint. Before calling a result negative, complete the sentence: "If the target were present, this procedure would probably have detected it because ..." If that sentence cannot be supported, the result is not yet a licensed rule-out.

Thought experiment: The silent diver

A deep-sea diver communicates through a hand-squeeze code. During an emergency, the diver stops squeezing. A monitor shows no hand movement, but the suit's audio system may have failed and one glove is mechanically jammed.

Stage 1. Specify the target claim. Is it hearing the command, understanding it, intending to respond, being conscious, or all four?

Stage 2. Identify which parts of the assessment tuple are unstable.

Stage 3. Describe a result that would be indeterminate and one that would be unlicensed.

Stage 4. Design a low-demand alternative route and state what must be validated before it can support a negative conclusion.

The diver case makes a general point: a missing output can be evidence about consciousness only through a chain of auxiliary conditions. Making those conditions explicit is not special pleading. It is the difference between detecting absence and failing to detect.

4. Cause profiles make unresolved results productive

A bare "uncertain" result does little scientific work. The framework therefore attaches a cause profile:

$$P_R(A) = \{e, s, m, b, t\}.$$

e is **evidential scarcity**: valid data are too imprecise or sparse. ***s*** is **surrogate discordance**: credible indicators conflict. ***m*** is **model dependence**: live inferential models yield different classifications. ***b*** is **boundary or grain sensitivity**: reasonable candidate definitions reverse the result. ***t*** is **transport uncertainty**: the evidence-to-target relation may not carry to a new population or substrate.

These entries diagnose the inference, not dimensions of experience. A case with high boundary sensitivity is not an experience with “more boundary.” It is a case whose classification changes when reasonable system definitions change.

Notation warning. Symbols in this research program are locally scoped. In this profile, ***s*** means surrogate discordance and ***m*** means model dependence. Elsewhere, the same letters may denote an availability selectivity threshold or a minimum recipient count. Define local notation when it first appears instead of assuming global meanings.

Each cause suggests a different response:

- Scarcity calls for more power, better precision, broader temporal sampling, or a more sensitive measure.
- Discordance calls for construct decomposition, replication, causal adjudication, or investigation of auxiliary demands.
- Model dependence calls for a diagnostic prediction or an explicit justification for weighting models.
- Boundary sensitivity calls for consciousness-independent individuation and boundary perturbation.
- Transport uncertainty calls for construct, causal-role, calibration, boundary, and regime preservation tests.

This yields a useful asymmetry. Indeterminate evidence may become decisive through more informative evidence. An unlicensed assessment must first be repaired. Repeating an invalid test with more trials can produce a highly reproducible nonanswer.

Thought experiment: The translation machine

A neural measure predicts reports in awake adult humans with strong held-out accuracy. A team applies the same numerical pipeline to a synthetic agent built from unfamiliar components. The agent’s score falls deep in the human positive range. The team has no evidence that the measured feature plays the same causal role, that the candidate boundary is comparable, or that the agent’s internal timing lies within the measure’s validity range.

Stage 0. Before inspecting the agent score, state the exact source-domain claim being transported: target, candidate, construct, calibration population, validity envelope, thresholds, and strongest licensed conclusion. A number cannot be transported responsibly until the interpretation attached to it has been frozen.

Stage 1. Classify the human result and the agent result separately. Do not assume that a shared number has a shared meaning.

Stage 2. Build a cause profile for the agent assessment.

Stage 3. Propose a staged transport program. Which evidence would test construct preservation, causal-role preservation, calibration, and boundary/regime preservation?

Stage 4. Suppose the team has ethical reasons to avoid harming the agent during validation. Explain how a precautionary policy could be justified without changing the scientific status.

5. Model and boundary disagreement are results, not noise

Suppose several live inferential models are frozen in advance. We can display their outputs as:

$$O_M(A) = \{D_m(A) : m \in M\},$$

where M is the declared model family. Similarly, for a mapped family of reasonable candidate boundaries B :

$$O_B(A) = \{D(A; B) : B \in B\}.$$

If these sets contain more than one outcome, comparative model or boundary indeterminacy remains even when each individual analysis is internally decisive. The sets are not probability distributions. They are transparent displays of how conclusions vary across declared alternatives. Analysts must not remove a model or boundary after learning that it produces an inconvenient result.

This makes the framework theory-plural without pretending to be assumption-free. Integrated Information Theory, workspace theories, recurrent-processing theories, higher-order theories, and CØ / N* can assign different roles to the same observation. Their disagreements become scientifically useful when they yield frozen, diagnostic predictions. Merely counting theories or averaging their conclusions can hide rather than resolve the conflict.

Method note. Record the version, date, target, and operational commitments of every rival.

“Workspace theory predicts X” is not sufficiently precise when different workspace formulations disagree about report, broadcasting, anatomy, or timing.

The four-outcome framework can itself fail. If trained reviewers cannot reliably distinguish indeterminate from unlicensed cases, if the extra category improves neither calibration nor error detection, if unresolved outputs absorb adverse findings only after inspection, or if protocol burden exceeds epistemic benefit, the framework needs revision. A preregistered **failure budget** can specify the unresolved-output rate or pattern that triggers methodological repair.

6. Coverage, selective risk, and the temptation to hide hard cases

An assessment should report more than accuracy among the cases it classifies. **Coverage** is the proportion receiving positive or negative outcomes. Let r_I be the indeterminate rate and r_U the unlicensed rate. **Selective risk** is the error rate among classified cases, when a defensible independent reference exists.

Suppose Tool A attempts 100 assessments and returns 30 positive, 20 negative, 25 indeterminate, and 25 unlicensed. Coverage is .50, $r_I = .25$, and $r_U = .25$. If five of the 50 classified cases disagree with a defensible reference, selective risk is .10. Tool B classifies 20 easy cases with no observed errors and abstains or fails on the remaining 80. Its measured selective risk is lower, but its coverage is only .20. It is not automatically the better general tool.

Why it matters. A test can look excellent by avoiding difficult cases. Coverage and unresolved-output rates reveal where its apparent accuracy comes from.

Confidence intervals are also needed. An unresolved rate based on 12 cases is not the same evidential object as the same rate based on 1,200 cases. Outcomes should be stratified by population, boundary, severity, and transport domain, because a single average may hide systematic failure in precisely the cases for which the test is needed.

7. Scientific status and practical action must remain separate

Let D be the scientific status and let Λ represent a loss structure: the costs of false denial, false attribution, delay, burden, and irreversible action. A practical decision uses both. High stakes, asymmetrical errors, and a low-cost reversible option can justify precaution under indeterminacy.

For example, imagine an indeterminate patient assessment. A short, low-burden alternative communication trial costs little. Missing covert command following would be serious. Trying the communication route can be rational even though the scientific classification remains indeterminate. Conversely, a high-risk intervention with low expected benefit may not be justified by the same evidence.

Do not infer. Precaution does not transform $D = \text{Indeterminate}$ into $D = \text{Positive}$. The practical policy is a conclusion from evidence plus values and consequences, not a hidden consciousness verdict.

This distinction matters in animal welfare, organoid research, artificial systems, and clinical care. It allows protective action without corrupting the scientific record. It also prevents a negative practical recommendation from masquerading as evidence of absence.

Source note. *Indeterminacy as a Scientific Result* treats science-action separation as a guardrail. The paper's contribution is methodological; it does not supply calibrated probabilities or one universal policy threshold.

8. Evidence, limits, and rival approaches

Three-way decision methods and classifiers with a reject option already show that abstention can improve error control. The four-outcome framework adds a distinction between valid nondiscrimination and failed interpretability. A reject option usually assumes the classifier applies and declines because confidence is low; that resembles indeterminate. Unlicensed covers cases in which the score's intended interpretation has not been established.

Bayesian approaches can represent uncertainty continuously, but a precise posterior can still rest on a failed target, boundary, likelihood, or transport assumption. Probabilities do not remove the need for applicability checks. Conversely, threshold categories should not erase useful probability distributions. The approaches can be combined: report calibrated uncertainty within the licensed part of the model, along with categorical status and cause profile.

Evidence supports this framework if independent reviewers apply it reliably, cause profiles predict efficient repair strategies, coverage and unresolved rates improve calibration and transparency, and preregistration reduces opportunistic reclassification. Evidence weakens it if the categories cannot be distinguished, add no predictive or practical value, or become asymmetrical escape routes for favored theories.

The framework's strongest protection against misuse is prospective commitment. Freeze applicability criteria, quality rules, conflict handling, thresholds, model and boundary families, quarantine rules, and failure budgets before inspecting critical outcomes. When a result is adverse, the response should be the one already attached to that cause - not an improvised new category.

Practice studio

Quick checks

1. Why can two identical scores license different conclusions?
2. What is the difference between indeterminate and unlicensed?
3. State one condition required for a licensed negative result.
4. What does the cause-profile entry t diagnose?
5. Why is O_M not a posterior probability distribution?
6. How can a tool have low selective risk but poor general usefulness?
7. Why may precaution be rational without a positive scientific result?

Individual transfer task: Audit a published or hypothetical test

Choose one consciousness indicator or create a carefully specified hypothetical one. Write its assessment tuple. Define its positive and negative license conditions, validity envelope, quality rule, conflict rule, and thresholds. Construct four cases - one for each scientific outcome - without changing the target. For every unresolved case, provide a cause profile and the smallest credible repair. End by identifying one way the framework itself could be misused to protect the test from falsification.

Collaborative exercise: Evidence tribunal

Materials and frozen rule. All cases are synthetic. The positive threshold is $\tau_+ = .70$ and the negative threshold is $\tau_- = .30$. Applicability and execution quality must pass before an interval is interpreted, and credible surrogate conflict takes precedence over either threshold. Every packet concerns the same severe-injury patient, the same stable patient-level referent, and minimal phenomenal presence during one specified 30-second trial.

Assign one packet to each team:

- **Packet A:** applicability and quality pass; $L = .78[.73, .83]$; two independently validated surrogates agree; the candidate referent is stable and the positive bearer is resolved.
- **Packet B:** applicability and quality pass; $L = .55[.47, .63]$; there is no credible surrogate conflict.
- **Packet C:** recording quality and injury-class calibration pass; $L = .18[.12, .24]$; severe hearing loss remains unresolved even though the task uses auditory commands.
- **Packet D:** applicability passes with sensitivity .93 in this injury class; quality, sensory route, detection opportunity, auxiliary capacities, and candidate stability pass; $L = .19[.13, .25]$.

For a separate action decision, a reversible visual communication trial costs one teaching-loss unit, while missing genuine command following costs 20. The packet supplies no probability of genuine command following. Teams must therefore state any decision assumption after freezing the scientific outcome; the action recommendation may not alter that outcome.

Teams of four use these roles:

- The **positive advocate** presents the strongest admissible rule-in argument.
- The **negative advocate** presents the strongest admissible rule-out argument.
- The **validity officer** audits target, applicability, execution quality, boundary, auxiliary demands, and transport before either argument is weighed.
- The **chair-recorder** applies the frozen precedence rule, records dissent, and separates the scientific verdict from the action recommendation.

After the first verdict, teams exchange packets and rotate roles. In the second round, every participant must defend a different outcome from the one initially favored, using only evidence that passes the validity officer's audit. The final product is a comparison table with packet, applicability, quality, conflict status, interval relation, scientific outcome, cause-profile emphasis, smallest repair, prohibited inference, and separate action recommendation. The table must show which changed fact, not which rhetorical style, changed the license.

Collaborative discussion: Does abstention expose or protect a theory?

Form two panels. One argues that indeterminate and unlicensed outputs create escape routes; the other argues that forced binary classification creates invalid certainty. Each panel must specify a prospective pattern that would count against its own position. Then form mixed groups with a methods referee. Produce a failure budget: a maximum unresolved rate, concentration of unresolved cases, reviewer-disagreement rate, or repair failure that would make the method scientifically unhelpful. Explain why the threshold is defensible rather than convenient.

Counterexample challenge

Construct a case in which the four-outcome rule seems to classify too cautiously, and a second case in which it seems not cautious enough. For each, keep the target and evidence fixed while varying only one rule: applicability, quality, conflict precedence, or threshold placement. State whether the problem calls for recalibration, a revised validity envelope, a different model family, or abandonment of the categorical framework.

Exit ticket

Write one sentence that begins, “More data would help because ...” for an indeterminate case, and one that begins, “More of the same data would not help because ...” for an unlicensed case.

Chapter summary

Scientific classification should follow inferential license rather than numerical appearance. The assessment tuple fixes the system, target, test, evidence, model, and validity envelope. Applicability and quality precede conflict handling; conflict precedes thresholds. Positive, negative, indeterminate, and unlicensed are evidential outcomes, not amounts of experience. Positive and negative claims have asymmetric burdens, and unresolved cases should receive cause profiles that direct distinct repairs. Coverage and unresolved-output rates reveal what accuracy alone can hide. Practical action adds stakes and error costs but must not rewrite the scientific result.

Key terms

- **Assessment tuple A :** the indexed system, target, test, evidence, model, and validity envelope.
- **Cause profile:** a record of evidential scarcity, surrogate discordance, model dependence, boundary sensitivity, and transport uncertainty.
- **Coverage:** the proportion of attempted cases receiving positive or negative classifications.
- **Indeterminate:** a valid, informative assessment that does not discriminate sufficiently.
- **Negative License Principle:** nondetection supports absence only under validated sensitivity and a fair detection opportunity.
- **Positive License Principle:** an impressive marker supports attribution only when relevance, specificity, applicability, and quality are established.
- **Selective risk:** error among the cases a test chooses to classify.
- **Unlicensed:** a meta-output indicating that conditions required to interpret the assessment failed.
- **Validity envelope:** the domain in which the evidence-to-target interpretation is supported.

Looking ahead

The four-outcome framework tells us what a body of evidence licenses. It does not yet show that every term in a conjunctive theory is doing nonredundant work. Chapter 8 asks how to test a stronger claim: does each component earn its place?

Chapter 8 - Does Every Conjunct Earn Its Place?

Opening puzzle. A security system opens a vault only when four indicators are positive: electricity is available, the keycard is valid, the fingerprint matches, and the code is correct. In every recorded success, all four indicators are positive. In every recorded failure, all four are negative because the building lost power during emergency drills. The four-indicator rule predicts the dataset perfectly. So does the keycard alone. So does the fingerprint alone. So do many two-indicator rules. Has the dataset shown that each indicator is necessary?

The CØ / N* theory has the same structural problem. Its profile rule is conjunctive: viability V , integration N_1 , report-independent system-wide availability N_2 , and recurrent stability N_3 must all pass. A conjunction advertises more than a useful bundle-level association. It says every term is needed. Yet data in which the components rise and fall together cannot identify the necessity of any particular term.

Why it matters. Writing “and” between four variables is easy. Showing that nature requires every “and” is a separate research program.

Chapter map

This chapter teaches you to:

- distinguish logical, incremental empirical, and causal necessity;
- derive the exact cases on which a full conjunction and a leave-one-out rival disagree;
- explain why diagnostic coverage is a result rather than a nuisance;
- separate criterion, mechanism, and measurement ablations;
- apply an anti-circularity firewall when no consciousness oracle exists;
- propagate component rules through the candidate lattice before making bearer-level claims; and
- classify an audit as indispensable, redundant, causally coupled, indeterminate, or unlicensed.

The main lesson is diagnostic: a model can predict ordinary cases well while its internal factorization remains almost completely untested.

1. Three different meanings of “necessary”

The word *necessary* can name three claims.

Logical necessity means a term appears in the written formula. If the rule is $A \wedge B$, then the formula returns false when A is false. Inspection of the syntax establishes this. It tells us what the theory says, not whether the world requires A .

Incremental empirical necessity means the full model predicts independent held-out anchors better than a declared rival that omits the term, specifically in cases where the two rules disagree. This conclusion is relative to the anchor family, population, operationalization, and rival class.

Causal necessity means selectively changing the term’s physical realization changes independent anchors while the retained components, candidate referent, and operating regime remain equivalent. This is stronger than criterion comparison, but it remains relative to the mechanism and domain tested.

Plain language. A component can be required by a spreadsheet formula, improve predictions in a dataset, and play a causal role in a physical system. Those are three achievements, not three ways of saying the same thing.

Analogy: ingredients in a recipe. A printed bread recipe may list salt. Removing the word “salt” changes the recipe. A taste test may show that the original predicts preferred loaves better than a no-salt recipe. A controlled baking experiment may show that salt changes gluten behavior while flour, water, yeast, temperature, and proofing are held stable. These correspond roughly to logical inclusion, empirical value, and causal contribution.

Where the analogy breaks. Recipe goals are chosen and multidimensional; phenomenal presence is not a flavor score. Ingredients also interact, and removing one can change the whole process - precisely the problem that makes selective mechanism ablation difficult. The analogy clarifies levels of necessity but does not provide a consciousness anchor.

This distinction prevents common overclaims. A lesion that reduces behavior does not automatically show causal necessity for phenomenal presence. A leave-one-out model that predicts worse does not show metaphysical necessity against every imaginable theory. And a component’s appearance in the definition cannot be cited as empirical evidence for its own indispensability.

Source note. “Ablating N*: Does Every Conjunct Earn Its Place?” develops this three-way distinction and treats component necessity as a held-out, anchor-relative audit rather than an automatic consequence of the conjunction.

2. The diagnostic-set result

For case x , write the component vector as:

$$X(x) = (X_0, X_1, X_2, X_3) = (V, N_1, N_2, N_3).$$

The full profile rule is:

$$M_*(x) = \bigwedge_{j=0}^3 X_j(x),$$

meaning that every component must pass. The rule with component k removed is:

$$M_{-k}(x) = \bigwedge_{j \neq k} X_j(x).$$

The two rules disagree only in the diagnostic set:

$$D_k = \left\{ x : X_k(x) = 0 \wedge \bigwedge_{j \neq k} X_j(x) = 1 \right\}.$$

Read the formula in words: the omitted component fails while every retained component passes. Why is this the only disagreement cell? If the full conjunction passes, then the ablated conjunction must pass because it asks for fewer things. If any retained component fails, both rules fail. Only a single failure in the omitted term lets the ablated rule pass while the full rule fails.

For N_2 , the exact profile is $(V, N_1, N_2, N_3) = (1, 1, 0, 1)$. Cases at $(1, 1, 1, 1)$ cannot distinguish the rules because both predict positive. Cases at $(0, 0, 0, 0)$ cannot distinguish them because both predict negative. More examples from those corners increase precision about bundle-level prediction but add no direct information about whether N_2 earns its place. For the deterministic Boolean profile rules just defined, a study containing no members of D_k gives the two rules identical casewise predictions. Any apparent performance difference must then arise from additional estimation, weighting, or modeling choices that require separate justification.

Checkpoint. Before comparing two models, write the set of cases on which their predictions differ. If the study contains none, a performance tie is guaranteed and a performance “win” usually comes from something other than the declared rules.

Analogy: testing bridge supports only on calm days. Suppose a bridge has four support systems. Engineers inspect it only when every support is intact or when a flood disables all four. Those observations may show that the complete bridge works in normal conditions and fails in catastrophe. They do not show what each support contributes. To test Support 3, engineers need cases where Support 3 changes while the others remain functional.

Where the analogy breaks. Ethical engineering does not require damaging a bridge with people on it, and naturally occurring near-misses may substitute for intervention. Consciousness audits face an additional difficulty: there is no universally accepted direct outcome measure. The analogy explains diagnostic variation, not how to construct independent anchors.

Empirical component estimates have uncertainty. A conservative diagnostic region requires the upper bound for the omitted component to fall below its threshold and the lower bounds for retained components to clear theirs. A point estimate just below threshold does not qualify if its interval straddles. The **diagnostic margin** is the distance to the nearest relevant threshold. Larger margins reduce ambiguity about membership, although they do not guarantee that the case is representative.

3. Coverage determines the scope of the result

Let Γ_k denote profile-level diagnostic coverage: the target-distribution weight in the region where the full and ablated rules disagree. An engineered case can produce a dramatic full-model advantage while representing only a tiny fraction of the intended population. The local result may be real, but the broad claim remains weak.

Consider a study of N_2 with threshold .60. One case has an N_2 upper bound of .42, while lower bounds for V , N_1 , and N_3 are .73, .80, and .69. Its nearest-threshold margin is:

$$m_2 = \min(.60 - .42, .73 - .60, .80 - .60, .69 - .60) = .09.$$

This is a conservative N_2 diagnostic case. Suppose the full model outperforms the ablated rival on held-out anchors, but diagnostic cases carry only five percent of the population weight. The study supports local incremental value. It does not establish that N_2 frequently discriminates the models across the population.

Coverage also has a bearer-level counterpart. The profile rules operate on each frozen candidate referent, but the theory's final positive subject attribution uses a candidate lattice and inclusion-minimality. For model m :

$$B_m^+(c) = \text{Min}_{\subseteq} \{ S \in P_{\Theta} : M_m(S, c) = 1 \}.$$

P_{Θ} is the frozen family of causally admissible candidates under specification Θ . The operator $\text{Min}_{\subseteq}$ retains candidates that have no qualifying all-pass proper subsystem. A diagnostic case somewhere in the lattice is necessary for the full and ablated bearer sets to differ, but it is not sufficient. The diagnostic candidate may be nonminimal under both rules. Removing one component can also make a smaller candidate pass, causing a larger candidate to disappear from the ablated minimal set.

Do not infer. A profile-level component result is not automatically a complete claim about where the subject is. Recompute the candidate lattice under every rival, then apply the same mapping, stability, and separability licenses.

This distinction prevents two opposite errors. Freezing the original positive bearer after ablation protects the theory from relocation or loss. Treating every all-pass candidate as a subject ignores minimality. The fair comparison freezes candidate generation and rules but allows each model's bearer output to change.

4. An anti-circularity firewall without a consciousness oracle

The audit cannot define “conscious” as “passes $V \wedge N^*$ ” and then show that the full model predicts those labels better than an ablated rival. That would be syntactic self-confirmation. Instead, separate two evidence streams.

The **component stream** X estimates V , N_1 , N_2 , and N_3 . The **anchor stream** Y contains independently constructed evidence that may include calibrated reports in appropriate cases, low-demand discrimination, metacognitive structure, physiological relations, perturbational signatures validated outside the critical comparison, and theory-light clinical outcomes. No anchor is an infallible consciousness oracle. The aim is comparative prediction across partially independent evidence.

The firewall requires:

1. treating the identity claim as a hypothesis, never a database definition;
2. separating component construction from anchor construction;
3. splitting development, calibration, and final adjudication data;
4. prohibiting one measure from constructing and validating the same component in the same fold;
5. freezing boundary family, grain, content, regime, interval, thresholds, interventions, anchor battery, translation rule, scoring rule, and reasonable-specification family; and
6. comparing full and ablated models on held-out diagnostic cases while recomputing each bearer set.

Method note. Draw a lineage diagram for every variable. If two differently named quantities share the same sensor, preprocessing step, learned labels, or selection rule, declare the overlap, run a leave-channel-out analysis, and downgrade the conclusion if it does not survive.

Thought experiment: The circular scholarship test

A university defines “excellent scholarship” as having four features: clarity, novelty, rigor, and impact. It trains a model to score the features. The final excellence label in its database is then set to 1 exactly when all four scores pass. Researchers remove the novelty term and report that prediction of “excellent scholarship” becomes worse.

Stage 1. Identify the circular step.

Stage 2. Design an anchor vector that preserves disagreement and is not constructed from the four feature scores.

Stage 3. Specify development, calibration, and adjudication splits.

Stage 4. Explain why even a clean held-out advantage would remain relative to the chosen anchors rather than proving the true essence of scholarship.

The consciousness audit has the same logic but higher stakes. Independence does not provide certainty. It provides a comparison that can genuinely go against the theory.

5. Three ablations ask three questions

The word *ablation* often hides important differences.

A **criterion ablation** removes a term from the decision rule. It asks whether the term adds held-out predictive discrimination. No physical system must be damaged.

A **mechanism ablation** changes a proposed physical realization. For a target in $\{N_1, N_2, N_3\}$, it asks about causal contribution only if the manipulation selectively changes that target while preserving retained components, viability, referent identity, and mapping. A V -targeted mechanism audit requires the separate logic stated below.

A **measurement ablation** removes a sensor, estimator, feature family, or preprocessing route. It asks whether the component classification is robust to measurement choices. It says nothing directly about what would happen if the physical component were deleted.

These operations can yield different results. A component may be empirically indispensable while no selective physical intervention is currently possible. A mechanism intervention may change several components together, leading to a causally coupled status. A measurement route may be removable without showing that the underlying component is redundant.

For an intervention ι_k intended to reduce component k , define a matched-control change vector:

$$\delta(\iota_k) = (\Delta s_0, \Delta s_1, \Delta s_2, \Delta s_3),$$

where the four scores correspond to (V, N_1, N_2, N_3) . For $k \in \{1, 2, 3\}$, a selectivity license requires the full uncertainty interval for the target reduction to meet the preregistered direction and magnitude, $\Delta s_k \leq -\eta_k$; every retained change interval must lie inside its equivalence band; every retained post-intervention lower bound must still clear its gate; and V must remain above its gate. The validity of the measurements and the frozen candidate referent's autonomy and cross-condition mapping must also remain licensed. Positive-bearer membership is deliberately not a condition of selectivity; it is an outcome to recompute.

A V -targeted audit must distinguish two questions that the phrase “viability remains licensed” can otherwise blur. The first is whether the V gate passes; the second is whether the V estimator, candidate mapping, anchor battery, and operating-regime interpretation remain valid. In a diagnostic V case, the gate may fail while those assessment licenses remain intact. If reducing V also destroys the validity envelope or makes the referent unmappable, the mechanism result is unlicensed rather than evidence that V is causally necessary. Criterion-level comparison of the full rule with a V -ablated rival remains possible on independently valid cases, but it does not inherit causal status from the label “viability intervention.”

For a preregistered scalar summary $g(Y)$ of the independent anchors, ψ_k is the target-versus-control anchor contrast among licensed diagnostic cases. It is an evidential-anchor contrast, not an average treatment effect on directly observed consciousness.

Plain language. Calling an intervention “the recurrence lesion” does not prove that it changed only recurrence. Names describe intentions. Measurements and uncertainty intervals establish selectivity.

Thought experiment: The disappearing orchestra

An engineer mutes a feedback channel in a musical robot to test N_3 . After the change, recurrence falls, but so do integration, power stability, content fidelity, and communication among modules. The original candidate boundary also becomes unstable because two modules operate independently.

Stage 1. Classify the intervention as selective, coupled, or unlicensed for a causal N_3 claim, and list the evidence needed for your classification.

Stage 2. Separate what the manipulation may show about the physical mechanism from what it can show about phenomenal presence.

Stage 3. Redesign the intervention or propose a criterion-level audit that does not require physical selectivity.

Stage 4. Describe the possible post-intervention bearer outputs: loss, replacement, plurality, or unresolved attribution.

6. Held-out risk and five audit outcomes

Let $R_k(m)$ be the held-out risk of model m on the independent anchor pattern for diagnostic cases. Lower risk is better. Define the incremental value of component k as:

$$\Delta_k = R_k(-k) - R_k(*).$$

A positive Δ_k means the ablated model has higher risk; a negative value favors removal. But a point estimate is not a verdict. Before declaring a component indispensable, the lower uncertainty bound should exceed a preregistered meaningful-effect margin δ_k^{min} , diagnostic coverage must be adequate, flexibility must be matched, evidence streams must be independent, and reasonable specifications should yield a stable conclusion.

The audit returns five possible statuses:

- **Indispensable:** adequate diagnostic coverage and a stable, meaningful held-out advantage; a selective intervention is additionally required for a causal-necessity claim.
- **Redundant:** adequate valid diagnostic coverage exists; the upper uncertainty bound for Δ_k lies below δ_k^{min} ; and the ablated rival transports at least as well under matched flexibility and without compensatory retuning.
- **Causally coupled:** a valid intervention cannot move the target component without moving retained components.
- **Indeterminate:** coverage is weak, intervals straddle, anchors conflict, or rivals tie.
- **Unlicensed:** boundary, viability, measurement, intervention, independence, or transport fails.

“Causally coupled” is not a consolation prize. It may reveal that the proposed factorization separates variables that cannot be independently realized in the tested system. That can motivate a combined component, a new intervention, or a change in the claim from universal factorization to biological implementation.

Measurement-route robustness, Ω_k , summarizes how often a component status survives the removal of plausible sensors or analysis routes. Positive-specification support, Π_k^+ , summarizes how often the lower risk-advantage bound clears the meaningful-effect criterion across a frozen family of boundaries, thresholds, causal metrics, and preprocessing choices. Neither frequency is a posterior probability of metaphysical necessity. Always report the distribution and the choices that reverse status.

7. Grouped ablations and strong minimality

One-at-a-time deletion tests weak minimality. Components may be jointly redundant or interact. For any nonempty removal set A :

$$M_{-A}(x) = \bigwedge_{j \notin A} X_j(x),$$

and the grouped diagnostic set is:

$$D_A = \left\{ x : \bigwedge_{j \notin A} X_j(x) = 1 \wedge \bigvee_{k \in A} X_k(x) = 0 \right\}.$$

The full and grouped rivals disagree when every retained term passes and at least one removed term fails. With four terms there are $2^4 - 1 = 15$ nonempty removal sets. Testing every set raises multiplicity and feasibility problems, so a mature program should prioritize scientifically motivated single deletions and coupled sets, then expand toward all proper submodels.

Strong empirical minimality means the full model outperforms every preregistered proper submodel under matched conditions. Even that would not establish sufficiency against unrelated nonconjunctive theories. A latent composite of integration and recurrence, for example, is not simply a grouped deletion; it is a revised factorization that deserves its own predictions.

Selection into diagnostic cases can introduce bias. If measurement error or a common cause affects both component scores and anchors, choosing cases because one score is low can enrich misleading anchor patterns. Safeguards include defining diagnostic membership without viewing held-out anchors, cross-fitting, negative controls, probabilistic membership near thresholds, and unconditional sensitivity analyses.

Do not infer. A rare synthetic near-miss may be excellent for construct separation while saying little about biological prevalence. Diagnostic informativeness and representativeness are distinct virtues.

8. Evidence, limits, and rival methodologies

The component audit resembles model selection, adversarial collaboration, causal intervention, and robustness analysis, but it combines them around a conjunctive identity claim. Ordinary lesion logic often asks whether a performance drops. The present audit asks whether the manipulated component was selectively changed, whether independent anchors behaved as predicted, and whether the candidate bearer was recomputed. Virtual lesions in artificial systems can help establish engineering separability, but they do not provide consciousness ground truth.

Evidence favors a conjunct when diagnostic cases exist, the full model gains stable held-out value over matched rivals, selective interventions and rescues produce the preregistered anchor pattern, and the result transports across defensible measurement routes and domains. Evidence against a conjunct includes valid redundancy across adequate diagnostic coverage, a negative risk difference favoring the ablated model, or repeated selective ablations that leave independent anchors unchanged.

An empty diagnostic set is not evidence for necessity. It is underidentification. A globally disruptive manipulation is not a clean falsifier. It is a coupled or unlicensed test. A failed operationalization may motivate a better estimator, but repeated post hoc replacement of estimators after adverse outcomes undermines the research program. Prospective revision criteria are therefore as important as favorable results.

Source note. The exact diagnostic-set theorem, the anti-circularity firewall, and the five statuses come from “Ablating N: *Does Every Conjunct Earn Its Place?*”. *The bearer-level propagation described here follows a cross-paper synthesis with Where Is the Conscious Subject?**

Practice studio

Quick checks

1. Write the exact N_2 diagnostic profile.
2. Why do all-pass and all-fail cases not identify individual conjuncts?
3. Distinguish criterion, mechanism, and measurement ablation.
4. What does diagnostic coverage add to an effect-size report?
5. Why must the anchor vector be independent of the component stream?
6. Name one way profile-level and bearer-level conclusions can diverge.
7. What does “causally coupled” tell researchers to reconsider?

Individual transfer task: Design a near-miss atlas entry

Choose one component. Specify a hypothetical case whose uncertainty intervals place it inside that component’s conservative diagnostic region. Define the target population weight, diagnostic margin, anchor vector, independence lineage, full and ablated rivals, and meaningful-effect threshold. Then add a candidate lattice with at least one proper subsystem. Explain what additional evidence is needed to estimate bearer-output disagreement. Do not calculate a favorable result by assumption; instead write prospective rules for every possible outcome.

Collaborative exercise: Component-audit laboratory

Teams of four use these roles:

- The **component theorist** writes the full, leave-one-out, and one scientifically motivated grouped rival.
- The **statistics lead** defines conservative diagnostic membership, margins, coverage, risk, and uncertainty rules.
- The **intervention lead** specifies target change, retained-component equivalence, candidate mapping, rescue, and bearer recomputation.
- The **anti-circularity auditor** traces each measure through development, calibration, selection, and held-out adjudication.

Teams first design a criterion audit without physical intervention. Next they add a mechanism test. Finally, another team gives them one complication: dual-use evidence, post-anchor case selection, off-manifold enrichment, referent-mapping failure, or coupled change. They must revise the status, not explain the complication away. The final claim must name the component, type of necessity, anchor family, population, specification neighborhood, rival, coverage, and unresolved limitation.

Collaborative discussion: Can necessity be studied without an oracle?

Assign oracle skeptic, comparative-method defender, causal-inference critic, and scope referee. The skeptic argues that no anchor vector can settle consciousness. The defender explains what comparative evidence can establish without pretending it is infallible. The critic searches for leakage, selection bias, and causal coupling. The referee marks every move from anchor-relative prediction to metaphysical language. After ten minutes, switch positions. Produce two lists: conclusions the audit can earn and conclusions it cannot earn even under ideal execution.

Counterexample challenge

Build a dataset in which all four one-at-a-time ablations favor the full model, yet a grouped or revised-factorization rival exposes redundancy. Then build a second dataset in which profile-level diagnostic coverage is high but bearer-output disagreement is nearly absent because the diagnostic candidates remain nonminimal. Describe both conceptually or numerically. State which advertised claim each dataset defeats.

Exit ticket

Complete the sentence: “A conjunction earns one conjunct only when ...” Include diagnostic disagreement, independent anchors, and scope in your answer.

Chapter summary

A conjunctive theory cannot establish the necessity of its parts from diagonal all-pass and all-fail data. The full and leave-one-out rules disagree only when the omitted component fails and every retained component passes. Those diagnostic cases must have adequate coverage and independent held-out anchors. Criterion, mechanism, and measurement ablations ask different questions; causal claims require selective change and preserved referent identity. Profile rules must also be propagated through the candidate lattice before making bearer-level claims. Five audit outcomes preserve the difference between support, redundancy, coupling, underidentification, and invalid inference. Grouped ablations test stronger forms of minimality but still do not establish sufficiency against every rival.

Key terms

- **Anti-circularity firewall:** prospective separation of hypotheses, component measures, anchors, data splits, and adjudication rules.
- **Bearer-output disagreement coverage:** target-distribution weight on which rival rules produce different licensed minimal-bearer sets.
- **Causal necessity:** anchor change following a selective change to a component’s physical realization.
- **Criterion ablation:** removal of a term from a decision rule.
- **Diagnostic coverage Γ_k :** target-distribution weight in the profile region where full and ablated rules disagree.
- **Diagnostic set D_k :** cases in which component k fails while every retained component passes.
- **Incremental empirical necessity:** held-out anchor advantage of the full model over a declared ablated rival in diagnostic cases.
- **Measurement ablation:** removal of an estimator, sensor, feature family, or pipeline.
- **Strong empirical minimality:** superiority of the full model over every preregistered proper submodel under matched conditions.

Looking ahead

The component audit concerns whether the model earns a gate for phenomenal presence. Even if that gate survives, it does not tell us why one present episode feels unlike another. Chapter 9 develops a separate, presence-gated bridge to phenomenal character.

Chapter 9 - From Phenomenal Presence to Phenomenal Character

Opening puzzle. Suppose two visual states both satisfy every licensed $CØ / N^*$ presence condition. One is an experience of red; the other is an experience of green. The presence model says that both episodes are there for the candidate system. It does not yet explain their difference. Now suppose a machine-learning model produces a beautiful two-dimensional plot in which red and green stimuli form separate clusters. Has the plot explained why the experiences differ?

Two mistakes tempt us. The first is to ask a presence theory to do work it was not designed to do. The second is to rename any structured neural or computational embedding a “quality space” and treat its geometry as phenomenal. The character extension avoids both by using a two-layer architecture. A licensed positive presence gate is established independently. Then a content-specific, perturbation-derived causal geometry is mapped to independently anchored phenomenal relations through a bridge that must predict held-out contents and intervention-driven changes.

Why it matters. Presence asks whether there is something it is like. Character asks what it is like. These questions interact, but their evidence and failure modes must not be merged.

Chapter map

This chapter will help you:

- preserve the two-layer separation between phenomenal presence and phenomenal character;
- distinguish causal-trajectory geometry from a decoder embedding or activation snapshot;
- build phenomenal anchors independently of the proposed causal geometry;
- test a bridge through held-out prediction, nuisance comparison, invariance, and deformation;
- interpret character up to the symmetries the evidence leaves unbroken;
- represent modality, relational location, intensity, valence, and temporal path without forcing one Euclidean space; and
- explain why bridge failure need not reverse a presence result.

The guiding question is not “Can we draw the states?” but “Can a frozen causal structure predict independent phenomenal relations, including how those relations change under intervention?”

1. Presence is a gate, not another quality coordinate

Let $P(S, c; \Theta_p)$ be the preregistered empirical presence decision for system S , content c , and frozen presence specification Θ_p . The value $P = 1$ represents a licensed positive result. Negative, indeterminate, or unlicensed results do not enter an ordinary phenomenal-character attribution in this framework. For a nonpositive presence result, write $\chi(S, c) = \perp_p$. The symbol means “character attribution not licensed by the presence gate”; it does not mean a zero vector, blank quale, or measured absence of quality.

In the bearer-complete version of $CØ / N^*$, a positive implementation requires an evidentially resolved boundary and membership in the estimated inclusion-minimal positive-bearer set. This is an evidence-relative decision. Researchers do not directly observe ontic membership - the mind-independent bearer relation posited by the theory; they estimate it under declared rules.

The second layer can be written:

$$\text{phenomenal character} = Q_{\Phi}(G_c, \Delta).$$

G_c is the content-specific causal-trajectory geometry, Δ is the evaluation interval, and Φ is the complete bridge specification: boundary, grain, perturbation family, features, anchor construction, model family, nuisance controls, data splits, invariances, and thresholds. Q_{Φ} is the character readout licensed by that bridge.

Two non-implications protect the architecture:

$$P(c_i) = P(c_j) = 1 \Rightarrow \not\Rightarrow \chi(c_i) = \chi(c_j),$$

and

$$G_{c_i} \cong G_{c_j} \Rightarrow \not\Rightarrow P(c_i) = P(c_j) = 1.$$

The first says that equally present states can differ in character. The second says that similar or isomorphic representational geometries do not manufacture presence. A nonconscious process can have rich structure.

Plain language. Passing the presence gate says an episode is admitted to the character question. It does not place the episode at “more red” or “less painful,” and an attractive map cannot open the gate from the other side.

Analogy: admission to a museum and location within it. A ticket determines whether a visitor may enter the museum. Once inside, a floor plan locates the visitor among galleries. Admission is not one coordinate on the floor plan, and drawing someone at the sculpture gallery does not prove they entered through a valid gate.

Where the analogy breaks. Museum admission is an institutional rule, while the presence gate is an empirical and constitutive hypothesis. A visitor exists independently of the ticket, whereas the theory concerns whether a physical organization is phenomenally present at all. The analogy only separates gating from differentiation.

Source note. *From Phenomenal Presence to Phenomenal Character* makes the two non-implications explicit and treats bridge outcomes as modular with respect to the presence result.

2. What a causal-trajectory geometry contains

G_c is not a single activation vector, an unconstrained latent space, or any embedding that separates stimulus labels. It is the neighborhood of content c within a structure derived from causal trajectories under a frozen family of admissible perturbations. Depending on the domain, it may include:

- distances among distributions of content-specific responses;
- directed transitions and their asymmetries;
- timing, direction, recurrence, and occupancy;
- route sensitivity under intervention; and
- local topology or stratification where one global metric fails.

The structure need not be Euclidean. A metric requires nonnegative, symmetric distances, identity of indiscernibles, and the triangle inequality. Real phenomenal or causal relations may be asymmetric, partially undefined, context-sensitive, or organized into modality-specific strata. A directed graph, pseudometric, ordinal structure, or stratified manifold may be more faithful than points on a flat plane.

The causal neighborhood must also be bounded. If every remote possible state were treated as constitutive of the current quality, an intervention on a never-visited distant region could change the predicted experience without changing any locally instantiated relation. The bridge specification should therefore freeze a neighborhood radius, occupancy rule, and time scale. A whole-structure version and a locally instantiated-relations version can then be compared by selectively changing remote geometry while preserving the local neighborhood.

Checkpoint. Ask of any proposed “quality geometry”: What interventions generated its relations? Which contents and transitions are included? Which metric assumptions were tested rather than plotted by default?

Analogy: a city map versus a photograph of traffic. A photograph records where cars happen to be at one moment. A useful transport map records routes, travel times, one-way streets, bottlenecks, and how closures change possible journeys. An activation snapshot resembles the photograph; a causal-trajectory geometry aims to resemble the route structure.

Where the analogy breaks. City routes are designed and externally surveyed, while neural or artificial causal relations must be estimated under intervention. Phenomenal relations may also involve valence, intensity, and meaning that are not reducible to travel distance. The analogy highlights counterfactual structure, not a literal spatial model of experience.

Decoding accuracy is therefore weaker than bridge fit. A decoder can exploit stimulus artifacts, response preparation, task difficulty, or a feature that separates labels without tracking experienced similarity. A causal geometry must survive controls for those nuisance structures and predict phenomenal relations that were not used to construct it.

3. Build the phenomenal target independently

The target quality structure, call it Q , should be estimated in anchor cases using multiple converging methods. Possible sources include similarity judgments, discrimination thresholds, confusion matrices, generalization, adaptation, cross-context matching, and judgments of transitions. Reports can contribute in calibrated report-capable participants without becoming the minimal presence criterion. The separation is between evidence streams and inferential roles, not an absolute ban on all reports.

No single measure defines Q . Similarity ratings may reflect task strategy. Discrimination thresholds can be asymmetric or context-sensitive. Confusion matrices mix perceptual and decisional processes. Adaptation may alter early encoding or attention. Convergence across partially independent routes is stronger; conflict beyond a preregistered tolerance produces an indeterminate target geometry.

The feature operator $\psi_\phi(G_c, \Delta)$ exposes exactly which distances, transitions, topological properties, occupancy patterns, and temporal paths the bridge may use. Its dimensionality and preprocessing are frozen inside Φ . A bridge F_ϕ maps those causal features to the independently estimated quality structure:

$$G_c \xrightarrow{\psi_\phi} \text{frozen causal features} \xrightarrow{F_\phi} \text{predicted phenomenal relations}.$$

Training, validation, and testing must remain separate. A high in-sample correlation can always reward flexibility. The stronger test is whether a bridge trained on some contents predicts the neighborhood of a held-out content and the direction and magnitude of phenomenal change after a geometry-changing intervention.

Method note. Draw two data-lineage streams. One runs from interventions to causal trajectories to G_c . The other runs from independently validated anchor procedures to Q . Mark the first point where they meet. If they meet before the frozen bridge, inspect for leakage.

Thought experiment: The perfect color decoder

A classifier identifies red, orange, green, and blue stimuli with 99.9% accuracy from neural data. Its embedding places the four classes in a neat circle. The classifier was trained on response labels collected during the same task. Independent similarity judgments place orange near red, but the decoder places orange equally far from every other color. A selective adaptation changes experienced similarity between orange and red while leaving decoder accuracy unchanged.

Stage 1. List the accomplishments the decoder has genuinely achieved.

Stage 2. Identify which facts prevent its embedding from functioning as an independently validated character bridge.

Stage 3. Propose a causal-trajectory geometry and a held-out test that could do more.

Stage 4. State what would follow if the decoder remained excellent while the bridge failed.

The exercise trains scope discipline. Label prediction may be scientifically useful even when it does not explain phenomenal relations.

4. The deformation test is the causal center

A supported bridge should do more than fit a static relation. A preregistered intervention changes G_c in a specified way. The frozen bridge predicts which phenomenal distances should contract or expand, which transition order should change, or which path feature should be altered. The phenomenal deformation is then measured independently.

For example, fit a bridge using red, orange, and green while holding blue out. Baseline success requires predicting blue's neighborhood from causal relations without training on blue's phenomenal anchors. A stronger test changes one causal distance through adaptation or perturbation and predicts the corresponding change in an independently measured phenomenal relation. Stimulus distance, semantics, arousal, attention, motor preparation, and task demand should be compared as nuisance models.

A bridge is supported only relative to explicit rules. It should:

1. predict held-out phenomenal distances or neighborhoods better than declared nulls;
2. preserve validated transition structure and asymmetries;
3. survive coordinate rotations, harmless sensor relabeling, rescaling, and reasonable preprocessing;
4. outperform stimulus, semantic, arousal, and response-preparation nuisances;
5. predict intervention-driven deformation; and
6. transport only under a separate bridge license.

Baseline fit and intervention fit should be reported separately. Otherwise, a flexible model can hide a failed causal prediction behind an impressive static correlation.

Do not infer. If a selective change in G_c produces no predicted phenomenal change under valid anchors, the tested bridge is threatened. The positive presence result need not change. Modularity is not optional bookkeeping; it identifies which claim failed.

Thought experiment: The geometry transplant

Two systems have matched local causal geometries for four contents. In System H, the bridge was calibrated using multiple human phenomenal anchors. In System A, the same geometry appears in a materially different artificial substrate. No phenomenal anchors are available from A, but A has a frozen, licensed positive presence result, including a resolved minimal-bearer output.

Stage 1. Separate the presence claim, within-H character claim, and cross-system character claim.

Stage 2. Specify what a bridge-transport license would require beyond geometric isomorphism.

Stage 3. Consider a remote-region intervention that changes the whole geometry while preserving each content's occupied local neighborhood. What would whole-space and local-relation bridges predict?

Stage 4. State the strongest conditional character attribution available if transport remains unresolved.

This thought experiment exposes a common shortcut: role-preserving transport of the presence profile does not automatically transport every calibrated character relation.

5. Symmetry limits absolute labels

Suppose a transformation preserves every measured distance, transition, valence relation, memory relation, and causal role in Q . The evidence cannot distinguish a point q from its transformed counterpart. The warranted attribution is therefore an equivalence class:

$$Q_{\phi}(G_c, \Delta) = [F_{\phi}(\psi_{\phi}(G_c, \Delta))]_{Aut(Q)}.$$

$Aut(Q)$ is the set of automorphisms: transformations that preserve all warranted relations in the quality structure. The brackets mean that the character is identified only up to those surviving symmetries.

An unlabeled equilateral triangle makes the point vivid. Any permutation of its three vertices preserves all pairwise distances. Purely relational evidence cannot say which vertex has an absolute name. Add a validated asymmetry - perhaps one vertex has a distinctive negative valence or directed transition - and fewer permutations remain.

This is a disciplined response to inverted-spectrum concerns. A global red-green swap that preserves every measured relation remains underdetermined. The theory should not assign an absolute label by assertion. Additional asymmetrical anchors can shrink the equivalence class, but only if they are independently warranted.

Analogy: rotating a constellation chart. A chart can be rotated while preserving distances and shapes among stars. If no “up” direction is independently marked, the orientation is underdetermined. A coastline or compass direction breaks the symmetry. Likewise, relational character evidence fixes only what its validated asymmetries can distinguish.

Where the analogy breaks. Stars have observer-independent spatial coordinates within physical frames, while the character problem concerns the relation between causal and phenomenal structures. The analogy illustrates symmetry and identifiability, not the ontology of qualia.

Plain language. The theory should claim no more individuality than its relations can identify. Underdetermination is a result, not a blank check for arbitrary labels.

6. Character may require more than one geometry

A structured character profile can be written:

$$\chi_c = (m_c, [q_c], I_c, A_c, \Pi_c).$$

m_c is a modality stratum; $[q_c]$ is a symmetry-qualified relational location; I_c is intensity; A_c is a valence field or vector; and Π_c is the temporal path profile. The tuple is not assumed to be exhaustive.

Color examples are useful but can distort the research agenda. Audition involves pitch, timbre, loudness, rhythm, and directed temporal order. Olfaction includes mixture effects and context dependence that may violate simple Euclidean relations. Pain separates intensity, unpleasantness, location, urgency, controllability, and temporal pattern. Interoceptive states may combine bodily location with action tendency and value. Cross-modal relations should be posited only where independent anchors justify them.

For pain, a one-dimensional scale can hide important differences. Two episodes may have equal reported intensity but differ in unpleasantness or urgency. N_3 asks whether the pain content is recurrently present. Π_c distinguishes steady, pulsing, spreading, or anticipatory temporal paths. A credible bridge may need directed graphs, stratified spaces, and value fields rather than a universal metric.

Checkpoint. Before forcing data into a common Euclidean space, ask which relations are undefined, asymmetric, context-sensitive, or modality-specific. A graph with honest gaps can be more informative than a smooth but invented map.

7. Four bridge outcomes and a modular error matrix

The bridge can be **supported**, **refuted**, **indeterminate**, or **unlicensed**. Combined with the presence result, these statuses prevent cross-layer overreach.

If presence is positive and the bridge is supported, the tested relational character claim is supported. If presence is positive and the bridge is refuted, the episode remains present under the model, but this mapping fails. If the bridge is indeterminate, the character relation is unresolved. If the bridge is unlicensed, its procedure cannot be interpreted.

If the presence model makes a licensed negative prediction, this framework licenses neither ordinary character attribution nor a specific claim about which character is absent; measured representational geometry nevertheless remains scientifically meaningful. If presence is indeterminate, representational and bridge evidence may be studied conditionally without converting it into an unconditional phenomenal claim. If presence is unlicensed, repair the gate before combined inference.

Do not infer. A failed bridge does not show that no phenomenal character exists. It shows that the proposed mapping did not earn the tested relation. A beautiful bridge cannot rescue an invalid presence assessment.

Specification stability matters. Let $N(\Phi_0)$ be a frozen family varying boundary, grain, causal metric, anchor combination, preprocessing, and bridge family. The weights assigned to its alternative specifications must also be fixed before bridge outcomes are inspected. A robustness summary can report how often the bridge status matches the primary specification across those weighted alternatives. This is a stability frequency, not a posterior probability that the character identity is true. Report which choices reverse the outcome.

8. Rival lenses, evidence, and explanatory limits

Quality-space theories emphasize the relational organization of perceptual qualities. Representational-geometry and neural-manifold methods provide tools for estimating relations among population states. CØ / N* borrows the pressure toward relational structure but adds a presence gate, perturbation-derived causal geometry, independent phenomenal anchors, and a frozen bridge. It rejects the inference from “structured representation” to “phenomenal structure” without validation.

Integrated Information Theory also treats intrinsic causal structure as important, but the present character program does not simply identify an IIT structure with experience. Exact comparison requires a dated IIT version and matched objects of analysis. Workspace and higher-order theories may explain how character information becomes globally usable, reportable, or represented at a higher order; the present bridge targets phenomenal relations after minimal presence is licensed. Representationalism and structuralism offer philosophical interpretations of why a successful relation might be constitutive, predictive, or explanatory.

A successful bridge would not by itself solve the “hard problem” or prove that every intrinsic phenomenal fact is captured. It could nevertheless achieve substantial scientific explanation: held-out relational prediction, causal deformation, invariance, fault localization, and transport under explicit licenses. The difference between predictive success and constitutive identity should remain visible.

Evidence directly challenges the bridge if selective changes in G_c fail to produce predicted phenomenal changes, phenomenal geometry changes while G_c remains stable across reasonable specifications, held-out performance falls to nuisance parity, a simpler stimulus model wins after causal controls, claimed transport fails, or small reasonable specification changes reverse status. The response should be revision or rejection of Q_ϕ , not post hoc redefinition of the geometry or anchors.

Source note. The character paper presents causal deformation and symmetry-qualified attribution as central safeguards against merely renaming an embedding. It explicitly leaves open whether valence, meaning, or bodily urgency require richer structures.

Practice studio

Quick checks

1. State the two non-implications that separate presence and character.
2. Why is a decoder embedding not automatically G_c ?
3. What must be independent when constructing Q ?
4. Why is a deformation test stronger than baseline correlation?
5. What does $Aut(Q)$ represent?
6. Give one reason not to force a Euclidean quality space.
7. What remains intact when a bridge is refuted under a positive presence result?

Individual transfer task: Design a character bridge

Choose a domain other than simple color. Define four or more contents, a perturbation family, the causal relations included in G_c , the independent phenomenal anchors, one held-out content, and a geometry-changing intervention. Select an appropriate structure - metric, pseudometric, directed graph, ordinal relation, or stratified space - and justify it. Name at least three nuisance models and two invariances. Finish with outcome rules for supported, refuted, indeterminate, and unlicensed bridge results.

Collaborative exercise: Build and try to break a bridge

Materials. Use Session 9, “Character-Bridge Break Test,” in the companion *Student Session Resource Pack*. The student materials supply a frozen positive presence gate, baseline causal and phenomenal relations, a held-out content, and a nuisance comparison. Use only the instructor-assigned card when it is released; keep every unreleased classroom item inaccessible until the team freezes its prediction and bridge rules.

Teams use four roles:

- The **causal-model lead** constructs G_c from intervention-conditioned trajectories.
- The **phenomenal-anchor lead** builds Q and documents the evidence lineage.
- The **bridge analyst** freezes training, validation, held-out, null, invariance, and deformation predictions.
- The **leakage and nuisance auditor** searches for stimulus structure, report reuse, task difficulty, attention, arousal, preprocessing freedom, and response preparation.

The team must lock a positive presence result and may not reopen it to improve the character fit. It then exchanges its bridge specification with another team, which designs the strongest valid mismatch by varying only declared intervention or nuisance relations - never by inventing convenient outcome data - and one harmless coordinate transformation. The original team revises only according to prospective rules. The final two-stream diagram must show where causal and phenomenal evidence first meet and what remains true under each possible bridge outcome.

Collaborative discussion: Does relational prediction explain character?

Form a fishbowl with a structural-identity advocate, predictive-only advocate, inverted-spectrum critic, and phenomenological pluralist. Outer observers track four distinctions: prediction versus explanation, absolute labels versus equivalence classes, geometric versus nongeometric structure, and presence versus character. Rotate roles. Produce an argument map with common ground, deepest disagreement, an empirical discriminant, and a residual philosophical issue. No group may use “it is just correlation” or “the structures match” without specifying which intervention or identity claim is at issue.

Counterexample challenge

Construct a case in which causal and phenomenal geometries correlate perfectly at baseline for the wrong reason. Then add an intervention that exposes the mismatch. Construct a second case in which the bridge predicts every measured relation but a large automorphism group leaves absolute labels underdetermined. Explain what each case refutes and what it leaves open.

Exit ticket

Write two sentences: “Decoding is weaker than a character bridge because ...” and “A refuted bridge need not defeat presence because ...”

Chapter summary

Phenomenal presence and phenomenal character require two linked but modular layers. A positive presence gate licenses the character question; structured geometry alone does not open the gate. G_c must be a content-specific causal-trajectory structure rather than an attractive activation embedding. The phenomenal target Q is independently anchored, and a frozen bridge must predict held-out relations, outperform nuisances, survive invariances, and forecast intervention-driven deformation. Character is identified only up to the symmetries left unbroken by warranted relations. Richer modalities may require directed, stratified, temporal, and value-sensitive structures. Bridge support, refutation, indeterminacy, and invalidity should never be allowed to rewrite the presence result.

Key terms

- **Automorphism:** a transformation that preserves all warranted relations in a structure.
- **Causal-trajectory geometry G_c :** the intervention-derived relational neighborhood and paths of content c .
- **Character bridge F_ϕ :** a frozen mapping from causal features to independently anchored phenomenal relations.
- **Deformation test:** an intervention test in which a predicted change in causal geometry is compared with an independently measured phenomenal change.
- **Equivalence class $[q_c]$:** character location identified only up to surviving symmetries.
- **Phenomenal character:** what a present episode is like - its relational location, modality, intensity, valence, and temporal path.
- **Phenomenal quality structure Q :** independently estimated relations among experiences in anchor cases.
- **Presence gate P :** the separately validated decision that licenses ordinary phenomenal-character attribution.
- **Specification Φ :** the frozen boundary, causal geometry, anchors, bridge family, controls, splits, invariances, and thresholds.

Looking ahead

If the presence and character models become sufficiently reliable, researchers may be able to read a structured role profile from a schematic. Chapter 10 asks what that claim would mean - and why readability would still fall short of a proof.

Transition notebook: from bridge to schematic

Draw three boxes labeled **causal organization**, **phenomenal relations**, and **phenomenal presence**. Add only the arrows licensed by this chapter, and mark where independent evidence enters. Exchange diagrams with a partner, who removes one arrow or validation stream. State whether the change defeats a character prediction, the presence gate, the constitutive interpretation, or only confidence in the study. Bring the revised diagram to Chapter 10; it will become the starting point for separating a readable representation, its running realization, and the identity claim applied to it.

Transition diagram workspace

Chapter 10 - Reading Consciousness from the Schematic

Opening puzzle. A trained engineer looks at an electrical schematic and says, “With the source connected, this switch closed, and these component values, the motor will turn.” A novice sees ink, lines, and unfamiliar symbols. The motor’s motion is not pictured inside the ink, yet the engineer can derive it from learned component meanings, operating conditions, and counterfactual dependencies. Suppose CØ / N* survives demanding empirical tests. Could a trained researcher inspect a role-annotated causal schematic and read phenomenal presence from it in a comparably structured way?

The question goes beyond prediction. A marker might correlate reliably with consciousness evidence. A black-box law might predict presence across held-out cases. A **constitutive schematic** would add an interpretable route: the reader reconstructs a bearer, realized operation, unity, system-level causal presence, and temporal presence, then applies the explicit identity hypothesis that this role-complete organization is minimal phenomenal presence.

Do not infer. A readable diagram does not experience anything, and readability does not prove the identity. The actual system must realize the causal organization. Independent evidence must still bear on whether the identity is true.

Chapter map

This chapter will enable you to:

- distinguish a representation, a causal realization, and a possible bearer;
- reconstruct the CØ / N* role vector from component results;
- compare correlational-marker, black-box-law, and constitutive-schematic readings;
- apply the four legibility conditions: decomposable semantics, compositional readout, fault localization, and evidential independence;
- explain what a schematic fault predicts at profile and bearer levels;
- assess objections involving circularity, zombies, external observation, and functionalism; and
- design a blind readout and transport study that tests teachability without confusing agreement with truth.

The chapter is intentionally conditional. It asks what explanatory form the theory could achieve if its operational and empirical program succeeds.

1. Three objects that must never be collapsed

The first object is the **representation**: a printed diagram, equation, graph, or digital schematic. It carries symbols and arrows whose meanings are learned. The second is the **realization**: the running physical system with effective causal dynamics at a specified grain, regime, and interval. The third is the possible **bearer**: a boundary-qualified, causally admissible candidate that may realize a unified episode.

Confusing these objects produces several errors. A diagram that depicts recurrent loops is not itself recurrently maintaining a content in the theoretical sense. Software that simulates a network is not automatically a causal realization at the modeled grain. A large system that contains all relevant components is not automatically the minimal bearer if a proper subsystem also passes. Conversely, a physical realization need not resemble the visual layout of its schematic.

Plain language. The map, the running machine, and the thing whose state the map describes are related, but they are not interchangeable.

Analogy: a musical score, performance, and ensemble. A score represents timed relations among parts. A performance realizes those relations in sound. The ensemble is the organized group whose performance is being described. A score can be copied without producing music; a performance can use different instruments while preserving key relations; and it can be unclear whether a guest player belongs to the ensemble or functions as external support.

Where the analogy breaks. Musical membership is partly conventional, and performances are interpreted through human practices. Candidate-bearer individuation is supposed to rest on causal autonomy, mapping, minimality, and separability. The analogy separates representation, realization, and bearer, but it does not validate phenomenal attribution.

Let the role-annotated schematic be S_{N^*} . The proposed readout has two arrows:

$$S_{N^*} \Rightarrow \rho_{C_0} \Rightarrow C_0.$$

The first arrow is mechanistic interpretation: use public component meanings and results to reconstruct a role profile. The second arrow applies the exposed constitutive identity hypothesis. That second arrow must remain visible. Hiding it inside a label such as “consciousness score” turns a substantive claim into an apparent measurement fact.

Source note. *Reading Consciousness from the Schematic* frames the schematic as a representation from which a trained reader reconstructs an intermediate role profile. It repeatedly distinguishes this explanatory achievement from validation of the identity.

2. The role vector makes the theory inspectable

The intermediate profile is:

$$\rho_{C_0} = \langle \text{bearer, realized operation, unity, system-level presence, temporal presence} \rangle.$$

Each entry has a distinct source.

- A stable member of the causally admissible candidate family P_{Θ} supplies a licensed referent for profile testing. Resolved inclusion-minimal positive-bearer membership supplies the theory-relative subject bearer.
- V supplies realized operation in the declared regime. An inert diagram, failed preparation, or out-of-domain state does not qualify.
- N_1 supplies unity: the content is irreducibly joint rather than a pile of independent events.
- N_2 supplies system-level causal presence: the content can modulate functionally diverse recipients without report being required.
- N_3 supplies bounded temporal presence: the content is recurrently or metastably maintained rather than an evanescent feedforward transition or indefinite fixation.

These meanings should not be defined by repeating “consciousness.” Integration is operationalized through excess joint causal contribution, availability through content-specific counterfactual reach, and recurrence through return-dependent temporal stability. The constitutive claim then says that a viable, bearer-complete organization with these roles is minimal phenomenal presence.

Checkpoint. Can you describe each role and its characteristic failure without using the words *conscious*, *experience*, or *phenomenal*? If not, decomposable semantics may be circular or too vague.

This role vector also makes limitations visible. It addresses minimal presence, not necessarily attention, introspection, autobiographical selfhood, report, intelligence, or full character. The character extension adds a second readout after the gate. A change in G_c with the presence roles intact may change predicted character rather than presence.

Analogy: a medical pathway rather than a single lab value. A clinician may interpret several independently meaningful findings - circulation, oxygenation, organ function, and causal response - to reconstruct a physiological state. The interpretation is more informative than a one-number score because different failures suggest different mechanisms and responses.

Where the analogy breaks. Medical states often have mature reference standards, long validation histories, and intervention outcomes that consciousness science lacks. The analogy illustrates compositional and fault-localizing explanation, not equal evidential maturity.

3. Marker, black-box law, or constitutive schematic?

Three explanatory levels should be distinguished.

A **correlational marker** covaries with independent consciousness evidence. It can be useful for detection while its relation to the target remains incidental, indirect, or confounded.

A **black-box law** says that licensed positive-bearer membership built from the full profile reliably guarantees C_0 in the declared domain. This is a sufficiency claim; the biconditional's necessity direction requires separate evidence from well-supported C_0 cases outside the positive-bearer set. Yet the transition from organization to phenomenal presence may remain opaque.

A **constitutive schematic** adds stable role semantics, compositional reconstruction, counterfactual fault structure, and the identity claim that the role-complete organization is the physical form of minimal phenomenal presence. It aims to make the proposal understandable even to someone who rejects it.

The difference between the last two can be written:

black-box sufficiency : $X \Rightarrow C_0$,

structurally legible sufficiency : $X \Rightarrow \rho \Rightarrow C_0$.

The intermediate ρ is not decorative prose. Its parts should support distinct intervention predictions, partial failures, and transport tests.

Why it matters. Accurate prediction and explanation are related achievements, not synonyms. Adding a readable mechanism does not erase the need for accuracy; achieving accuracy does not guarantee a readable mechanism.

Thought experiment: Two high-performing classifiers

Classifier A receives a thousand neural features and returns “present” or “absent” with excellent held-out performance. Its internal representation is not stable across retraining, and no one can state which failure should follow from changing any feature family. Classifier B has slightly lower held-out accuracy but reconstructs the five-role profile, abstains when the bearer is unresolved, and predicts different anchor changes under selective N_1 , N_2 , and N_3 faults.

Stage 1. Classify each system as marker, black-box law, constitutive schematic, or some intermediate achievement.

Stage 2. Identify what evidence would be required before calling either classifier a validated consciousness test.

Stage 3. Design a prospective fault test on which their predictions differ.

Stage 4. Discuss whether B’s interpretability could compensate for lower accuracy, and specify the decision context in which your answer changes.

The exercise resists a false choice. A mature program should seek calibration and structural legibility while reporting tradeoffs rather than awarding automatic priority to either.

4. Four conditions for schematic legibility

Let M be a structured model, o the target state, and K the publicly teachable component and representational knowledge.

Notation note. Symbols are locally scoped. Here D, C, F, E name the four legibility conditions; they are not the scientific-status D , target C , or evidence E used in Chapter 7.

Define:

$$\text{Legible}(M, o / K) \Leftrightarrow D \wedge C \wedge F \wedge E.$$

The four conditions are:

1. **Decomposable semantics (D).** Elements have stable role meanings that do not merely repeat the target outcome.
2. **Compositional readout (C).** A trained reader can derive an intermediate organization profile and apply an explicit rule.
3. **Counterfactual fault localization (F).** Licensed changes support differentiated predictions about which role fails or changes.
4. **Evidential independence (E).** The readout procedure is not counted as evidence that its own constitutive rule is true.

The conditions form a conjunctive minimum for strong legibility, but each can be studied in a graded way. Readers may agree on component meanings yet fail to combine them. They may combine them but make only generic “system failed” predictions. They may localize faults reproducibly while validation labels were circularly generated. Reporting the pattern is more informative than a single interpretability score.

Method note. Audit D and C at the level of the common law sheet or representational system. Audit F for each licensed profile or contrast. Audit E for each validation stream. A boundary failure in one case does not erase stable semantics for every case, while circular labels in one study can invalidate its confirmation claim without making the schematic unreadable.

Analogy: learning to read a circuit. The learner first acquires symbol meanings, then learns how component laws compose, then predicts what happens when a switch opens or a resistor fails, and finally compares those predictions with measurements not created by the diagram itself. This sequence parallels D , C , F , and E .

Where the analogy breaks. Electrical theory has mature laws, reliable instruments, and direct operating-state checks. Consciousness science has contested targets, imperfect anchors, boundary uncertainty, and substantial theory disagreement. The circuit analogy proposes a target epistemic form, not current parity.

Legibility can fail while prediction remains useful. If a one-number score predicts well but its elements have no stable meanings, D fails. If roles are meaningful but no rule combines them, C fails. If all interventions collapse the system globally, F remains unresolved. If outcome labels are derived from N^* itself, E fails. None of these failures proves that the system is nonconscious; they identify limits in the explanatory and evidential architecture.

5. Faults operate at profile and bearer levels

The schematic's fault map is one of its main explanatory claims. Different failures should have different interpretations.

If the frozen P_{ϕ} referent loses autonomy, causal admissibility, or cross-condition mapping, component inference becomes unlicensed because the entity under comparison is unstable. A viability failure means the operating regime no longer supports the relevant dynamics. On a stable referent, a selective N_1 failure is a profile-level loss of unity. A selective N_2 failure is profile-level causal isolation. A selective N_3 failure is profile-level temporal evanescence. A change in G_c under an intact presence gate changes the predicted character relation.

After a component fault, recompute the all-pass set across the frozen candidate lattice. The final bearer-level result may be:

- loss of the former bearer;
- emergence of a smaller or alternative bearer;
- several separable minimal bearers;
- unresolved multiplicity; or
- a licensed negative prediction if a decisive failure or minimality result applies on a resolved domain.

Do not infer. A profile fault in the original candidate does not by itself prove that consciousness vanished everywhere in the larger system. Nor does the persistence of some all-pass subsystem prove that the original subject remained unchanged.

Thought experiment: The modular ship

A research vessel contains two tightly coupled control modules. Under baseline conditions, the combined candidate is stable and inclusion-minimal. An intervention selectively blocks content-specific routing between modules while leaving each module viable and internally recurrent. Afterward, one module passes all profile conditions for content c , the other passes for content d , and separability evidence is incomplete.

Stage 1. State the profile-level fault in the original combined candidate.

Stage 2. Reconstruct the possible bearer-level outputs without choosing one by intuition.

Stage 3. Identify the mapping and separability evidence needed to distinguish relocation, plurality, and unresolved multiplicity.

Stage 4. Explain which parts of a schematic readout remain legible even if the final bearer attribution is indeterminate.

The case shows why schematic competence includes correct abstention. A reader who announces one subject merely because a diagram is detailed has not demonstrated legibility; they have ignored the boundary rule.

6. Teachability is testable, but agreement is not confirmation

A public schematic should be learnable by independent investigators. A three-stage protocol can test this.

First, train readers on component semantics, candidate and bearer distinctions, worked profiles, and correct abstention. Second, test blind readout on unseen cases: readers reconstruct roles, classify outcomes, localize faults, and mark the step that depends on the constitutive identity. Third, test transfer to a materially different realization whose role organization is claimed to be preserved.

Measure inter-reader agreement, calibration, fault specificity, abstention quality, and transport accuracy. A reader who refuses to classify an unstable-boundary case may be more competent than one who always issues a verdict. Disagreements should be coded by source: semantics, combination rule, boundary mapping, uncertainty policy, fault attribution, or identity stance.

High agreement establishes that the readout is teachable under K . It does not establish that C_0 is identical to the role profile. Two readers can reconstruct the same organization and rationally disagree about the constitutive claim. This is a virtue: the theory's empirical and philosophical commitments remain exposed.

Checkpoint. Every teachability report should contain the sentence: "Agreement on the readout is evidence for public interpretability; it is not independent evidence that the identity is true."

Five empirical payoffs should be tracked separately:

- **Role-preserving transport:** does the same role profile survive a materially different realization?
- **Prospective fault prediction:** do component-specific changes forecast differentiated failures?
- **Estimator convergence:** do independent measures recover the same intermediate roles?
- **Compression:** does the role profile omit causal detail while preserving rival-distinguishing predictions?
- **Teachability:** can independent readers reconstruct, abstain, and localize faults on held-out cases?

Success on one payoff does not validate all five. A schematic can be teachable but fail transport, compressive but poorly calibrated, or fault-localizing while the identity remains empirically unsupported.

7. Objections clarify what the claim does and does not say

Circularity

An objector may argue that the roles appear consciousness-like only because the theory describes them using words such as *unity* and *presence*. The reply must be operational, not rhetorical. Each role should be stated through consciousness-independent causal tests. If the role meanings cannot be stated without assuming the conclusion, *D* fails. If the roles are readable but readout-generated labels are reused to validate the identity, *E* fails. Passing both conditions leaves the identity substantive and contestable rather than definitional.

The zombie challenge

A physical duplicate with the complete organization but no experience would defeat sufficiency if independently established. Merely describing such a duplicate as conceivable does not provide an empirical case, but it does identify the deepest constitutive disagreement. The schematic thesis does not dissolve that disagreement by making the roles vivid. It exposes exactly where the opponent denies the second arrow.

External observation

A researcher can observe that a system realizes the proposed organization without undergoing its token experience. Under a separately supported identity hypothesis, observing that organization licenses attribution of C_0 ; this remains distinct from inferring token character or occupying the bearer's first-person position. Scientific observation need not reproduce its object inside the observer. Whether organization is sufficient for warranted attribution remains the empirical and constitutive question.

Functionalism and implementation

The proposal is a physicalist information-dynamical functionalism, not unrestricted input-output equivalence. Static lookup tables, behaviorally equivalent scripts, and merely isomorphic diagrams do not automatically qualify. Effective causal realization, correct grain, stable boundary, timing, and viability do real work. A materially different implementation may qualify only if the public role profile and its counterfactual organization transport.

Why it matters. The strongest objections do not show that the schematic is meaningless. They locate disagreements about operational independence, causal realization, and constitutive identity. A theory becomes easier to test when opponents can say exactly which arrow they reject.

Source note. The schematic paper treats the zombie and external-observation objections as challenges to the constitutive identity, not as reasons to abandon public causal description. Its conclusion remains explicitly conditional.

8. Evidence, limits, and rival explanatory ideals

Mechanistic explanation emphasizes organized entities, activities, dependencies, and counterfactuals. $C\emptyset / N^*$ seeks this structure through its role vector and fault map. Black-box predictive approaches may outperform it in some settings, especially where mechanisms are poorly understood. Correlational biomarkers can be clinically useful without constitutive interpretation. The levels should be compared on their actual aims rather than ranked by label.

Evidence favors schematic legibility when independent readers learn stable meanings, reconstruct roles on held-out cases, abstain appropriately, predict distinct faults prospectively, converge across estimators, and transfer to new implementations under preserved causal roles. Evidence against legibility includes persistent semantic disagreement, absence of a compositional rule, generic rather than differentiated fault predictions, circular validation, failure of transport, or compression that discards rival-distinguishing information.

Evidence for legibility is not yet evidence for the identity. The constitutive claim additionally faces the full empirical program: sufficiency challenges from licensed all-pass cases with strong independent evidence against presence, necessity challenges from well-supported conscious cases outside the positive-bearer set, component audits, rival comparisons, and transport tests. Structural intelligibility can coexist with empirical failure.

The analogy to electrical schematics has a strict limit. Mature electrical models are embedded in well-confirmed theory and measurement practices. $C\emptyset / N^*$ is a new, adversarial research program. The analogy sets a standard - public semantics, composition, counterfactuals, fault localization, and independent validation - rather than awarding the model the authority of engineering.

Practice studio

Quick checks

1. Distinguish representation, realization, and bearer.
2. What are the five entries in ρ_{C_0} ?
3. How does a black-box law differ from a constitutive schematic?
4. State the four legibility conditions D, C, F, E .
5. Why can a boundary failure block fault localization without erasing component semantics?
6. What does correct abstention show in a blind readout study?
7. Why is inter-reader agreement not confirmation of the identity?

Individual transfer task: Create a fault-localizing readout

Design five unseen profiles: one licensed all-pass case, one boundary failure, one selective N_1 failure, one selective N_2 or N_3 failure, and one coupled fault. For each, specify the representation, claimed physical realization, candidate lattice, component evidence, uncertainty status, and expected role-level readout. Write no consciousness verdict into the component labels. Then create a scoring rubric that separately assesses semantic reconstruction, composition, fault localization, bearer recomputation, abstention, and identification of the constitutive step.

Collaborative exercise: Blind schematic-readout trial

Materials. Use Session 10, “Blind Schematic Readout,” in the companion *Student Session Resource Pack*. Study only the common component-law sheet and training cases before the blind round. The adjudicator or instructor distributes each unseen profile separately after readers have committed their forms; keep every unreleased classroom item inaccessible until adjudication.

Work in groups of four:

- The **training designer** teaches only the frozen component and bearer rules using two practice cases.
- **Blind Reader A** and **Blind Reader B** independently reconstruct unseen profiles, assign evidential outcomes, localize faults, and flag the identity-dependent step.
- The **adjudicator** compares their forms, codes disagreement sources, and audits evidential independence.

The designer may not answer questions about whether a case “seems conscious.” Readers may abstain when boundary, mapping, coupling, or evidence quality blocks inference. After the core profiles, exchange one profile with another group that uses a materially different implementation but claims the same role structure. Report exact-outcome agreement, role agreement, fault-family agreement, abstention calibration, and transport errors separately.

Collaborative discussion: Constitutive-identity tribunal

Assign identity theorist, correlationist, conceivability critic, and evidential referee. The tribunal receives a system that passes every frozen role test and supports accurate prospective fault predictions. It must issue four separate verdicts: marker status, lawlike predictive status, warrant for constitutive identity, and the remaining explanatory or perspectival gap. The referee stops every argument that moves from “intelligible” to “confirmed” or from “not first-personally accessible” to “scientifically unobservable” without an additional premise. Rotate roles and ask what evidence would change each verdict.

Counterexample challenge

Construct one schematic that is highly legible but empirically false, and another that predicts accurately but is not legible under $D \wedge C \wedge F \wedge E$. For each, specify which achievement remains and which fails. Then propose the smallest study that discriminates the two explanatory profiles without treating readability as an outcome label.

Exit ticket

Complete the distinction: “The first arrow from S_{N^*} to ρ_{C_0} is ...; the second arrow from ρ_{C_0} to C_0 is ...; evidence for the first does not by itself establish the second because ...”

Chapter summary

Schematic legibility separates the representation, running causal realization, and possible bearer. A trained reader reconstructs a five-part role profile - bearer, realized operation, unity, system-level causal presence, and temporal presence - before applying the exposed constitutive hypothesis. This differs from both a correlational marker and an opaque predictive law. Strong legibility requires decomposable semantics, compositional readout, differentiated fault localization, and evidential independence. Faults must be interpreted first at the profile level and then propagated through the candidate lattice. Teachability, agreement, transport, compression, and fault prediction are empirical achievements, but none alone proves that the role-complete organization is phenomenal presence. The electrical analogy names a scientific ideal, not an attained evidential parity.

Key terms

- **Black-box law:** a reliable predictive relation whose transition from organization to target remains uninterpreted.
- **Compositional readout (C):** derivation of a whole role profile from stable component meanings and an explicit combination rule.
- **Constitutive schematic:** a role-complete, fault-localizing model paired with an explicit identity claim.
- **Counterfactual fault localization (F):** differentiated predictions about which role changes under licensed component changes.
- **Decomposable semantics (D):** stable role meanings that do not merely repeat the outcome.
- **Evidential independence (E):** validation that does not use the schematic’s own output as its truth label.
- **Realization:** the running physical system that instantiates effective causal dynamics.
- **Representation:** the diagram, equation, or model presented for interpretation.
- **Role vector ρ_{C_0} :** bearer, operation, unity, system-level presence, and temporal presence.
- **Schematic legibility:** publicly teachable reconstruction, fault reasoning, and evidence separation under $D \wedge C \wedge F \wedge E$.

Looking ahead

The schematic chapter completes the transition from component tests to an explanatory ideal. The next phase of the textbook turns toward application: biological and clinical systems force the framework to confront damaged routes, unstable regimes, changing boundaries, and decisions made before evidence is complete.

Chapter 11 - Biological and Clinical Applications

Opening puzzle: the silent patient

A patient opens her eyes but does not follow commands. On Monday, an auditory imagery task produces no usable signal; the recording is noisy and no one has confirmed that she can hear the instructions. On Thursday, during a higher-arousal period, a motor-imagery pattern appears repeatedly even though she still makes no movement. On Sunday, the pattern disappears during a low-arousal recording. Has consciousness appeared, vanished, and returned? Was it present all week? Or has the study not yet earned any of those conclusions?

The puzzle is difficult because behavior, measurement quality, physiological state, and phenomenal presence can all vary - and they need not vary together. $CØ / N^*$ asks us to separate them. Here $CØ$ means minimal phenomenal presence: there is something it is like for the candidate system during the specified interval. N^* is the conjunction $N_1 \wedge N_2 \wedge N_3$: integration or synergy, system-wide causal availability, and recurrent stability. V is a separate viability condition. The theory's core hypothesis is conditional and exposed to failure:

$$C_0(S, c, \tau) \leftrightarrow V(S, \tau) \wedge N_1(S, c, \tau) \wedge N_2(S, c, \tau) \wedge N_3(S, c, \tau).$$

This compact equation states the source profile conjecture and suppresses bearer selection. Empirically, $H_*(S, c) = V \wedge N^*$ may be estimated for a frozen referent $S \in P_\Theta$, but final subject attribution additionally requires $Resolved_B(c; \Theta)$ and $S \in B_{min, \Theta}^+(c)$. Hatted estimates are evidence outputs, not the underlying ontic properties.

The symbols force a useful discipline. The claim is always about a candidate system S , a content or target c , and an interval τ . A positive result for one content during one minute is not automatically a verdict about every experience the person had that day.

Plain language. A person can be unable to show consciousness while still being conscious. A test can also fail to find consciousness because the test was not capable of finding it in that moment.

Chapter map

By the end of this chapter, you should be able to:

1. apply the $CØ / N^*$ workflow to waking perception, sleep, anesthesia, disorders of consciousness, blindsight, rivalry, and split systems;
2. distinguish a target-specific scientific result from a whole-person conclusion;
3. explain why a negative observation may be negative, indeterminate, or unlicensed;
4. identify common routes to false positives and false negatives;
5. keep empirical classification separate from clinical and ethical action; and
6. design repeated, controlled measurements that could improve an uncertain case.

1. Begin with the question, not the instrument

Clinical urgency tempts us to begin with a device: an EEG score, an fMRI task, a bedside scale, or a complexity index. CØ / N* begins one step earlier. What exactly are we trying to determine?

The fixed inference order is:

1. define the target C ;
2. declare the candidate system, grain, operating regime, and interval;
3. audit whether the boundary of that candidate is resolved;
4. assess viability V independently;
5. estimate N_1 , N_2 , and N_3 for the same candidate and target;
6. apply conflict, uncertainty, and validity rules;
7. state the model-relative scientific result; and
8. make any clinical or ethical recommendation separately.

The four possible scientific outcomes are **positive**, **negative**, **indeterminate**, and **unlicensed**. Positive and negative are not simply high and low scores. A negative requires a validated rule-out region and a fair opportunity for detection. Indeterminate means that relevant, valid evidence exists but does not converge strongly enough. Unlicensed means that a prerequisite failed: perhaps the sensory route was blocked, the candidate boundary shifted, the instrument was not calibrated for this population, or artifact made the recording uninterpretable.

Why it matters. Without this four-way vocabulary, “the test did not find it” easily becomes “it was not there.” In vulnerable patients, that linguistic slide can have enormous consequences.

Think of the workflow as an airport departure board. A flight may be listed as departed, cancelled, delayed, or unable to report because the data feed is down. “Unable to report” is not a hidden form of “cancelled.” Likewise, an unlicensed test result is not a disguised negative.

Where the analogy breaks. Airlines can check a centralized operational record. Consciousness research has no universal gold-standard record, so the classification rules and anchor evidence remain theory-laden and fallible.

Checkpoint. For the opening patient, write one target that is narrower than “Is she conscious?” Then specify an interval over which that target could be assessed.

An assessment record can be written as

$$A = \{ S, C, T, E, M, V \},$$

where S is the candidate system, C the exact target, T the test, E the evidence, M the model, and V the validity envelope. If the result is indeterminate, record why:

$$P_R(A) = \{ e, s, m, b, t \}.$$

The five entries represent evidence scarcity, surrogate discordance, model dependence, boundary or grain sensitivity, and transport uncertainty. The tuple is a cause profile, not another consciousness score. A component can be absent, mild, or dominant; the point is to direct the next study toward the actual obstacle.

2. Waking perception is a calibration case, not a definition

Healthy waking adults offer abundant converging evidence. A dominant visual percept, for example, normally occurs in a viable system with integrated processing, causal availability to multiple functions, and recurrent temporal organization. Reports, discrimination, memory, eye movements, physiology, and perturbational responses can be compared. This makes waking perception useful for developing and stress-testing measures.

But the easy case can distort the theory. If every measure is chosen because it correlates with button presses in alert adults, report-related motor planning and executive demand may be built into the operational definition. Such a measure will then misclassify conditions in which experience remains but reporting fails.

Do not infer. A useful waking calibration does not show that wakefulness, responsiveness, or accurate report constitutes phenomenal presence.

Thought experiment: The perfect mute

Imagine a reversible intervention that blocks all voluntary muscles and speech while leaving perception, memory, attention, internal reasoning, and the N* organization intact.

Stage 1. Before any alternative communication channel is available, what would a behavior-only test conclude?

Stage 2. A brain-computer interface later reveals reliable answers to novel questions. Which earlier inference changes: the patient's state, the evidence, or both?

Stage 3. Suppose the interface itself requires substantial executive control. What narrower claim can it support, and what broader claims remain open?

The case highlights a central distinction: evidence for presence can depend on a reporting route even if presence itself does not.

3. Sleep and dreaming: internal availability

Sleep makes the index τ impossible to ignore. A night is not one homogeneous state. REM periods, non-REM stages, brief awakenings, local sleep-like events, and transitions may support different component profiles.

During REM dreaming, motor output is greatly reduced and external interaction is limited. Yet dream contents may affect attention, emotion, memory formation, imagery, and other internal recipient classes. On this model, N_2 does not require outward broadcast. It requires content-specific causal availability across functionally different recipients within the candidate system. The relevant recipients and timescales may differ from those of waking task performance.

Retrospective dream report is useful but displaced. It occurs after the target interval and depends on encoding, preservation, awakening, comprehension, and willingness to report. A negative report therefore has multiple possible explanations.

Deep non-REM sleep may include sparse, episodic, or content-specific presence; it may also include periods that genuinely lack it. The model does not settle that question by stipulation. Repeated sampling, well-timed awakening, physiological characterization, and perturbational evidence must do the work. Perturbational measures can reduce reliance on behavior, but their interpretation still requires population- and regime-specific validation (Casali et al., 2013).

Method note. Report the distribution of outcomes across sampled intervals. “Seven licensed negatives, two positives, four indeterminate windows, and three unlicensed recordings” contains more information than a single label for the whole night.

An analogy is a wildlife survey conducted in a large forest. One empty camera-trap image does not establish that no animal was present anywhere that night. Repeated, well-placed sensors improve coverage, and a failed camera tells us less than an empty image from a functioning one.

Where the analogy breaks. Animals leave publicly measurable traces and can sometimes be directly counted. Phenomenal presence is inferred through imperfect anchors, so even excellent sampling does not remove the theory-relative character of the conclusion.

Thought experiment: The dream with no archive

Suppose a person has a rich dream, but a temporary memory-blocking process prevents any trace from surviving awakening.

Stage 1. Would the later absence of report count against the dream’s occurrence, or only against its retrieval?

Stage 2. During sleep, a content-specific perturbation changes several internal processes in a coordinated way. What role could that evidence play in estimating N_2 and N_3 ?

Stage 3. The same perturbation also partially wakes the sleeper. How does that threaten the inference?

4. Anesthesia is not one mechanism or one moment

“Under anesthesia” can hide differences among drug, dose, timing, physiology, sensory input, and individual response. Some agents alter thalamocortical organization, some reduce recurrent connectivity, and some change neuromodulatory conditions. Induction, maintenance, emergence, and recovery are different regimes.

A licensed negative under CØ / N* requires more than low responsiveness. Investigators must show that the target candidate had a fair detection opportunity, that relevant measures were valid under the agent and regime, and that the N* bundle failed for the specified content. If the candidate boundary reorganizes, positive-bearer status must be recomputed; one cannot assume that the pre-anesthetic bearer persists unchanged.

Near misses are especially informative. Dreaming-like experiences during anesthesia, disconnected responsiveness, and uneven recovery can separate motor output from internal dynamics. Different agents can also reveal whether a measure tracks consciousness-relevant organization or merely a pharmacological signature.

Plain language. Global physiological disruption is easy to produce but hard to interpret. A selective failure teaches more because fewer explanations change at once.

Do not infer. A low value on a consciousness-associated measure is not a licensed absence verdict unless that measure has a validated negative region in the same kind of state.

5. Disorders of consciousness and the negative-license checklist

Locked-in syndrome is the clearest warning against equating motor silence with experiential absence. A person may have a stable, viable N* organization but no ordinary path to speech or movement. Cognitive motor dissociation presents a related challenge: some behaviorally unresponsive patients show task-specific neural evidence of command following.

Clinical labels such as **unresponsive wakefulness syndrome** and **minimally conscious state** summarize observed behavioral patterns; they are not direct readings of phenomenal presence. In either group, arousal, sensory access, motor capacity, and test quality can fluctuate. Assessment should therefore be repeated across suitable windows, tied to a declared content or capacity, and reported with its negative license. One failed command-following trial is weak evidence of absence, especially when hearing, comprehension, or signal quality has not been established. Repeated behavioral assessment and recent work on cognitive motor dissociation illustrate why single observations can underclassify patients (Wannez et al., 2017; Bodien et al., 2024).

Such evidence is important but narrower than it first appears. Successful command following supports hearing or other sensory access, comprehension, task maintenance, and a content-specific response. It does not automatically establish continuous awareness, rich experience, or the exact boundary of the subject. Conversely, failure on the task is interpretable only when those auxiliary capacities had a fair opportunity to operate.

Before treating non-detection as a negative result, ask:

1. Was the target explicit and appropriate?
2. Was the candidate system stable enough to assess?
3. Did the patient receive the stimulus through a working route?
4. Were arousal, attention, comprehension, memory, and task demands adequate?
5. Was the measure validated in this injury class and operating regime?
6. Did recording quality and artifact controls pass?
7. Did the result enter a preregistered rule-out region?
8. Were plausible alternative causes of non-detection controlled?

If a required item fails, the appropriate result may be unlicensed. If all pass but credible measures conflict, indeterminate may be correct. Repeated assessment matters because both the patient's state and the test's quality can fluctuate.

Thought experiment: Three windows, one person

Consider a fictional patient, Mira.

Window A. Low arousal, heavy artifact, and unconfirmed hearing accompany no command response.

Window B. During high arousal, two valid surrogates disagree while the candidate boundary remains stable.

Window C. Repeated, content-specific imagery modulation appears with route controls and a resolved minimal bearer, but there is no motor response.

For each window, ask:

1. Which of the four outcomes is even available under the evidence rules?
2. What is the dominant uncertainty source? Record the cause profile $P_R(A) = \{e, s, m, b, t\}$ rather than naming only one vague doubt.
3. What single next measurement would most directly address that source?
4. What broader conclusion would exceed the target?

Checkpoint. Referring to the same patient across windows does not by itself establish persistence of the same minimal conscious bearer. Track the patient-level mapping separately from the time-indexed scientific classification and from any claim that one mapped minimal bearer persists.

6. Blindsight and binocular rivalry: performance is not presence

In blindsight, a person can make above-chance forced choices about stimuli in a region they report as unseen. The performance establishes information sensitivity; it does not by itself settle whether the blind-field content is phenomenally present. Cø / N* directs attention to the content-specific component profile. Unconscious information may guide behavior while failing one or more of integration, diverse causal availability, and recurrent stability.

Binocular rivalry offers a complementary design. Physical stimulation can remain stable while the dominant percept alternates. Eye movements and other involuntary markers may help time those changes without requiring a report on every trial. The model predicts that content-specific N* should track the dominant content, but the markers are anchors or timing aids, not constituents by definition. Physical-alternation controls and report/no-report contrasts help separate perception from introspection and motor preparation (Frässle et al., 2014; Tsuchiya et al., 2015).

Why it matters. These cases replace the weak contrast “can perform versus cannot perform” with the sharper question “which causal role profile supports this content?”

7. Split systems: count candidates before counting subjects

Split-brain cases invite a dramatic question: one subject or two? The boundary-first approach slows the question down. Investigators should evaluate left, right, and residual joint candidates under a declared task and temporal horizon.

A subsystem qualifies as a candidate only if it shows sufficient autonomy, role completeness, stability, and separability. Shared subcortical regulation or longer-timescale coordination may support a larger candidate even when task-specific processing is lateralized. Results may differ across a one-second perceptual horizon and a day-long regulatory horizon without making the method arbitrary. What matters is that the horizons and candidate-generation rules were fixed before the preferred metaphysical answer was known.

Method note. Do not compare pre- and post-disconnection component scores as properties of one unchanged bearer unless boundary identity has been established. The intervention may change what the candidate system is.

Think of a university that divides into two campuses. Each campus may handle admissions, teaching, payroll, and governance independently, or the two may still depend on one central administration. Counting buildings does not settle how many institutions exist; functional autonomy and role completeness matter.

Where the analogy breaks. Universities are legal and social entities whose boundaries can be stipulated. Candidate conscious systems are proposed causal units, so their individuation must be supported by intervention and dynamical evidence rather than paperwork.

8. Evidence and limits

CØ / N* turns clinical and biological cases into structured hypotheses. Repeated assessment, task-based neuroimaging, perturbational measures, and no-report contrasts can provide anchors. They do not independently validate the complete theory. A measure may support one work package - such as boundary stability, availability, or bundle discrimination - without establishing constitutive identity.

Source note. This chapter synthesizes the core model with *Indeterminacy as a Scientific Result* and uses clinical context from Wannez et al. (2017), Bodien et al. (2024), and Casali et al. (2013). These external studies motivate measurement and repeated-assessment problems; they do not validate the full CØ / N* identity claim.

Fair rival comparison

A report-centered clinical classifier and CØ / N* may agree in an alert communicating patient but diverge in locked-in syndrome or an invalid task window. That divergence is fair only if the report-centered model actually treats report as necessary; many workspace and higher-order theories allow consciousness without immediate motor output. Compare a named version and the same target. Similarly, a perturbational-complexity account may issue a state-level prediction where CØ / N* issues a content-specific one. A disagreement requires matched grain and interval rather than two labels placed side by side.

The strongest comparison asks each view to predict a near miss prospectively: motor output lost while the internal bundle persists; task performance absent because the sensory route fails; or an N* component selectively altered while arousal and the candidate remain stable. Both sides should state what invalidates the case and what result would revise their view.

The deepest limit is the absence of a universal consciousness oracle. Investigators rely on converging, partly theory-laden evidence. That does not make testing pointless; it makes independence, preregistration, held-out validation, and calibrated abstention especially important.

Clinical action introduces another layer. A scientific result combines evidence and inference rules. A practical recommendation also combines possible harms, reversibility, urgency, alternatives, and values. The same indeterminate scientific result can rationally support low-burden communication trials or continued monitoring without being relabeled “probably conscious” in the scientific record.

Do not infer. Nothing in this chapter is a clinical protocol or treatment recommendation. Classroom cases train inference, not bedside decision-making.

Practice studio

Quick checks

1. Why is a negative awakening report weaker than a licensed negative about a specified dream interval?
2. Name two auxiliary capacities required by a command-following task that are not identical to phenomenal presence.
3. What is the difference between indeterminate and unlicensed?
4. Why might two anesthesia agents require different validation studies?
5. In a split system, why must positive-bearer status be recomputed after disconnection?

Individual transfer

Choose one case - REM dreaming, deep non-REM sleep, locked-in syndrome, cognitive motor dissociation, blindsight, rivalry, or split brain. Complete a one-page inference record with these fields: target; candidate; grain; regime; interval; boundary status; V ; N_1 ; N_2 ; N_3 ; strongest anchor; strongest confound; current four-way result; smallest next study; and one conclusion the evidence does not license.

Collaborative exercise: evolving multidisciplinary case conference

Form teams of five. Assign these roles:

- **Boundary specialist:** freezes the candidate and horizon.
- **Component specialist:** tracks V , N_1 , N_2 , N_3 and their uncertainty.
- **Clinical-methods lead:** audits routes, comprehension, timing, task burden, and repeated sampling.
- **Epistemic auditor:** enforces outcome precedence and records uncertainty causes.
- **Patient-and-ethics liaison:** writes a separate practical recommendation from declared harms and reversibility.

Use the three-window Mira case. First, each role writes an independent preliminary assessment. Then the team discusses Window A and commits to one status before moving to Window B. Earlier entries may be revised but never erased; record what changed and why. For every window, request exactly one next measurement. Finish with two statements: a target-specific scientific conclusion and a broader practical recommendation. Label the evidential premises and value premises separately.

Discussion prompts. Which disagreement was empirical, which was about target scope, and which was about acceptable risk? Did urgency tempt the team to change the scientific label? Which role most often detected overreach?

Counterexample challenge

Construct the strongest case you can in which a candidate appears to satisfy V and all of N^* while independent anchors strongly suggest absent phenomenal presence. Then construct the reverse: strong presence-relevant anchors with one genuinely failed component. State which direction attacks sufficiency and which attacks necessity. Do not repair either case by changing the target, boundary, or threshold after seeing the result.

Exit ticket

Write one sentence completing each stem:

- A behavioral non-response becomes a licensed negative only if ...
- A positive result for one content does not show ...
- The most informative next measurement in my chosen case is ... because ...

Chapter summary

- Biological and clinical applications require a fixed order: target, candidate, boundary, viability, N* components, conflict rules, model output, then practical action.
- Positive, negative, indeterminate, and unlicensed are different results, not degrees on one scale.
- Waking report-rich adults are valuable calibration cases but do not define consciousness.
- Sleep, anesthesia, injury, and split systems require time-indexed and target-specific conclusions.
- Motor silence, task failure, and low scores can arise from auxiliary failures rather than absent phenomenal presence.
- Repeated sampling and selective interventions are more informative than one global contrast.
- Empirical classification must remain separate from clinical and ethical recommendation.

Key terms

Anchor evidence: evidence used to evaluate a model prediction without simply reusing the model's own defining measure.

Cognitive motor dissociation: evidence of command-related cognitive activity despite absent bedside motor response.

Four-way outcome: positive, negative, indeterminate, or unlicensed.

Negative license: the validity and fair-opportunity conditions required before non-detection can support absence.

Operating regime: the relevant state conditions, such as drug, arousal, task, or sleep stage.

Repeated assessment: sampling a variable state and test across multiple suitable windows.

Target-specific conclusion: a result restricted to the content, capacity, candidate, and interval actually tested.

Looking ahead

Clinical cases remain within one familiar biological species and institutional setting. The next chapter asks a harder transport question: when, if ever, can the same constructs and measures apply to another animal, a developing organism, an organoid, or an artificial system?

Chapter 12 - Nonhuman Animals, Organoids, and Artificial Systems

Opening puzzle: the same score in three worlds

An adult human, an octopus, and a neuron-computer organoid hybrid receive the same numerical score on a perturbational-complexity measure. A tool-using artificial agent receives a mathematically similar score from a network analysis. May we carry the human interpretation across all four cases?

Numerical equality feels objective, but the meaning of a number depends on what was measured, what causal role the measured feature plays, how the test was calibrated, and which system the score belongs to. A thermometer reading of 38 can mean a fever in degrees Celsius, an ordinary afternoon in degrees Fahrenheit, or nothing at all if the sensor is measuring a circuit board. Cross-substrate consciousness research faces the same interpretive problem, amplified by the absence of a universal gold standard.

Where the analogy breaks. Temperature scales have mature conversion rules and independently understood physical quantities. Consciousness measures lack an agreed cross-substrate reference standard, so matching values cannot be converted by a simple formula.

Plain language. “Substrate-independent” does not mean “anything with the right-looking score qualifies.” It means that biology is not required by definition if the relevant causal organization can be shown in another material.

Chapter map

By the end of this chapter, you should be able to:

1. explain **constrained substrate independence**;
2. perform a four-part transport audit covering construct, causal role, calibration, and boundary/regime;
3. distinguish neural chauvinism from empty functionalism;
4. analyze animals, infants, organoids, hybrid systems, and artificial agents without importing human assumptions silently;
5. separate a transport license from a CØ / N* verdict;
6. distinguish constituent, interface, enabling support, and environment; and
7. recommend proportional precautions without turning policy into evidence.

1. Constrained substrate independence

CØ / N* is stated in causal-dynamical terms. It does not define phenomenal presence as “having human neurons.” In principle, a nonhuman or nonbiological system could realize viability V , integration N_1 , system-wide causal availability N_2 , and recurrent stability N_3 . But the words “in principle” do not supply an empirical mapping.

The theory therefore occupies a middle position between two errors:

- **Neural chauvinism:** denying phenomenal presence to any non-neural system without asking whether the same relevant causal roles are realized.
- **Empty functionalism:** granting phenomenal presence whenever an abstract diagram, behavior, or input-output function resembles a conscious case.

The proposed organization must be actually realized in the running system. Investigators must preserve relevant directionality, effective coupling, temporal grain, recurrent organization, recipient structure, operating ranges, and a substrate-appropriate form of viability. **Effective connectivity** means modeled directed influence - how changing one part would change another - not merely an anatomical link or statistical correlation.

Imagine moving a musical score from piano to string quartet. The melody can survive a change of instrument, but only if the new arrangement preserves the relevant relations of pitch, timing, harmony, and phrasing. Copying the page numbers or matching the total number of notes would not be enough.

Where the analogy breaks. A musical work has public performance criteria and human judges who already understand its identity. The phenomenal interpretation of a causal role is itself under investigation, so transport requires independent empirical support rather than aesthetic recognition.

Why it matters. A theory that permits every convenient mapping cannot be tested. A theory that forbids every unfamiliar realization by definition cannot discover anything surprising.

2. The four supports of a transport bridge

Let T be a declared mapping from a source-domain realization to a target-domain realization. The source might be a human neural case in which a measure has some validation. The target might be an octopus, infant, organoid, or artificial agent. Four supports must hold.

2.1 Construct preservation

Does the target measure estimate the same property? A human EEG complexity index may combine anatomical, dynamical, and measurement assumptions that do not hold for an organoid or silicon network. Even when the same formula can be computed, the construct may change.

2.2 Causal-role preservation

Does the measured feature make the relevant kind of difference inside the target? For N_2 , apparent “broadcast” must support content-specific influence over multiple functionally different recipient classes. For N_3 , repeated activity must contribute to recurrent stability rather than merely reflect an externally clocked loop. Homologous anatomy can help, but causal-role evidence is ultimately the important link.

2.3 Calibration preservation

Do thresholds, error rates, rule-in regions, rule-out regions, and uncertainty estimates remain defensible? A threshold chosen in healthy adults may fail in a species with different timescales or in hardware with nearly noiseless transmission. Retuning after seeing a favorable case creates an illusion of transport.

2.4 Boundary and regime preservation

Which target system receives the attribution, at what grain and horizon? For an organoid, candidates may include tissue alone, tissue plus electrodes, or tissue plus an adaptive controller. For an AI service, candidates may include one model call, the model plus context, an agent loop, a distributed service, or a human-machine team. The mapping must also survive relevant operating-regime differences.

Method note. Transport is a license condition, not a fifth consciousness outcome. First ask whether the source interpretation can be applied. Only then use the ordinary positive, negative, indeterminate, or unlicensed rules on target evidence.

A transport bridge resembles an actual bridge with four load-bearing supports. A roadway may look intact from above, yet removal of one pier makes crossing unsafe. Likewise, construct similarity without calibration - or calibration without a stable boundary - does not license the inference.

Where the analogy breaks. Physical bridge failure is often binary and directly observable. Transport supports admit degrees, uncertainty, and interacting failures; investigators may need to report which support is weakest rather than declare one dramatic collapse.

Checkpoint. Return to the opening puzzle. For each target, identify the transport support most likely to require new evidence before the identical score can be interpreted.

3. Animals and infants: ecological fairness

Human-designed tasks often assume human sensory priorities, motor capacities, motivation, and training tolerance. An animal's failure can therefore reflect an unfair interface rather than absence of a target capacity. An octopus may use touch and chemical sensing differently from a primate. A bird may express flexible control through navigation rather than manual response. An infant may lack language and stable motor output while still supporting rich sensory organization. Multidimensional animal-consciousness frameworks help expose these task and capacity mismatches (Birch et al., 2020).

Constructing recipient classes for N_2 must be species- and stage-appropriate. Candidate recipients might include orienting, memory updating, value-guided action, internal regulation, or flexible cross-modal learning. Investigators should preregister why the classes are functionally distinct rather than merely count anatomical sites.

Evidence can draw on several sources:

- conserved mechanisms where genuine homology exists;
- causal interventions and pathway-specific effects;
- flexible behavior under species-appropriate conditions;
- motivational tradeoffs and generalization;
- known sensory and motor dependencies;
- perturbational and physiological measures; and
- developmental patterns that support or weaken a proposed bridge.

None of these alone is decisive. The goal is a convergent, independently disciplined mapping.

Thought experiment: The alien aquarium

Researchers encounter an aquatic organism whose nervous system has no familiar cortex. It learns complex spatial routes, flexibly trades safety against food, and changes internal processing after a content-specific intervention.

Stage 1. A human-calibrated cortical marker is absent. Is that a negative result or an out-of-domain test?

Stage 2. A different dynamical measure reveals integration and recurrence but was fitted after researchers saw the organism's behavior. What circularity risk appears?

Stage 3. A preregistered intervention shows that one content can modulate navigation, memory updating, and defensive control through distinct routes. Which part of N* does this most directly bear on, and what remains untested?

Do not infer. Species-appropriate adaptation is not permission to make every criterion easy to pass. The new operationalization must still create risky failures and validated negative regions.

4. Organoids and hybrid systems: where does the candidate end?

Neural organoids can display electrical activity, oscillations, complexity, plasticity, and responses to perturbation. Those achievements are scientifically interesting. Yet measures developed in organisms may rely on differentiated anatomy, sensory history, vascular and regulatory support, or behavior that an organoid lacks. Work on embodied neural cultures and organoid intelligence motivates these questions without settling phenomenal attribution (Kagan et al., 2022; Smirnova et al., 2023).

Before asking whether an organoid has CØ, ask what the candidate system is. Suppose tissue is connected to electrodes and an adaptive digital controller. Several possibilities arise:

1. **Tissue-only candidate:** the controller is a replaceable interface or support.
2. **Hybrid candidate:** tissue and controller jointly realize indispensable roles.
3. **Controller-dominated assembly:** the digital system imposes the trajectory, leaving the tissue insufficiently autonomous.
4. **Boundary-indeterminate case:** available interventions cannot distinguish the possibilities.

Removal, matched replacement, latency perturbation, and generic substitution help. If a controller can be replaced by any device delivering the same simple input, it looks more like an interface. If its specific internal memory and feedback are indispensable to role completeness, it may be constitutive at the declared horizon. If it scripts every trajectory, the assembly may fail the autonomy conditions needed for a candidate.

Plain language. A wire connected to a system is not automatically part of the subject. Neither is an external service automatically outside it. Constitution is a causal question indexed to time and task.

Thought experiment: The organoid's borrowed memory

An organoid produces adaptive responses only when a digital controller stores the previous ten trials.

Stage 1. Remove the controller; the adaptive pattern vanishes. Does indispensability alone show that the controller is a constituent?

Stage 2. Replace it with a simple matched memory buffer; performance returns. How does this affect the boundary hypothesis?

Stage 3. Extend the horizon from one second to one day. The controller now maintains long-term state that no generic buffer preserves. Should the candidate family change, and what must be frozen before testing consciousness-relevant properties?

Method note. Boundary results are not consciousness verdicts. They identify the S for which V and the N_i conditions can later be assessed.

5. Artificial systems: inspect the running architecture

Product labels such as “large language model,” “agent,” or “multi-agent system” are too coarse. A deployed service may include a base model, prompt, current context, retrieval system, persistent memory, scheduler, tool interfaces, evaluators, other models, monitoring software, and human operators. Different subsets may form candidates at different horizons - or none may meet the boundary criteria.

A stateless model call can perform a complex transformation while lacking persistence across calls. An agent loop can add recurrence, memory, and wide availability yet remain externally scripted. A multi-agent service may contain several semi-autonomous candidates, one larger workspace-like candidate, or a collection held together only by a scheduler.

Useful interventions include:

- resetting hidden or persistent state;
- substituting memory stores;
- ablating schedulers or tools;
- delaying or interrupting message routes;
- removing human selection and repair;
- replacing services with matched generic controllers;
- intervening on latent or token-level content; and
- testing whether results survive replication and implementation changes.

Fluent language is not a substitute for these tests. Nor are logs, mirrored replicas, idle subscribers, or large node counts automatically evidence of functional diversity or integration. Theory-derived indicator proposals for artificial systems are best read as research programs with explicit assumptions, not turnkey consciousness detectors (Butlin et al., 2023).

Why it matters. Artificial systems are unusually useful construct laboratories. Components can sometimes be manipulated with precision, allowing researchers to test whether proposed N_i measures distinguish the intended causal roles. Success validates construct behavior, not phenomenal presence by itself.

An analogy is a theater production. The audience sees one coherent performance, but the causal organization may include actors, stage manager, lighting board, script, understudies, and remote producer. A polished show does not tell us which parts form an autonomous production unit. Removing and replacing parts reveals the organization.

Where the analogy breaks. Theater roles are designed by humans and interpreted against an explicit plan. Artificial systems can acquire unplanned dependencies, and the consciousness-relevant causal roles remain theoretical proposals rather than occupational descriptions.

Thought experiment: The day-long agent

An AI assistant seems consistent across a day. It uses a base model, retrieval service, persistent memory, scheduler, tools, and periodic human approval.

Stage 1. Over a single model call, which elements are causally active, and which are merely enabling conditions?

Stage 2. Over a day, replacing persistent memory destroys goal continuity while replacing the scheduler with a matched scheduler does not. How should the candidate boundary change?

Stage 3. A human operator selects every goal and repairs every failure. Does the larger human-machine assembly meet autonomy and role-completeness conditions? What intervention would bear on that question?

Stage 4. The system verbally claims to be conscious after a prompt requests such a claim. How much evidential weight should that carry without content-specific causal and calibration evidence?

6. Artificial viability is a research problem

Biological viability involves maintaining an organized living regime, but an artificial analogue cannot simply be assumed. Several candidates should be compared prospectively:

Candidate criterion	What it captures	A likely error
Uptime	continued execution	includes passive scripts maintained from outside
Endogenous fault recovery	restoration after bounded disruption	may include a narrow retry service
Operating-variable regulation	maintenance of timing, memory, resources, or state integrity	may be supplied entirely by a platform

Candidate criterion	What it captures	A likely error
Functional-organization maintenance	preservation of the relevant causal roles	may depend on constant operator reconstruction
Goal-preserving adaptation	flexible recovery of task organization	sophisticated control need not be phenomenal

The choice affects the core inference because V is independently required. It cannot be defined as “whatever keeps a favored AI candidate in the positive set.” Researchers should state the criterion, test it in nonconscious benchmark systems, and explore how it behaves under bounded perturbation.

Checkpoint. Propose one artificial system that uptime would falsely include and one adaptive system that a biology-specific viability criterion would falsely exclude.

7. Evidence, limits, and transport outcomes

Transport can fail in several informative ways. The construct may not survive; the causal role may differ; thresholds may lack calibration; or the candidate boundary may remain unresolved. These are not equivalent and should be recorded separately.

When transport is licensed, target evidence can still be positive, negative, or indeterminate under $C\emptyset / N^*$. When the transport required for a particular source-domain interpretation fails, that interpretation is unlicensed in the target domain. The target is not thereby negative: independently validated target-domain measures or another licensed bridge may remain available. In addition, a theory-conditional positive can coexist with field-level model indeterminacy if serious rival theories interpret the preserved organization differently.

This layered reporting protects two facts at once: a theory can issue a clear prediction, and the broader field may remain divided about its interpretation.

The principal limitation is that cross-substrate validation is young and case-specific. A detailed protocol is not evidence that a bridge has already been built. Source-domain support, replication, and held-out transport must be distinguished. The farther a target is from the calibration domain, the more explicit the mapping burden becomes.

Source note. The transport framework combines the boundary and indeterminacy papers with comparative context from Birch et al. (2020), Butlin et al. (2023), and Kagan et al. (2022). Those contextual sources motivate candidate cases and precautions; none supplies a finished transport license for this theory.

Fair rival comparison

Two comparison dimensions are useful even before an exact rival is selected. A **biology-essentialist dimension** asks whether a feature of living organization is constitutive rather than one realization of viability and the N^* roles. A **liberal computational-functional dimension** asks whether an abstract transition structure is sufficient where $C\emptyset / N^*$ instead demands realized causal dynamics, calibrated operating ranges, and a boundary audit. These are dimensions, not complete theory versions. An adversarial study must attach each to a named, sourced view and specify both the biological requirement and the level of computational equivalence.

A fair diagnostic case would preserve the strongest rival's requirements while varying the disputed condition: a nonbiological system with prospectively demonstrated causal-role preservation but no metabolism, or two behaviorally equivalent systems that differ in autonomy and effective coupling. Both views must declare how transport failure, measurement failure, and genuine theoretical challenge are distinguished.

8. Precaution without epistemic inflation

Animals, organoids, and artificial systems raise ethical questions before science becomes decisive. Proportional precaution may consider possible severity, reversibility, alternatives, monitoring burden, evidence strength, and uncertainty. Low-cost safeguards can be reasonable under indeterminacy.

But action and evidence must remain separate. A laboratory might limit a potentially harmful organoid protocol because the downside is serious and the safeguard is cheap. That decision does not convert an unlicensed consciousness assessment into a positive one. Conversely, an unlicensed test does not prove that no protection is warranted.

Do not infer. “Treat as if conscious for this policy” and “scientifically classified as conscious” are different statements with different premises.

Practice studio

Quick checks

1. What is constrained substrate independence?
2. Name the four transport supports.
3. Why can the same formula measure different constructs in two substrates?
4. What is the difference between a constituent and an enabling support?
5. Why is an AI's fluent self-report not automatically N* evidence?
6. How can an unlicensed result still support a precautionary action?

Individual transfer

Choose one target: cephalopod, corvid, infant, organoid, organoid-controller hybrid, stateless language model, tool-using agent, or multi-agent service. Draw the actual causal organization rather than the product label. Identify a source-domain measure, the proposed mapping T , one tolerance that must be frozen, the weakest transport support, one boundary alternative, and the smallest study that could discriminate it.

Collaborative exercise: cross-substrate transport jigsaw

Create groups of five and assign roles:

- **Construct specialist:** asks whether the target variable means the same thing.
- **Causal-role specialist:** identifies difference-making in the running target.
- **Calibration specialist:** audits thresholds, error, rule-in, and rule-out regions.
- **Boundary specialist:** compares candidates at a declared grain and horizon.
- **Ethics-and-action specialist:** proposes proportional action without changing the scientific label.

Each group receives one source measure and one target dossier. During the first round, complete a transport ledger independently. During the second round, same-role specialists from different groups meet to compare failure modes. During the third round, return and revise the ledger. Record three separate outputs: transport license; within-theory four-way result; and any field-level model-indeterminacy qualifier.

Use one of these self-contained dossiers. All numbers are instructional and intentionally incomplete; the missing support is part of the exercise.

- **Animal or infant dossier:** source measure is an adult-human perturbational index with a frozen rule-in threshold. The target has a different dominant sensory modality and slower response times. Perturbations elicit repeatable differentiated responses, but no target-specific rule-out validation exists. Candidate anatomy is stable; behavioral anchors use a species-appropriate orienting task.
- **Organoid-hybrid dossier:** source measure is human recurrent effective connectivity. The tissue-plus-controller score clears the human threshold. Tissue-only recurrence is unresolved, the controller supplies patterned feedback, and generic replacement has not been tested. Calibration and boundary evidence are therefore distinct open questions.
- **Short-horizon AI dossier:** source measure is content-specific availability over one 800-ms human perceptual trial. During a single model call, three services change after token intervention, but two are mirrored loggers and the routing graph was chosen after inspection. The service boundary is stable for the call; source-domain thresholds have not been transported.
- **Long-horizon agent dossier:** source measure is bounded recurrent stability over a day. Persistent memory supports goal continuity across 12 hours, generic scheduler replacement preserves it, and human repair occurs in 18% of episodes. The one-call boundary and day-long boundary differ; the anchor battery was calibrated only on human reports.

Discussion prompts. Did a numerical resemblance survive causal-role inspection? Which role found the largest hidden human assumption? Would changing the temporal horizon alter the candidate? Which precaution relied on values rather than new evidence?

Collaborative discussion: constituent, interface, support, or environment?

Analyze an external persistent-memory service for an artificial agent. Assign system architect, interventionist, boundary theorist, skeptical reviewer, and recorder. Consider removal, matched replacement, increased latency, state substitution, and horizon change. Submit a boundary verdict for a one-minute horizon and a one-day horizon. Include one dissent and one intervention that would change the dissenter's mind.

Counterexample challenge

Design two cases. First, construct a nonbiological system that genuinely preserves every declared causal role but fails only because an operationalization is tied to human anatomy. Second, construct a system that perfectly imitates behavior and numerical scores while failing a crucial causal-role or autonomy condition. Explain how your pair challenges both neural chauvinism and empty functionalism.

Exit ticket

Complete the sentence: “The strongest transport claim I can currently license for my target is [write here] because [write here]; I must not infer [write here].” Then name one low-burden precaution that could be justified without altering that scientific result.

Chapter summary

- C $\emptyset$ / N* is substrate-independent in form but demanding in application.
- Cross-substrate inference requires preservation of construct, causal role, calibration, and boundary/regime.
- A transport license permits interpretation; it does not itself provide a positive or negative consciousness verdict.
- Species-appropriate measures must avoid importing human sensory, motor, and motivational assumptions.
- Organoid and artificial cases require explicit candidate-boundary tests.
- Running architecture matters more than product labels, benchmarks, or verbal fluency.
- Artificial viability must be operationalized and validated rather than chosen to preserve a favored verdict.
- Precaution can be rational under uncertainty, but policy is not evidence.

Key terms

Boundary/regime preservation: maintaining an interpretable target candidate and operating conditions across a mapping.

Calibration preservation: retaining defensible thresholds, error rates, and uncertainty rules.

Causal-role preservation: showing that a mapped property performs the same relevant difference-making function.

Constrained substrate independence: openness to multiple materials combined with strict causal and validation requirements.

Construct preservation: showing that a measure estimates the same property in source and target.

Constituent: a part of the candidate system at the declared grain and horizon.

Enabling support: a necessary condition that does not belong to the candidate’s constitutive organization.

Transport license: authorization to apply a source-domain interpretation to a target domain under declared evidence and tolerances.

Looking ahead

Transport asks whether a claim travels. The next chapter asks whether the theory itself earns confidence: what distinguishes C $\emptyset$ / N* from neighboring theories, what evidence validates each branch of the program, and which objections remain genuinely dangerous?

Chapter 13 - Differentiation, Validation, and the Strongest Objections

Opening puzzle: a theory made of familiar parts

Integration is central to integrated information theories. Global availability is central to workspace theories. Recurrence is central to recurrent processing theories. Viability appears in biological and dynamical approaches. If CØ / N* combines these familiar ideas, is it a new theory or only a list?

The question cannot be answered by counting novel nouns. A scientific proposal can be distinctive because it assigns familiar elements a new division of labor, joins them under restrictive rules, or predicts a pattern that none of its sources requires. But mere packaging is not enough. The conjunction must generate risky contrasts, disciplined measurements, and revision under failure.

Plain language. Novelty is not a certificate issued for unusual vocabulary. The important question is whether the architecture makes a difference to what we should observe and what would force us to change our minds.

Chapter map

By the end of this chapter, you should be able to:

1. state the distinctive commitments of CØ / N* without claiming that every component is new;
2. compare exact versions of rival theories on matched targets;
3. distinguish validation branches from a single all-or-nothing score;
4. classify support by maturity level;
5. steelman major objections and identify their residual pressure;
6. distinguish progressive revision from result-driven rescue; and
7. propose observations that would meaningfully raise or lower confidence.

1. What the package commits us to

The distinctive claim lies in a set of joint restrictions:

1. The explanandum is minimal phenomenal presence, not report, intelligence, self-reflection, or access as ordinarily defined.
2. One frozen candidate system must realize every required role for the same target in the same interval.
3. Integration N_1 , report-independent system-wide causal availability N_2 , and recurrent stability N_3 are jointly necessary and individually insufficient.
4. Viability V is assessed independently and cannot be repaired after seeing the desired verdict.
5. Unresolved credible conflict blocks positive or negative certainty rather than disappearing into a score.

6. The biconditional can fail in two directions: a resolved inclusion-minimal all-pass bearer accompanied by strong, licensed independent evidence against C_0 threatens sufficiency; a well-supported C_0 case whose recomputed frozen positive-bearer set excludes S threatens necessity.
7. Every conjunct must compete with leave-one-out and grouped rivals on cases where their predictions differ.
8. Phenomenal character is a second layer requiring a positive presence gate and an independently tested bridge.
9. Boundary selection, transport, and schematic legibility are separate work packages with explicit licenses.

That collection is not what IIT, global neuronal workspace theory (GNWT), or recurrent processing theory (RPT) individually asserts. Yet this difference on paper matters only if it creates diagnostic cases.

Why it matters. A conjunction can look precise while remaining empirically idle. If the proposed components always rise and fall together, no study can determine whether all are needed.

Think of a four-lock vault. The manufacturer claims that opening requires the correct key for every lock. Observing all four unlocked whenever the vault is open does not prove that each lock is necessary; one may be decorative or mechanically linked to another. We need cases in which one lock can be tested while the others remain correctly set.

Where the analogy breaks. Vault opening is publicly observable and the mechanism can be inspected directly. Phenomenal presence lacks a universal external indicator, so tests require held-out anchors and remain vulnerable to anchor assumptions.

2. Compare theories only after matching the target

Theory comparisons often become slogan contests: “IIT is integration,” “GNWT is global broadcasting,” or “RPT is recurrence.” Each family contains multiple versions, and neighboring theories may explain different targets. One may aim at phenomenal presence, another at conscious access, another at metacognitive report, another at the structure of experience, and another at the concept of consciousness itself.

Before comparing, prepare a **version card**:

- exact explanandum;
- proposed bearer;
- necessary and sufficient commitments;
- role of report;
- temporal commitment;
- boundary assumptions;
- source passage; and
- prospective prediction for the current case.

Then follow a comparison protocol:

1. Match targets or make the bridge between them explicit.
2. Freeze the theory versions and candidate boundaries.
3. Identify an ordinary agreement case.
4. Build a near miss on which commitments diverge.
5. State observations, validity requirements, and revision rules for both sides.
6. Score the result without retuning either theory.

Do not infer. If a GNWT version addresses conscious access and CØ / N* addresses minimal presence, different labels for one case do not yet create a contradiction. The link between access and presence must itself be specified.

Thought experiment: The three near misses

Consider three idealized candidates, all viable and measured with adequate validity.

Case A: The integrated island. Activity is strongly synergistic and recurrent but remains causally isolated from diverse recipient functions.

Case B: The reducible broadcast. A content reaches many recipient classes, yet the effects can be decomposed into independent channels with little integration.

Case C: The recurrent alcove. A stable local sensory loop persists but does not gain broad causal availability.

For each case:

1. What does CØ / N* predict?
2. Which exact version of IIT, GNWT, or RPT might disagree?
3. Is the apparent disagreement about the same explanandum?
4. What observation would adjudicate without using either theory's output as the ground truth?

Checkpoint. If your answer begins "IIT would say...", add the exact version and target before continuing.

3. Fair rival lenses

The following comparisons identify pressure points, not final verdicts.

Integrated information approaches

IIT-style theories emphasize intrinsic causal structure, complexes, integration, and exclusion. In this vocabulary, a **complex** is the candidate substrate selected by the theory's maximally irreducible cause-effect criterion; **exclusion** requires a definite substrate and grain rather than treating every overlapping candidate as a simultaneous experience. CØ / N* overlaps in treating reducible organization as insufficient, but it adds separate requirements for functionally diverse availability and bounded recurrence. A candidate that clears an IIT-specific intrinsic-structure criterion while remaining local or temporally unstable may therefore divide the theories. Fair comparison requires the exact IIT formulation, its complex-selection rule, and matched boundaries (Albantakis et al., 2023).

Global workspace and global neuronal workspace approaches

Workspace theories emphasize global availability or ignition. CØ / N* shares concern for widespread causal availability but defines N_2 without requiring report or a privileged executive recipient. It also separately requires synergy and recurrence. A case with broad report-free reach but low N_1 is a useful contrast. Yet “global” and “available” must be operationalized in both theories rather than assumed equivalent (Mashour et al., 2020).

Recurrent processing approaches

RPT versions often treat recurrent sensory processing as central and, on some formulations, sufficient for phenomenal consciousness. CØ / N* treats recurrence as necessary but insufficient without integration and diverse availability. A stable local loop with weak reach is therefore an important near miss. The temporal grain and meaning of recurrence must be held constant across the comparison (Lamme, 2006).

Higher-order theories

Higher-order theories connect consciousness to a suitable representation of a first-order state. CØ / N* does not require explicit metarepresentation for minimal presence. A system satisfying N* without a licensed higher-order state could differentiate them - but only if the higher-order theory really targets the same minimal explanandum (Lau & Rosenthal, 2011).

Predictive, active-inference, embodied, and enactive approaches

These families often emphasize hierarchical prediction, action-oriented regulation, embodiment, or organism-environment coupling. CØ / N* can include predictive and bodily dynamics inside its components and candidate boundary, but it does not currently make them separate universal conjuncts. Evidence that affective regulation or organismic coupling is repeatedly necessary in otherwise diagnostic cases could motivate revision.

Attention-schema, illusionist, and fundamentalist views

Attention-schema accounts may explain a system’s model of attention and its self-attributions (Graziano & Webb, 2015). Illusionist views may reject phenomenal properties as ordinarily conceived (Frankish, 2016). Panpsychist or other fundamentalist views may locate experience at a more basic metaphysical level. These positions may disagree with CØ / N* about explanandum or metaphysics rather than a case-level neural prediction. Forcing all of them into one score table creates false precision.

Method note. A fair comparison can end with “no direct diagnostic contrast at this level.” That is more informative than pretending that every theory predicts a different brain signal.

4. Validation is a branching map

It is tempting to imagine one ladder from concept to proof. The CØ / N* program is better represented as a common foundation with several branches.

The common foundation includes conceptual clarity, construct separability, measurement validity, and a sufficiently stable candidate boundary. After that, different questions require different evidence:

Branch	Main achievement	What failure need not destroy
Bundle attribution	Licensed $V \wedge N^*$ profile and positive-bearer result for a case	The target concept or untested operationalizations
Component necessity	Full model beats proper submodels on diagnostic cases	Other components and some bundle-level findings
Boundary audit	Prospectively generated candidates survive autonomy, mapping, minimality, and separability tests	Component constructs evaluated on other licensed referents
Transport	Source-domain interpretation survives a new domain	The source-domain result
Character bridge	Causal geometry predicts held-out phenomenal relations after the presence gate	Presence classification
Schematic competence	Trained readers reconstruct, abstain, and fault-localize reproducibly	Empirical truth of the identity claim

Selective causal intervention is a cross-cutting strategy rather than a seventh branch: it can test a component, boundary, transport mapping, or character bridge when its selectivity and licenses are adequate. Failure propagates selectively. If the character bridge fails, the presence gate may remain viable. If one component proves redundant, the exact factorization must change, but the other constructs may retain value. If transport to organoids fails, a human source-domain result is not automatically erased.

Research-status note. The seven-paper corpus specifies a coordinated validation program, not a completed validation record. Each branch below remains at the maturity level supported by its own frozen evidence; progress in one branch does not upgrade another automatically.

This map resembles a house with a common foundation and several wings. A cracked conservatory does not show that the kitchen is unsupported, but a failed foundation threatens every wing that depends on it.

Where the analogy breaks. Research branches are not physically separate. Evidence and assumptions can overlap, and a problem in one branch may indirectly alter another through revised constructs or boundaries.

Maturity levels

Each branch can be described without hype:

1. **Not yet attempted:** a criterion or protocol exists only as a proposal.
2. **Construct demonstration:** a measure behaves as intended in synthetic or tightly controlled cases.
3. **Single-domain support:** frozen predictions succeed in one relevant domain.
4. **Replicated support:** independent teams reproduce the diagnostic result.
5. **Transported support:** construct, causal role, calibration, and boundary/regime survive a materially different domain.

A detailed protocol can improve testability while remaining at the proposal stage. The existence of equations or preregistration fields is not evidence that nature cooperates.

Plain language. A blueprint is valuable, but it is not a finished building.

Where the analogy breaks. Scientific programs are revised through interaction with evidence; a blueprint normally specifies the object before construction. In research, even the intended structure may change.

5. The strongest objections

Good objections identify a route by which the theory could appear successful without earning its conclusion. Each reply below therefore ends with residual pressure.

5.1 “This is a hybrid without novelty”

Reply. The exposed conjunction, fixed inference order, boundary-first procedure, four-way evidence policy, component audit, and separate character bridge create constraints not jointly imposed by source theories.

Residual pressure. The architecture must yield unique diagnostic predictions. If every apparent success is already required by one source theory, or if the conjunction never separates empirically, the novelty claim weakens.

5.2 “ N_2 smuggles access consciousness back in”

Reply. N_2 is defined as internal, content-specific, counterfactual causal availability to multiple functionally diverse recipients. Report and executive recipient classes are not required.

Residual pressure. If every credible operationalization ultimately relies on report, working memory, or executive control, the distinction may be verbal rather than empirical.

5.3 “Viability is an escape hatch”

Reply. V must be defined and assessed independently before the critical classification. Versioned criteria and nonconscious benchmark systems can reveal overinclusion.

Residual pressure. Viability may differ greatly across substrates. Repeated case-by-case adjustment can protect the theory from counterexamples and damage transport credibility.

5.4 “Thresholds manufacture the answer”

Reply. Explicit thresholds are better than hidden judgment. They should be calibrated, preregistered, and stress-tested across a specification neighborhood.

Residual pressure. If modest threshold changes repeatedly reverse conclusions, the sharp biconditional may need a probabilistic reformulation.

5.5 “The evidence is correlation, not constitution”

Reply. Selective interventions, route interruption, mechanism ablation, rescue, and independent held-out anchors can move beyond passive decoding.

Residual pressure. Even causal relevance does not logically establish metaphysical identity. The constitutive interpretation remains an integrative theoretical judgment.

5.6 “The hard problem remains”

Reply. The theory offers a mid-level constitutive hypothesis and a compositional account of unity, system-level presence, and temporal presence. It does not claim to derive why any physical organization is accompanied by experience from premises acceptable to every metaphysics.

Residual pressure. A conceivability or explanatory-gap objection remains live. Schematic intelligibility does not close it by itself.

5.7 “Embodiment and affect are underplayed”

Reply. Bodily and affective organization can enter candidate boundaries, viability, content dynamics, and the character profile.

Residual pressure. If licensed diagnostic cases show that organismic regulation or valence is independently necessary, the current conjunction may omit a component.

5.8 “The conjuncts cannot be separated”

Reply. The ablation framework includes a “causally coupled” outcome. It does not force an indispensable verdict when selective dissociation is impossible.

Residual pressure. Stable inseparability would undermine claims that the components have separately established necessity. The factorization may need to combine terms.

5.9 “There is no consciousness oracle”

Reply. Comparative predictions can be tested with plural, held-out anchors that are independently validated and not dual-used to define the model.

Residual pressure. Results remain anchor-relative. No finite anchor set eliminates the other-minds problem or every theoretical assumption.

Checkpoint. Choose the objection you think creates the greatest empirical risk. What result - not merely argument - would sharpen it?

6. Revision rules: change that increases constraint

A research program becomes progressive when failure leads to a principled change that creates more demanding future predictions. Possible revisions include:

- remove a redundant conjunct;
- combine components that are robustly coupled;
- narrow a transport claim;
- replace an invalid operationalization;
- reformulate a brittle sharp threshold probabilistically;
- add a component demanded by repeated diagnostic cases; or
- reject the bundle if licensed counterexamples persist.

A degenerating response moves boundaries after results, shifts thresholds without independent calibration, adds one-off exceptions, or ensures that no future result can count against the view.

Thought experiment: The moving target laboratory

A preregistered study finds a viable, integrated, recurrent candidate with low N_2 and strong positive anchors. The result challenges N_2 necessity.

Stage 1. The team raises the N^* threshold for every component so the case becomes “unlicensed.” Is this a principled revision?

Stage 2. The team discovers that the N_2 intervention also changed the candidate boundary. Which inference is blocked, and what follow-up becomes necessary?

Stage 3. Several independent studies find adequate selectivity and the same pattern. What revision options preserve maximum empirical content?

Stage 4. The team proposes a new recipient class after inspecting every dataset. What prospective constraint would distinguish discovery from rescue?

7. Evidence and limits

The strongest case for $C\emptyset$ / N^* would not be one high-performing classifier. It would combine construct separability, stable boundary determination, diagnostic component cases, selective intervention and rescue, independent anchors, preregistered conflict rules, replication, and cautious transport. Schematic readability would add explanatory organization, while a separately validated geometry bridge would extend the account toward character.

The current corpus mainly articulates a research architecture, methods, failure modes, and proposed empirical programs. That is a real contribution, but it must not be confused with completed empirical validation. A new theory should be presented at the maturity level its evidence supports.

Do not infer. Precision in a manuscript is not precision in a measurement. A formal rule can be fully explicit while its operationalization remains immature.

Practice studio

Quick checks

1. Why is “the components are familiar” not yet a decisive objection?
2. What must a theory version card contain?
3. Give one example of target mismatch in a theory comparison.
4. Why does a failed character bridge not automatically refute the presence gate?
5. What distinguishes progressive from degenerating revision?
6. Which objection most directly threatens the interpretation of N_2 ?

Individual transfer

Select one exact rival theory version from a primary source. Build a one-page card listing its explanandum, bearer, commitments, report role, temporal requirement, boundary assumptions, and source passage. Construct one ordinary agreement case and one near miss. State what each theory predicts before writing which view you favor.

Collaborative exercise: versioned rival-prediction laboratory

Form teams of four:

- **CØ / N* advocate:** reconstructs the target model and freezes its prediction.
- **Rival advocate:** reconstructs one exact alternative and has veto power over caricatures.
- **Comparison chair:** rejects target or boundary mismatches.
- **Source-and-evidence auditor:** checks source support and identifies adjudicating evidence.

Analyze the integrated island, reducible broadcast, and recurrent alcove. For each, complete four fields: ordinary agreement or diagnostic disagreement; exact target; rival predictions; decisive observation with licenses. Rotate advocates after each case. The deliverable is a 250-word adversarial-test abstract signed as fair by both advocates or accompanied by a precise unresolved objection.

Discussion prompts. Did the conjunction produce a unique prediction or only a new description? Where did “it depends” indicate a real version difference, and where did it hide insufficient preparation? Which theory had more freedom to reinterpret the case?

Collaborative discussion: research-program health board

Assign program proponent, rival theorist, empirical methodologist, philosophy-of-science chair, and recorder. Consider four failure cards: a redundant conjunct; repeated transport failure; unstable thresholds; and a resolved minimal positive bearer with strong negative anchors. For each card, choose among measure repair, scope restriction, factorization change, probabilistic reformulation, adding or removing a conjunct, or rejecting the program. Record one dissent and the future prediction created by the chosen revision.

Counterexample challenge

Design the strongest licensed counterexample to the **sufficiency** direction and the strongest licensed counterexample to the **necessity** direction. You may not use behavior alone, an unresolved candidate, an invalid transport, or evidence generated from N* itself. Then write the strongest fair reply to each without changing frozen specifications.

Exit ticket

State:

1. the most distinctive risky claim of CØ / N*;
2. one licensed result that would increase your confidence;
3. one licensed result that would decrease it; and
4. the revision you would demand after the negative result.

Chapter summary

- CØ / N* is distinctive, if at all, through its joint restrictions and diagnostic predictions rather than novel vocabulary.
- Fair comparisons require matched targets, exact theory versions, frozen boundaries, and prospective predictions.
- IIT, GNWT, RPT, higher-order, predictive, embodied, illusionist, and fundamentalist theories may disagree at different explanatory levels.
- Validation branches share foundations but have different additional requirements.
- Failure should propagate only to claims that depend on the failed branch.
- The strongest objections expose access smuggling, viability flexibility, threshold instability, circular anchors, metaphysical gaps, omitted embodiment, component coupling, and the lack of an oracle.
- Progressive revision increases future constraint; degenerating revision protects the theory after the fact.

Key terms

Adversarial test: a study whose diagnostic contrast and outcome rules are accepted as fair by serious proponents of competing views.

Diagnostic case: a case on which frozen models make different predictions.

Maturity level: the evidential status of a work package, from proposal through transported support.

Progressive revision: a principled change that narrows or alters a theory while creating sharper future predictions.

Residual pressure: the part of an objection that remains after the strongest available reply.

Target matching: ensuring that compared theories address the same explanandum or an explicit bridge between explananda.

Version card: a source-backed record of an exact theory formulation and its prospective commitments.

Looking ahead

The final chapter turns evaluation into design. You will build an adversarial capstone that freezes its target, rival predictions, measures, validity rules, and outcome-to-revision commitments before the critical result is known.

Chapter 14 - Capstone Adversarial Research Design

Opening puzzle: the study that cannot lose

A research team proposes to test whether recurrence is necessary for phenomenal presence. If recurrence and the anchor measures both decline, the team will call recurrence necessary. If neither declines, it will say the intervention was too weak. If recurrence declines but anchors do not, it will say another kind of recurrence must have remained. If anchors decline but the recurrence score does not, it will say the score missed the relevant mechanism. Every outcome receives an explanation, yet none can lower confidence.

The problem is not that explanations after data are always illegitimate. The problem is that the study has no exposed commitment. A strong capstone makes room for failure by deciding in advance which target is tested, where rival predictions differ, how measures are licensed, and what revision follows each outcome.

Plain language. A scientific plan should say not only “here is what we hope to find,” but also “here is what we will conclude if we do not find it.”

Chapter map

By the end of this chapter, you should be able to:

1. choose one tractable pressure point in the CØ / N* program;
2. define a target, candidate, grain, regime, interval, and diagnostic contrast;
3. construct fair, prospective predictions for the full model and a serious rival;
4. protect measurement validity and anchor independence;
5. specify selectivity, negative controls, equivalence, uncertainty, and missingness;
6. preregister positive, negative, indeterminate, and unlicensed outcomes;
7. map every result to a revision or next commitment; and
8. organize an adversarial team whose disagreement improves the study.

1. Choose a branch, not “consciousness in general”

The common foundation freezes the target, candidate family, constructs, measures, and validity rules. From that frozen foundation, choose one of six branches:

- case-level presence classification;
- component-necessity audit;
- boundary audit;
- transport audit;
- phenomenal-character bridge; or
- schematic-competence audit.

A phenomenal-character project has the additional gate:

positive presence output → phenomenal-character bridge.

A capstone should not attempt every branch. Good projects find one point where an interpretable result would alter the model. Examples include:

- an N_2 diagnostic case that preserves V, N_1, N_3 ;
- a feedforward but duration-matched adversary for N_3 ;
- a grouped ablation asking whether N_1 and N_3 are jointly redundant;
- a boundary intervention in a biological-digital hybrid;
- transport of one validated surrogate across species;
- a geometry-changing intervention for the character bridge; or
- a blinded reproducibility study of schematic readout.

Use the common workbook sections that provide genuine diagnostic leverage for the selected branch. A field may be marked **not applicable** only with a scientific justification and a replacement source of diagnostic leverage; it may not be skipped merely because it is difficult.

Selected branch	Required common workbook sections	Additional branch module	Branch-specific output to preserve
Case-level presence	Cover, A-F, I-L, P-R	none	Study disposition plus V, N_1, N_2, N_3 and bearer statuses
Component necessity	Cover, A-G, I-L, P-R	none	Study disposition plus five-status result for every audited component
Boundary audit	Cover, A-E, H-L, P-R	none	Study disposition plus candidate, minimality, separability, and bearer-license results
Transport audit	Cover, A-F, J-L, P-R	M	Study disposition plus transport license and target-domain result
Phenomenal character	Cover, A-F, J-L, N, P-R	N	Study disposition plus presence gate and four bridge outcomes
Schematic competence	Cover, A-F, J-L, O, P-R	O	Study disposition plus separate D, C, F, E statuses, competence result, and identity-confirmation limit

Source note. This capstone architecture integrates the complete seven-paper corpus. The branch modules preserve the fact that bearer resolution, component necessity, transport, phenomenal character, and schematic competence have different success conditions and different failure implications.

Why it matters. “Does anesthesia eliminate consciousness?” is usually too broad. “Does selective recurrent-feedback interruption remove a specified auditory content while preserving V, N_1, N_2 in this candidate and interval?” exposes a narrower and more interpretable contrast.

Choosing a capstone is like selecting one load-bearing joint in a bridge for stress testing. Testing every bolt, cable, and foundation at once may look comprehensive, but if the bridge flexes you will not know which element mattered.

Where the analogy breaks. Engineered bridges have known design intentions and public failure criteria. Consciousness constructs and anchors are partly contested, so a capstone must also test whether its instruments mean what the design assumes.

2. Freeze the specification

Every project needs a written analysis specification Θ containing at least:

- exact target and unit of analysis;
- candidate system or frozen candidate family;
- spatial and temporal grain;
- operating regime;
- interval τ ;
- content or contrast $\mathcal{C}$;
- component definitions and thresholds;
- boundary and viability rules;
- primary and rival models;
- data partitions and analysis order; and
- outcome precedence rules.

Freezing does not mean pretending that the first specification is certainly correct. It means recording a primary specification and a principled neighborhood of alternatives before critical inspection. Sensitivity analysis can then show whether the conclusion is stable or fragile.

Method note. The specification should be versioned. Amendments remain visible with dates, reasons, and whether critical data were already inspected.

Thought experiment: The expanding candidate

A team intervenes on a local network to reduce N_2 . After the result threatens the theory, it expands the candidate boundary to include a remote control loop that restores availability.

Stage 1. Why does retrospective expansion threaten the comparison?

Stage 2. Suppose boundary evidence collected before the experiment genuinely favored the larger candidate. How should the team proceed?

Stage 3. Suppose the intervention itself changes autonomy and creates a new candidate. Is the result negative, indeterminate, or unlicensed for the original comparison?

Checkpoint. Write one specification decision in your project that would be tempting to change after an unfavorable result. Add a prospective rule governing it.

3. State full and rival predictions before the result

An adversarial project needs at least one genuine rival. This may be another consciousness theory, a leave-one-out version of N^* , a grouped-ablation model, a stimulus-only account, or a boundary alternative.

For each model, state:

1. the same target;
2. the same candidate family where possible;
3. the cases on which predictions diverge;
4. the expected direction and meaningful size of the effect;
5. observations that count as support or challenge;
6. license conditions; and
7. revisions under every outcome.

A weak rival is one chosen to lose - for example, “the null theory says nothing matters.” A serious rival explains ordinary cases and disagrees only in a diagnostic region.

Worked example: interrupting availability

Suppose a visual content is locally stable and integrated. A reversible route intervention prevents that content from modulating several preregistered recipient classes while preserving source-content identity, measurement quality, V , N_1 , and N_3 .

- The full model predicts that independently constructed presence-relevant anchors should decline when N_2 genuinely fails.
- An N_2 -ablated rival predicts that those held-out anchors can remain stable.

The interrupted routes cannot also supply the decisive anchors. Report and motor output cannot be the only validators. The study must test whether the candidate boundary or operating regime changed. After intervention, positive-bearer status is recomputed rather than inherited from baseline.

This design resembles a court case in which prosecution and defense agree in advance on the identity of the disputed document, chain-of-custody requirements, and what evidence would bear on authenticity.

Where the analogy breaks. Courts must reach practical verdicts under legal standards and deadlines. A scientific study may properly return indeterminate or unlicensed, and its goal is not to defeat an opponent.

4. Build an anti-circularity firewall

The greatest design danger is allowing the same evidence to define a component, select diagnostic cases, and validate the theory. That is like writing an exam question and grading it with the student's own answer as the key.

Separate evidence streams:

1. **Development data** help design measures and hypotheses.
2. **Calibration data** establish construct behavior, thresholds, and error properties.
3. **Adjudication data** test the frozen predictions.

For each component, separate the measure family used to construct it from the anchors used to adjudicate its necessity. If literal disjointness is impossible, declare the overlap, run a leave-channel-out analysis, and downgrade the main conclusion if it does not survive. If a recurrence score defines N_3 , it cannot be the sole evidence that removing N_3 changed presence. Held-out anchors should be plural and supported by independent validation genealogies.

Where the analogy breaks. In real science, complete independence is rare. Measures can share sensors, preprocessing, task demands, and latent causes. The goal is to map and reduce dependence, not merely place variables in different files.

An independence ledger should ask:

- Does the anchor use the same raw sensor as a predictor?
- Does it share preprocessing or a training label?
- Does it depend on the same report or task path?
- Was it selected after seeing the diagnostic effect?
- Is its interpretation derived from the tested theory?
- Could a common cause move both predictor and anchor?

Do not infer. “Multimodal” is not synonymous with “independent.” EEG and fMRI measures may still share task demand, arousal, or a theory-derived selection rule.

5. Causal diagrams, selectivity, and controls

Draw a causal diagram containing at least:

- intended component intervention;
- retained components;
- candidate boundary;
- viability;
- independent anchors;
- report and task paths;
- common causes;
- measurement channels; and
- selection into the diagnostic set.

Ask four questions:

1. Can the intervention reach the anchors through a path other than the target component?
2. Can a common cause move both the component estimate and anchors?
3. Does selecting diagnostic cases create collider or selection bias? A **collider** is a variable jointly caused by two other variables; conditioning on it can create a false association between those causes even when none existed before selection.
4. Can the intervention change the bearer, content, regime, or viability itself?

Selectivity is not established because the target score changes more than the others. Retained components need equivalence margins: ranges within which a difference is small enough to count as practically preserved. The study also needs a minimum meaningful target effect, negative controls, and, when possible, pathway-specific rescue.

Negative controls should be chosen for their diagnostic role. A sham intervention may detect nonspecific disturbance. A content control may test selectivity. A matched route may expose general communication loss. A rescue can show that restoring the proposed route, rather than merely waiting, reverses the effect.

Thought experiment: The clean-looking lesion

A virtual lesion lowers an N_3 estimator by 40% and leaves point estimates of N_1 and N_2 almost unchanged.

Stage 1. Confidence intervals for retained components are wide. Has selectivity been established?

Stage 2. The lesion also slows every process. Could duration, rather than recurrence topology, explain the anchor change?

Stage 3. A duration-matched feedforward control preserves anchors. What does that add?

Stage 4. A pathway-specific rescue restores recurrence and anchors but also raises arousal. What additional control is needed?

6. Predeclare the full outcome space

The four-way classification applies to projects as well as cases.

- **Favorable/supporting:** the diagnostic contrast appears with adequate validity, selectivity, independence, and coverage in the predicted direction.
- **Disfavoring:** the rival prediction succeeds under equally adequate conditions, or a licensed counterexample challenges necessity or sufficiency.
- **Indeterminate:** valid evidence straddles, credible anchors conflict, or uncertainty remains too large.
- **Unlicensed:** applicability, execution quality, required validity, selectivity, transport needed for a cross-domain claim, or fair detection opportunity fails.

These labels should be project-specific. A null result can mean different things:

- redundancy, if the ablated rival performs equivalently with adequate diagnostic coverage;
- low power or imprecision, if equivalence cannot be established;
- inadequate diagnostic cases, if the models rarely disagree;
- invalid intervention, if selectivity fails; or
- genuine indeterminacy, if valid evidence remains conflicted.

Precedence rules matter. If applicability, execution quality, or required validity fails, the assessment is unlicensed. If several independently admissible boundaries or models yield different licensed results, classify the field-level result as indeterminate and record its cause profile. Boundary reversal should not be buried inside ordinary error bars. Anchor conflict should trigger the declared conflict policy rather than an average chosen for convenience.

Plain language. A study can fail to answer its question for a specific, informative reason. Naming that reason is better science than forcing every dataset into “worked” or “did not work.”

7. Quantitative readiness without false precision

A graduate capstone need not solve every power problem, but it should be executable. State:

- minimum meaningful effect;
- equivalence margins for retained components;
- expected diagnostic coverage;
- multiplicity plan;
- missing-data rule;
- stopping rule;
- primary model and robustness models;
- specification neighborhood; and
- simulated-data recovery check.

Three terms prevent false precision. An **estimand** is the exact quantity the analysis aims to estimate, such as a mean target-versus-control anchor difference in the diagnostic set. **Cross-fitting** rotates data through training and held-out roles so each final prediction is evaluated on observations not used to fit that prediction. A **multiplicity plan** controls or transparently organizes the extra false-positive opportunities created by testing many outcomes, models, subgroups, or time windows.

Diagnostic coverage deserves special attention. If full and ablated models differ on only 2% of observations, excellent average accuracy may conceal a useless necessity test. Report how many cases actually enter the comparison region and how selection affects generalization.

Missingness can also be causal. Artifact may increase precisely when a patient is unstable or an intervention is strongest. A complete-case analysis can therefore select an unrepresentative subset. Record missingness reasons and sensitivity analyses prospectively.

Method note. A simulation does not validate the theory. It tests whether the planned analysis can recover known patterns and classify the declared outcomes under controlled conditions.

8. Adversarial collaboration is governance

Good will is not enough. Proponent and rival representatives need symmetric authority over prediction statements, controls, exclusions, and result language. Use a disagreement log. If agreement cannot be reached, preregister both analyses or report the unresolved dispute.

Useful roles include:

- **Principal theorist:** reconstructs $C\emptyset / N^*$ and protects target fidelity.
- **Rival representative:** ensures the alternative is serious and fairly tested.
- **Causal-design lead:** freezes candidates and maps pathways and common causes.
- **Statistics and reproducibility lead:** specifies effects, equivalence, coverage, missingness, code, and data structure.
- **Independence and licensing auditor:** tracks evidence partitions, dual use, precedence, and transport.

At each checkpoint, one of these members also serves as the **rotating chair/facilitator**. The chair manages time, applies the charter's predeclared deadlock procedure, and ensures that dissent and accessibility needs are recorded. The chair does not receive unilateral authority to alter targets, thresholds, rival descriptions, or result language.

Blinking, scripted analysis, version control, and locked stopping rules can reduce result-driven negotiation. The rival need not endorse $C\emptyset / N^*$. The fairness agreement says only that the test would bear on the declared contrast if its validity conditions pass.

9. From every outcome to a next commitment

An outcome-to-revision table is the capstone's intellectual center. For every cell, distinguish:

- theory challenge;
- construct-operationalization failure;
- intervention-selectivity failure;
- anchor-validity failure;
- boundary reversal;
- transport failure; and
- inconclusive precision or coverage.

Then state the next commitment. A licensed challenge to N_2 necessity may require removing or narrowing the conjunct. A failed N_2 measure requires new construct validation, not immediate theory rejection. A changed candidate boundary requires a new boundary-frozen test. A transport failure narrows scope. Repeated licensed sufficiency counterexamples may require rejecting the bundle.

The capstone succeeds when readers can see how a result changes the research program. It need not promise a decisive metaphysical settlement.

10. Evidence and limits

Adversarial preregistration cannot eliminate judgment. Target choice, anchor interpretation, equivalence margins, and model versions remain contestable. Selective interventions may be technically impossible in biological systems, and synthetic systems may not transport phenomenally. A clean null can still leave multiple explanations.

The value of the workflow is narrower and substantial: it exposes assumptions, prevents easy reinterpretation, separates failure causes, and makes future disagreement productive. It turns “more research is needed” into a specific statement about what kind of research and why.

Do not infer. A preregistered study is not automatically a good study. Preregistration preserves decisions; it cannot repair an invalid construct, unfair rival, or incoherent target.

Practice studio

Quick checks

1. Why should a capstone choose one branch rather than test the entire theory?
2. What is a diagnostic region?
3. Why must positive-bearer status be recomputed after intervention?
4. Give two ways an anchor can fail to be independent.
5. What is the difference between a minimum meaningful effect and an equivalence margin?
6. When should a null be classified as unlicensed rather than disfavoring?
7. What does symmetric authority contribute to adversarial collaboration?

Individual transfer

Write a 300-word capstone seed containing: exact target; primary candidate; full and rival claims; one diagnostic case; decisive observation; strongest validity threat; primary negative control; independent anchor; and the revision you would accept if your favored view loses. Mark every phrase that would be tempting to change after inspecting data.

Collaborative exercise: adversarial preregistration studio

Form teams using the five roles listed in Section 8, rotate the chair, and begin with Appendix D’s Cover, A, and B worksheets. During the studio, complete the scientifically applicable parts of C–L. After the common-core review, complete any applicable branch module M–O; do not force irrelevant fields or leave them blank without justification. Complete the following sequence:

1. **Minutes 0–8:** freeze target, models, and the diagnostic cell.
2. **Minutes 8–18:** draw the causal diagram and freeze candidate, grain, regime, and interval.
3. **Minutes 18–28:** specify measures, anchors, selectivity, equivalence, negative controls, and rescue.
4. **Minutes 28–35:** complete the four-way outcome-to-revision table.
5. **Minutes 35–42:** exchange plans with a red team.

The red team must identify one circularity route, one selectivity or validity failure, and one result the plan cannot currently classify. The original team must visibly revise at least one challenged field while preserving an audit trail. The rival representative signs a fairness statement or records the unresolved reason for withholding agreement.

Before final submission, every team also completes the universally required ethics/action, red-team, and final-audit worksheets P-R.

Discussion prompts. Which criticism was fatal, repairable before data, or a limitation to report? Did the team ever try to redefine the target? What result would still be ambiguous after all controls, and why?

Collaborative discussion: outcome-and-revision tribunal

Assign proponent, rival representative, statistician, reproducibility officer, licensing auditor, and chair. Analyze one result card: positive effect with very low diagnostic coverage; null with adequate equivalence; apparent target effect plus boundary change; conflict among independent anchors; or transport failure. Submit exactly three statements titled **Result**, **Inference licensed**, and **Next commitment**. Include one minority report if consensus hides a principled disagreement.

Counterexample challenge

Redesign the “study that cannot lose” from the opening puzzle. Preserve its substantive question but add a serious rival, a diagnostic case, independent anchors, a selectivity test, a negative control, four-way outcome rules, and at least one result that requires theory revision. Then ask another team to find a remaining escape hatch.

Exit ticket

Complete both statements privately before discussion:

- The theory would gain my confidence if [write here] under license conditions [write here].
- The theory would lose my confidence if [write here] under license conditions [write here], and I would then require revision [write here].

Chapter summary

- A strong capstone tests one tractable branch or pressure point.
- Targets, candidate families, thresholds, data partitions, and outcome rules should be frozen and versioned.
- Full and rival models must make prospective predictions on a diagnostic case.
- Development, calibration, and adjudication evidence should be separated.
- Anchor independence requires a dependency audit, not merely multiple modalities.
- Causal diagrams reveal alternative paths, common causes, selection bias, and boundary changes.
- Selectivity requires retained-component equivalence, negative controls, and preferably rescue.

- Favorable, disfavoring, indeterminate, and unlicensed outcomes should all map to declared next commitments.
- Adversarial collaboration needs symmetric governance and an auditable disagreement process.

Key terms

Adjudication data: held-out data used to evaluate frozen predictions.

Diagnostic coverage: the proportion and diversity of cases on which compared models actually disagree.

Equivalence margin: a prospectively justified range within which a retained component counts as practically unchanged.

Independence ledger: a record of shared sensors, preprocessing, labels, tasks, and theoretical assumptions among predictors and anchors.

Minimum meaningful effect: the smallest target change that would matter for the theoretical contrast.

Negative control: a condition expected not to show the target-specific effect but able to reveal a confound.

Outcome-to-revision table: prospective mapping from every declared result class to its inferential consequence and next commitment.

Specification neighborhood: a justified set of nearby analytic choices used to assess robustness.

Symmetric authority: equal procedural input for serious rival and proponent representatives.

Where inquiry goes next

You now have the conceptual and methodological vocabulary to engage the Cø / N* program as a participant rather than a spectator. The appropriate posture is neither loyalty nor reflexive dismissal. It is disciplined curiosity: reconstruct the claim accurately, identify its strongest pressure point, design the fair test, preserve the limits of the result, and revise in proportion to what the evidence licenses.

Appendix A - Notation at a Glance

Notation is useful only when it reduces ambiguity. Read every symbol with its local prose label, especially when a letter is reused in different papers. Your first task is not calculation; it is identifying what a symbol commits the investigator to measure, hold fixed, or refrain from inferring.

The core claim

$$C_0(S, c, \tau) \leftrightarrow V(S, \tau) \wedge N_1(S, c, \tau) \wedge N_2(S, c, \tau) \wedge N_3(S, c, \tau).$$

Read this as a **hypothesis**, not a definition: minimal phenomenal presence for candidate S , content or content-bearing state c , and interval τ obtains if and only if the same candidate is viable and jointly realizes integration, system-wide causal availability, and recurrent stability in the declared specification.

This is the compact source profile conjecture. It suppresses bearer selection. Empirically, $H_*(S, c) = V \wedge N^*$ may be estimated only for a frozen $S \in P_\Theta$; final subject attribution additionally requires $Resolved_B(c; \Theta)$ and $S \in B_{min, \Theta}^+(c)$. Hatted values are estimates and statuses, not ontic properties themselves.

Symbol or term	Student reading	Guardrail
C_0 or $C\emptyset$	minimal phenomenal presence	not report, intelligence, wakefulness, richness, or self-reflection
C_1	enriched reflective or metacognitive consciousness	not required by the minimal gate
S	candidate system	must be generated and frozen before the desired verdict
c	target content or content-bearing state	conclusions can be content-relative
τ or Δ	declared interval or horizon	presence and boundaries can be time-indexed
g	spatial, functional, and temporal grain	cross-grain comparison needs a mapping
R	operating regime	includes state, task, drug, arousal, and relevant context
Θ	frozen analysis specification	collects target, candidate rules, grain, interval, estimators, thresholds, and mappings
V	independently assessed viability	not biological life alone or an after-the-fact rescue
N_1	integration or synergy	not mere activity, correlation, or complexity
N_2	report-independent system-wide causal availability	not diffusion, anatomical size, report, or node count
N_3	bounded recurrent stability	not mere delay, duration, or memory

Symbol or term	Student reading	Guardrail
N^*	$N_1 \wedge N_2 \wedge N_3$	a conjunction: one high component does not compensate for a failed one
$H_*(S, c)$	bundle profile $V \wedge N^*$ on a frozen referent	an all-pass profile is not yet final subject attribution if the bearer is unresolved

Boundary notation

Symbol or term	Student reading	Guardrail
Ω	declared observational universe	known omissions must be recorded
I	admissible intervention family	conclusions are relative to tested interventions
G_Θ	prospectively generated candidate family	candidacy is not consciousness
P_Θ	stable, causally admissible test referents	supports component tests, not automatically a final positive subject attribution
$B_{min, \Theta}^+(c)$	inclusion-minimal all-pass bearers for content c	plural nonseparable candidates can leave the bearer indeterminate
$Resolved_B(c; \Theta)$	a unique minimal bearer or a plurality passing the separability rule	a license for attribution, not a device for forcing one answer
Inclusion-minimality	no qualifying proper subsystem at the same specification	not “fewest nodes” or “highest score”
Separability	organization persists under reciprocal channel interruption	low separability does not by itself prove one larger subject
Constituent	removal causes role loss and generic replacement does not restore it	more than being connected or remotely necessary
Interface	bounded input or output compatible with autonomous role completion	contact does not imply inclusion
Enabling support	generic replacement restores the relevant role	classification can depend on horizon

Evidence and outcome notation

Term	Student reading	Guardrail
Positive	evidence licenses attribution of the exact target under the declared model	not certainty or complete explanation
Negative	valid, sensitive evidence enters a rule-out region	not simply failure to pass a positive threshold
Indeterminate	valid evidence does not discriminate enough	not “half conscious”
Unlicensed	an interpretation prerequisite failed	not a subtle negative

Term	Student reading	Guardrail
Anchor vector Y	held-out convergent evidence for model comparison	not an infallible consciousness oracle
Surrogate conflict	credible measures disagree beyond a preregistered tolerance	check before aggregation
Cause profile	structured record of evidence, surrogate, model, boundary, and transport limitations	not another consciousness score
Validity envelope V	domain in which an evidence-target interpretation has support	a formula does not carry meaning automatically

For one assessment, $A = \{ S, C, T, E, M, V \}$ records the candidate, target, test, evidence, model, and validity envelope. Its indeterminacy cause profile is $P_R(A) = \{ e, s, m, b, t \}$: evidence scarcity, surrogate discordance, model dependence, boundary/grain sensitivity, and transport uncertainty. Neither tuple is a consciousness score.

Ablation notation

Symbol or term	Student reading	Guardrail
D_k	diagnostic cases where component k fails and every retained component passes	only here do full and leave-one-out conjunctions disagree
I_k	diagnostic coverage in the target distribution	average accuracy can hide poor coverage
Criterion ablation	delete a term from the decision rule	tests incremental prediction, not a physical mechanism
Mechanism ablation	intervene on a proposed realization	requires selectivity and boundary checks
Measurement ablation	remove an estimator or data route	tests measurement robustness, not physical necessity
Indispensable	full model reliably beats the declared rival in diagnostic cases	relative to anchors, domain, and specification
Redundant	the rival performs equivalently under adequate conditions	do not infer from a nonsignificant difference alone
Causally coupled	tested interventions cannot vary components separately	may motivate a composite factorization

Character and legibility notation

Symbol or term	Student reading	Guardrail
$P(c)$	presence-gate indicator for content c	a gate, not a richness score
G_c	causal-trajectory neighborhood for a content	not an arbitrary latent embedding

Symbol or term	Student reading	Guardrail
Q	independently anchored empirical quality-space structure	need not be Euclidean or completely labeled
F_ϕ	frozen bridge from causal geometry to quality relations	in-sample fit is insufficient
χ_c	structured character profile	not delivered by N* alone
$Legible(M,o/K)$	trained readers can reconstruct and fault-localize output $\mathcal{O}$ from model M using public knowledge K	legibility does not validate the identity hypothesis

Three quantitative habits

1. **Conjunction:** $X_1 \wedge \cdots \wedge X_k = 1$ only if every term passes. Do not average a failure away unless the rival model explicitly proposes compensation.
2. **Intervals:** a point estimate above a threshold is not enough when its uncertainty interval straddles. Use the preregistered lower- and upper-bound rules.
3. **Equivalence:** “not statistically different” does not show that retained components are preserved. The full uncertainty interval should fall inside a justified equivalence margin.

Blank notation translation

Use this worksheet to translate an equation into an auditable scientific claim before calculating. Complete every field from the frozen source or protocol; if a value is unspecified, write **not specified** rather than inventing it. The final line is the guardrail: a correctly computed quantity still cannot license more than its construct and evidence bridge support.

Translation field	Response
Equation	
Target quantity	
Candidate system	
Content/state	
Grain and interval	
Operating regime	
Frozen parameters	
Observed inputs	
Output type	
License conditions	
What the equation does not establish	

Appendix B - How to Read an Empirical Claim

An empirical claim has more structure than “the score increased” or “the theory was supported.” Use the following sequence whenever you read a paper, hear a presentation, or write your own result.

1. Locate the exact claim

Underline the smallest sentence that states the result. Then identify its grammar:

- **Target:** What property or phenomenon is being attributed?
- **Bearer:** To which system or unit?
- **Scope:** Which population, content, regime, and time window?
- **Contrast:** Compared with what?
- **Modality:** Is the language causal, correlational, predictive, constitutive, or merely descriptive?
- **Qualifier:** Under which model, assumptions, and confidence?

“The classifier detected consciousness in patients” is too compressed. A more auditable form might be: “Under model M and the preregistered validity conditions, the classifier entered the positive region for covert command-following capacity in three high-arousal assessment windows.” The second sentence may sound less dramatic, but it tells readers what was actually tested.

2. Identify the inferential branch

Ask which work package the evidence bears on:

- construct separability;
- boundary validity;
- case-level bundle attribution;
- component necessity;
- causal mechanism;
- transport;
- phenomenal-character mapping; or
- schematic competence.

Do not let evidence migrate silently between branches. A synthetic network can show that an N_2 estimator distinguishes routing from diffusion. It cannot, by that fact alone, validate phenomenal presence. A successful human character bridge does not automatically transport to an animal or AI.

3. Reconstruct the evidence chain

Write the chain in separate lines:

1. **Observation:** what was recorded or manipulated?
2. **Preprocessing:** how was the raw record transformed?
3. **Estimator:** what score or category was produced?
4. **Construct bridge:** why is the estimator interpreted as the target construct?
5. **Model rule:** how does that construct enter the prediction?
6. **Anchor comparison:** what independent evidence evaluates the prediction?
7. **Conclusion:** what is licensed at the stated scope?

Every arrow is a possible failure point. A beautiful final graph does not repair an invalid construct bridge.

4. Check validity before direction

Before asking whether a result is positive or negative, ask whether the test had the right to speak. Check:

- target and candidate match;
- boundary and regime are stable;
- population and substrate fall within the validity envelope;
- detection had a fair opportunity;
- recording quality passes;
- intervention selectivity passes;
- calibration and thresholds were frozen;
- conflicts follow a declared policy; and
- missingness and exclusions do not create a favorable subset.

If applicability, execution quality, or required validity fails, classify the inference as unlicensed even if the number looks impressive. If several independently admissible boundaries or models yield different licensed results, classify the field-level result as indeterminate and record the relevant cause profile.

5. Separate point estimate, uncertainty, and decision

A point estimate summarizes data under a model. An uncertainty interval represents a set of values compatible with the data and assumptions. A decision rule maps that evidence to an outcome. These are different objects.

Ask:

- Does the full interval clear the rule-in region?
- Does it clear a validated rule-out region?
- Does it straddle a threshold?
- Were equality rules specified?
- Is the conclusion stable across a reasonable specification neighborhood?
- Was equivalence tested or merely a nonsignificant difference reported?

6. Audit alternative explanations

Create a two-column ledger. In the left column, write every path by which the intended construct could produce the observation. In the right, write alternative paths: arousal, report, attention, task difficulty, common input, sensor leakage, selection bias, global disruption, boundary change, or retuned thresholds. For each alternative, name a negative control, measurement, or design change that would distinguish it.

7. Ask what would count against the claim

A claim with no disfavoring outcome is not yet an exposed empirical claim. Record:

- licensed favorable result;
- licensed disfavoring result;
- indeterminate result;
- license failure;
- required revision under each; and
- stopping condition for repeated repair.

8. Keep action separate

Scientific results inform decisions but do not determine them alone. A practical action combines evidence with values, losses, reversibility, burden, and alternatives. Mark each premise accordingly.

Worksheet B1 - Science-action separation

Use this record whenever a scientific result informs a practical recommendation. State the evidence premise and value/loss premise separately, then explain why the action does not change the scientific classification. The guardrail prevents precaution or urgency from being counted as new target evidence.

Science-action field	Response
Scientific premise	
Value or loss premise	
Recommended action	
Why the action does not change the scientific label	

Empirical-claim annotation sheet

Use this two-part record when reading a paper, evaluating a result card, or preparing a seminar claim. Complete the source, target, and observation fields before moving to the license continuation; the split prevents a compressed response from hiding a missing inferential bridge.

Empirical-claim field	Response
Source and location	
Exact quoted or paraphrased claim	
Target	
Bearer	
Population/domain	
Content, regime, and interval	
Comparison	
Branch of the research program	
Observation/manipulation	
Estimator and uncertainty	

Worksheet B2 - Empirical-claim license continuation

Continue the same claim record by auditing validity, boundary, anchor independence, alternatives, and the strongest licensed conclusion. Keep both pages together in the submission and repeat the source citation here if they are separated.

Empirical-claim field	Response
Source and location repeated	
Construct-validity support	
Boundary status	
Independent anchor genealogy	
Primary alternative explanation	
Negative control	
Four-way outcome licensed	
Strongest conclusion	
Tempting but unlicensed conclusion	
Next discriminating study	

Appendix C - How to Argue Fairly Across Theories

Fair argument is not timid argument. It is exact enough that disagreement bears on the strongest available version of a claim. Use this appendix when leading a seminar, writing a comparative essay, or preparing an adversarial study.

This appendix uses six records in sequence: C1 matches the question, C2 audits symmetry, C3 preserves objections and replies, C4 builds a diagnostic case, C5 freezes a disagreement contract, and C6 rehearses the comparison with peer review. Treat this page as the route map; each record begins on its own page so it can be completed, exchanged, and retained independently.

1. State the common question

Two theories can use the word “consciousness” while targeting different phenomena. Resolve the target relationship before comparing them.

Worksheet C1 - Common-question record

Use this record to determine whether two theories answer the same question before comparing their predictions. Complete it with exact versions and source passages. If no shared target or defensible bridge exists, record that limit instead of forcing a verdict.

Comparison field	Response
Theory A exact version and source passage	
Theory A's explanandum	
Theory B exact version and source passage	
Theory B's explanandum	
Shared target, if any	
Bridge needed to compare them	
What a direct comparison would leave out	

If the targets cannot be matched, compare how the theories choose targets rather than pretending they make opposing case-level predictions.

2. Reconstruct before criticizing

For each theory, identify an exact version and a source passage. Separate what the source explicitly claims from your inference about its implications. Ask a knowledgeable advocate whether your reconstruction is recognizable.

A useful reconstruction contains:

1. target;
2. bearer;
3. necessary commitments;
4. sufficient commitments;
5. role of report and access;
6. temporal and boundary assumptions;
7. ordinary success cases;
8. diagnostic failure cases; and
9. explicit limitations.

Do not make a rival easier to defeat by replacing it with a family label.

3. Apply symmetric standards

If you demand a precise boundary from one theory, demand it from the other. If one view may appeal to measurement error, specify when the rival may do so. If a failed prediction triggers revision rather than rejection for your preferred theory, allow the same option under equally constraining rules for its competitor.

The next worksheet makes those burdens visible side by side.

Worksheet C2 - Symmetry audit

Use this table after both theory versions have been reconstructed. Apply the same burden to favorable, disfavoring, indeterminate, and invalidating results on both sides. A blank cell signals an unresolved fairness problem, not permission to improvise after the data.

Question	Theory A	Theory B
What is frozen before data?		
What counts as a favorable result?		
What counts as disfavoring?		
What counts as indeterminate?		
What invalidates the test?		
Which revisions are allowed?		
When would revision become degenerating?		

4. Distinguish kinds of disagreement

A disagreement may concern:

- **semantics:** what “phenomenal presence” means;
- **metaphysics:** what kind of fact consciousness is;
- **constitution:** which organization is identical with it;
- **mechanism:** how that organization is produced or maintained;
- **measurement:** which observable estimates a construct;
- **evidence policy:** what licenses positive, negative, or abstaining conclusions;
- **scope:** which systems and regimes are covered; or
- **practical action:** what to do under uncertainty.

Label the level before offering evidence. A neural intervention may adjudicate a mechanism disagreement while leaving a metaphysical disagreement intact.

5. Steelman, reply, and preserve residual pressure

Preserve the strongest objection, the best reply, and any pressure that survives the reply. Cite the source or advocate behind the objection, and distinguish a repaired premise from a merely renamed problem.

6. Use diagnostic cases, not only intuitive cases

Ordinary waking perception is compatible with many theories. A fair comparison seeks a near miss on which their commitments diverge while nuisance factors remain controlled.

7. Disagree in a way that improves the next study

End the comparison with a prospective research commitment rather than a vote. The next three records separate the objection, diagnostic case, and disagreement contract so each can be exchanged and reviewed independently.

Worksheet C3 - Objection map

Use this form to preserve the strongest objection, the best reply, and any pressure that survives the reply. Cite the source or advocate behind the objection. The guardrail is to distinguish a repaired premise from a merely renamed problem.

Use a six-part argument map:

Argument-map field	Response
Objection source or advocate	
Strongest objection	
Premises	
Hidden assumption	
Strongest reply	
Residual pressure after reply	
Observation that would adjudicate	

A reply that leaves no residual pressure is often too quick. The point is not to manufacture a draw; it is to identify which part of the objection has and has not been answered.

Worksheet C4 - Diagnostic-case record

Use this record to move from a shared success case to a controlled disagreement case. Freeze the validity conditions before predicting the observations. If both versions still agree, the case is not diagnostic and should not be reported as a theory test.

Record:

Diagnostic-case field	Response
Agreement case	
Diagnostic near miss	
Theory A's prediction	
Theory B's prediction	
Why predictions diverge	
Shared validity conditions	
Theory-specific validity conditions	
Observation favoring A	
Observation favoring B	
Result neither can yet interpret	

Worksheet C5 - Disagreement contract

Use this contract at the end of a comparison to convert disagreement into a prospective research commitment. Each advocate should approve the description of their own view or record a specific objection. Agreement is not required, but hidden target drift and asymmetric escape rules are prohibited.

End a debate with a **disagreement contract** rather than a vote:

Disagreement-contract field	Response
Agreed target	
What the present evidence shows	
Remaining disagreement	
Primary level: semantic / metaphysical / constitutive / mechanistic / measurement / scope / policy	
Evidence that would move Theory A's advocate	
Evidence that would move Theory B's advocate	
One fair study both sides would support	
One unresolved fairness concern	
Theory A advocate approval or specific objection	
Theory B advocate approval or specific objection	

Discussion norms for difficult topics

- Critique claims, operationalizations, and inferences rather than a person's intelligence or motives.
- Give participants time to formulate objections in writing before rapid discussion.
- Rotate advocacy roles so students learn reconstruction without requiring personal agreement.
- Make uncertainty an acceptable endpoint.
- Do not use patients, disabled people, animals, or disputed artificial systems merely as rhetorical props; preserve the distinction between a teaching case and a real individual.
- Cite the strongest accessible source for a theory version, not only a secondary caricature.
- Correct your own side's overstatement before exploiting an opponent's imprecision.

Worksheet C6 - Fair-comparison rehearsal record

Use this collaborative worksheet to test whether a proposed comparison is recognizable and risky to both sides. Complete the team record before exchanging it. The exercise succeeds through a fair diagnostic contrast, not through majority agreement.

Choose two exact theory versions and form groups of four: one advocate per theory, one target-matching auditor, and one recorder. In six minutes, agree on one diagnostic case and complete the five-field team record.

Rehearsal record	Team entry
Shared question or explicit bridge between different questions	
Theory A's prospective prediction	
Theory B's prospective prediction	
Validity condition either advocate could fairly invoke	
Result that would leave the comparison unresolved	

Worksheet C6b - Peer review and advocate approvals

Exchange the completed C6 team record with another group, then preserve the reviewer’s correction, residual disagreement, and both advocates’ judgments on this continuation page. A specific objection is an acceptable result; silence is not approval.

Swap the five-line team record with another group. The reviewing group may reject it only by naming a mismatched target, a caricatured commitment, an asymmetric license, or a nondiagnostic case. Return one accepted correction and one residual disagreement. This rehearsal is successful when both advocates regard the revised test as risky, not when the class reaches a vote.

Reviewer record	Entry
	Accepted correction
	Residual disagreement
	Theory A advocate: recognizable and risky? Approval or objection
	Theory B advocate: recognizable and risky? Approval or objection

Appendix D - Student Capstone Workbook

This workbook is intentionally key-free. It does not tell you which theory to favor or which result to expect. Its purpose is to make your commitments visible before data can reward one interpretation. Complete it in stages, retain every version, and mark amendments made after any critical data inspection.

Cover record

Use this page to identify the project, its current version, and the people responsible for each intellectual role. Update it at every checkpoint rather than replacing the prior copy. The rotating chair is a temporary facilitation duty, not a sixth authority over scientific decisions.

Cover field	Response
Project title	
Team members	
Primary theorist	
Rival representative	
Causal-design lead	
Statistics/reproducibility lead	
Independence/licensing auditor	
Chair/facilitator for this checkpoint	

Worksheet Cover 2 - Provenance and inspection record

Continue the cover record by locating every frozen artifact and declaring what critical information has already been seen. Keep prior versions rather than overwriting them so reviewers can reconstruct when each commitment was made.

Project-provenance field	Response
Registration location	
Data location	
Code and environment location	
Version number and date	
Critical data inspected? Yes / No	
If yes, what was visible?	

A. Collaboration charter

Use this worksheet before selecting measures or inspecting critical data. Record how roles rotate, how every member can participate, and how disagreement and missed work will remain visible. The charter governs process; it cannot authorize post hoc changes to a frozen scientific rule.

Collaboration field	Response
Each member's private prior expectation and storage location	
Accessibility and participation plan	
Missed-work and redistribution procedure	
Behavior-specific contribution log location	
Decisions requiring unanimous agreement	
Decisions the chair may resolve	
Method for preserving minority views	

Worksheet A2 - Role-rotation matrix

Record the actual chair and role changes at each checkpoint rather than merely promising to rotate them.

Checkpoint	Chair/facilitator	Other role assignments or changes
1 - question and branch		
2 - design and measures		
3 - frozen protocol		
4 - analysis review		
5 - interpretation and submission		

Worksheet A3 - Governance and fairness continuation

Continue the collaboration charter by freezing authority, authorship, response, and fairness rules. The rival representative must either sign the fairness rationale or record a specific objection; silence is not approval.

Governance and fairness field	Response
Authority to stop an invalid analysis	
Authorship and contribution record	
Response deadline for disputed changes	
Versioned disagreement-log location	
Rival representative's fairness rationale	
Rival signature or recorded objection	

B. Select one capstone branch

Use this worksheet to choose the one branch whose success and failure conditions the project can test well. Consult Chapter 14’s routing table, mark the applicable workbook sections, and justify every **not applicable** field by naming replacement diagnostic leverage. Do not choose a branch simply because its data are easiest to obtain.

Check one primary branch:

- ☐ Case-level presence classification
- ☐ Component-necessity audit
- ☐ Boundary audit
- ☐ Transport audit
- ☐ Phenomenal-character bridge
- ☐ Schematic-competence audit

Branch-decision field	Response
Why this branch is tractable within the available time, data, skills, and ethics constraints	
Neighboring branch placed out of scope, with rationale	
Applicable common sections	
Scientifically justified N/A sections and replacement diagnostic leverage	
Required branch module (M, N, or O)	

C. Exact target and scope

Use this worksheet to define the smallest claim the study can genuinely adjudicate. Freeze the bearer, content, grain, regime, interval, and excluded claims together. If a branch does not concern empirical phenomenal presence, state its own exact conceptual or competence target rather than borrowing Cø language.

Target-and-scope field	Response
Target property or explanandum	
What the target is not	
Unit of analysis	
Content or contrast c	
Candidate system S or candidate family	
Observational universe Ω	
Spatial/functional/temporal grain g	
Operating regime R	
Interval or horizon τ / Δ	
Population or substrate	

Worksheet Capstone C2 - Claim-license continuation

Continue the exact-target record by freezing exclusions and the strongest claim the completed design could license. State the required conditions and final-report wording prospectively so broader language cannot appear after an attractive result.

Claim-license field	Response
Target property or explanandum repeated	
Primary exclusions	
Strongest conclusion the design could license	
Conditions required for that conclusion	
Broader conclusion it cannot license	
Exact scope language for the final report	
Observation or failure that would force a narrower claim	
Rival reviewer confirmation or specific target-drift objection	

D. Boundary and viability record

Use this worksheet to generate candidates independently of the desired consciousness verdict and to define viability without circular rescue. Retain excluded candidates and failed tests in the ledger. For a justified noncandidate project, explain the scope boundary that replaces a bearer audit.

Complete one vertical record per prospectively generated candidate. Duplicate this page before writing when the candidate family contains more than one member; never erase an excluded candidate.

Candidate-record field	Response
Candidate label and scope	
Independent discovery basis	
Endogenous capacity	
External dependence	
Role completeness	
Stability across the frozen horizon	
Included in frozen family? If not, exact exclusion rule	

Worksheet D2 - Boundary and viability decisions

Continue the boundary record on this page after completing the candidate table. Use the fields to freeze minimality, separability, mapping, and viability decisions. Do not remove a candidate merely because its retained presence profile is inconvenient.

Boundary-decision field	Response
Primary boundary rule	
Inclusion-minimality test	
Separability test	
Boundary-change trigger	
Cross-horizon mapping	
Frozen candidate referent preserved after intervention?	
Recomputed post-intervention positive-bearer set	

Worksheet D3 - Bearer resolution and viability decisions

Continue D2 by turning the candidate comparison into an explicit bearer license and an independently testable viability rule. Preserve a nonconscious benchmark and the action required if intervention changes either the boundary or viability status.

Bearer-and-viability field	Response
Bearer-resolution result	
Resulting subject-attribution license	
Independent viability definition	
Viability estimator and validity evidence	
Nonconscious benchmark that could expose overinclusion	
Rule if boundary or viability changes during intervention	

E. Full and rival models

Use this worksheet to reconstruct the full claim and its strongest serious rival at matched target, grain, and interval. Cite an exact version for each and freeze allowed pre-data revisions. The diagnostic rows must contain a case on which the models actually disagree.

Matched-model field	Full model	Primary rival
Exact name/version and source passage		
Target match or bridge		
Formal or verbal claim		
Necessary commitments		
Sufficient commitments		
Expected ordinary success case		
Allowed revisions before data		

Worksheet E2 - Diagnostic predictions

Continue the model comparison by recording cases in which the frozen versions diverge. Use one vertical record per case and duplicate this page for additional cases. Ordinary agreement cases can calibrate measures, but they cannot establish incremental theoretical value.

Diagnostic-prediction field	Response
Diagnostic case	
Full-model prediction	
Rival prediction	
Why the predictions differ	
Observation that adjudicates	
Contribution to diagnostic coverage	

Diagnostic-program field	Response
Minimum diagnostic coverage	
Plan if coverage is below minimum	

F. Construct and measurement validity

Use this worksheet to connect every operative construct to an estimator, calibration domain, and known failure mode. Complete only the rows required by the selected branch, but justify omissions. A precise number from outside its validity envelope remains unlicensed.

Complete one copy for every operative construct and adjudicating anchor, including V , each required N_k , and every primary anchor. Duplicate this page before writing; a single compressed row cannot carry a construct-validity argument.

Construct-record field	Response
Construct or anchor and branch role	
Operational definition	
Estimator and scoring procedure	
Calibration domain and data	
Validity envelope and transport limit	
Main confound	
Stress test or negative control	
Required omission or replacement evidence, if not applicable	

Worksheet F2 - Measurement decision rules

Continue the validity record by freezing thresholds, uncertainty, fragility, and data-quality rules for one named construct-estimator pair. Duplicate this page for every operative measure. These choices determine when an estimator may speak; they must not be tuned to preserve a desired classification.

Measurement-rule field	Response
Construct and estimator identity	
Rule-in threshold	
Rule-in justification	
Rule-out threshold	
Rule-out justification	
Equality-at-threshold rule	
Primary uncertainty interval	
Specification neighborhood	
Fragility criterion	
Missingness mechanisms considered	
Data-quality failure rule	

G. Intervention and selectivity

Use this worksheet when the branch makes a causal claim or depends on a decisive perturbation. Predeclare the intended path, retained conditions, equivalence margins, and rescue logic. For a viability-targeting intervention, distinguish failure of the viability gate from failure of the estimator’s validity and candidate mapping.

Intervention-specification field	Response
Intended intervention or decisive observation	
Target component or construct	
Proposed realization	
Dose, timing, and route	
Minimum meaningful target effect	

Worksheet G2 - Preservation, selectivity, and rescue

Continue the intervention record by freezing what must remain equivalent, what would invalidate selectivity, and what rescue would identify the intended pathway. Use full-width records so a long equivalence margin or boundary condition cannot be squeezed into a token answer.

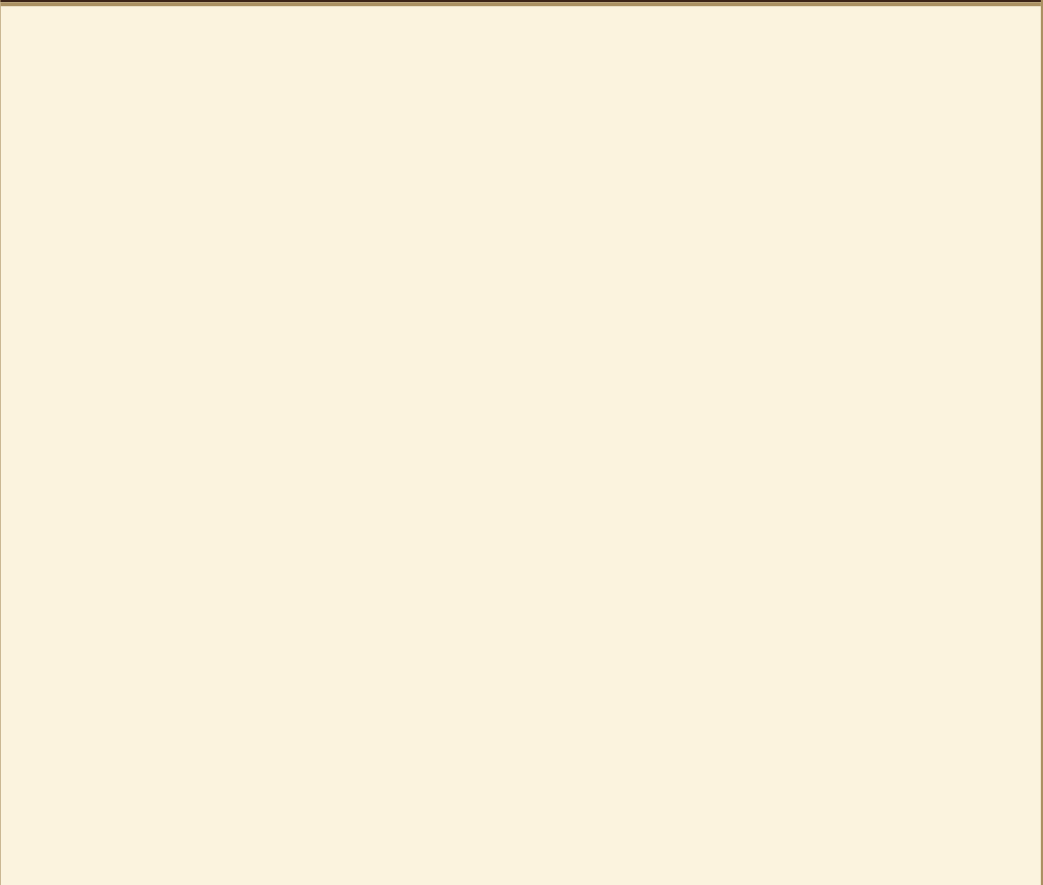
Selectivity-and-rescue field	Response
Retained components	
Closed equivalence margin for each retained component	
Candidate-boundary preservation check	
Viability preservation check	
Content-identity preservation check	
Operating-regime preservation check	
Selectivity failure rule	
Pathway-specific rescue, or reason it is infeasible	

H. Causal diagram audit

Use this worksheet to make alternative causal paths visible before interpreting an effect. Attach the diagram, then answer each audit question from the graph rather than from the hoped-for narrative. A missing or uncertain arrow must remain visible in the inference limit.

Draw or attach a diagram containing intervention, target component, retained components, boundary, viability, anchors, report/task paths, common causes, measurement channels, and selection into the diagnostic set.

Causal diagram - draw here or attach a labeled page



Worksheet H2 - Causal-path audit questions

Answer every question from the frozen graph rather than from the hoped-for narrative. If the answer differs by branch or regime, preserve the alternatives.

Causal-audit field	Response
Can the intervention reach an anchor outside the target path?	
Can a common cause move predictor and anchor?	
Can diagnostic selection create collider bias?	
Can the intervention change bearer, content, regime, or V ?	
Which arrow is least well supported?	
Which measured variable would most reduce that uncertainty?	

I. Negative controls and falsification criteria

Use this worksheet to state how confounds would reveal themselves and what observations would challenge, suspend, or leave the claim untouched. Tailor controls to the branch and document any infeasible control. A failed control cannot be relabeled as a negative theory result.

Audit each required control type, then complete one vertical record per implemented or infeasible control. Duplicate this page as needed.

- ☐ Sham intervention
- ☐ Content control
- ☐ Route control
- ☐ Duration/intensity match
- ☐ Sensor/preprocessing control
- ☐ Boundary/viability control
- ☐ Other branch-specific control

Control-record field	Response
Control name and branch role	
Confound it detects	
Expected pattern if the control passes	
Failure consequence	
Feasible? If not, reason and inferential cost	

Worksheet I2 - Falsification commitments

Continue the control audit by stating one licensed result in every direction and one failure that silences the test. Then state the repeated evidence that would force substantive revision. Do not count operational or bridge failure as direct refutation of a claim it never licensed.

Falsification-commitment field	Response
Licensed observation favoring the full model	
Licensed observation disfavoring the full model	
Result that remains indeterminate	
Failure that makes the inference unlicensed	
Repeated result forcing removal, combination, narrowing, probabilistic reformulation, or rejection	
Result that does not falsify the theory because it bears only on an operationalization or transport bridge	

J. Anchor-independence ledger

Use this worksheet to trace every path by which a component estimate and an adjudicating anchor might share data, labels, preprocessing, or theory assumptions. Record corrective action before final testing. An unavoidable overlap must weaken the conclusion and receive a leave-channel-out analysis where possible. Duplicate the dependency table when the study uses more than two anchors.

Anchor registry	Identity, source, and genealogy		
Anchor 1			
Anchor 2			
Additional anchor or continuation-page reference			

Dependency question	Anchor 1	Anchor 2	Corrective action
Shares raw sensor with component estimate?			
Shares preprocessing?			
Shares development labels?			
Depends on report or task route?			
Selected after seeing the target effect?			
Interpretation derived from tested theory?			
Common cause likely?			

Worksheet J2 - Data partition and blinding plan

Continue the independence ledger by freezing which records may inform development, calibration, and final adjudication. Name people or roles, not only datasets, when blinding is part of the license.

Partition-and-blinding field	Response
Development data	
Calibration data	
Final adjudication data	
Cross-fitting or held-out plan	
Who remains blind to which data	

K. Quantitative and reproducibility plan

Use this worksheet to make the estimand, uncertainty rules, missingness, stopping, code, and provenance reproducible. Freeze confirmatory choices before critical data inspection and label later analyses exploratory. Numerical detail is useful only when it corresponds to a scientifically justified decision.

Quantitative-plan field	Response
Primary estimand	
Primary statistical model	
Minimum meaningful effect	
Equivalence margins	
Expected diagnostic coverage	
Sample-size or precision rationale	
Multiplicity plan	
Sensitivity analyses	

Worksheet K2 - Missingness, simulation, and reproducibility

Continue the quantitative plan by freezing data-loss rules, stopping, recovery tests, software, and provenance. Every location must be specific enough for a reviewer to find the record.

Reproducibility-plan field	Response
Missing-data rule and mechanisms	
Exclusion rule	
Stopping rule	
Simulation scenarios	
Recovery success criterion	
Variable dictionary location	
Analysis code or pseudocode location	
Software and version lock	
File naming and data provenance plan	
Confirmatory versus exploratory labels	

L. Outcome vocabulary and pattern plan

Use this worksheet before writing any outcome-to-revision record. Freeze both levels of conclusion: the study-level disposition and every special component, bridge, boundary, transport, or competence status required by the selected branch. This order prevents a branch output from being invented after an observed pattern is seen.

Branch-status field	Response
Selected branch	
Required branch output vocabulary	
Definition and decision rule for each branch status	
Mapping from branch status to study-level disposition	
Rule preventing collapse of component or bridge results into a generic label	

Prepare a separate L2 record for every applicable pattern:

- ☐ Preregistered favorable pattern
- ☐ Preregistered rival-favorable pattern
- ☐ Threshold-straddling evidence
- ☐ Credible anchor conflict
- ☐ Low diagnostic coverage
- ☐ Selectivity failure
- ☐ Boundary reversal
- ☐ Transport failure
- ☐ Adequate-equivalence null
- ☐ Other unanticipated pattern

Worksheet L2 - Outcome-to-revision record

Complete one vertical record for each pattern selected above and duplicate this page as needed. Every record must preserve the study disposition and branch-specific output separately, and every abstaining or license-failure outcome must still name a next commitment.

Outcome-record field	Response
Observed or prospective pattern	
License checks	
Study disposition	
Branch-specific output or status	
Claim affected	
Required revision or next study	

M. Transport audit, if applicable

Use this module only when a source-domain interpretation is carried to a target domain. Test construct, causal-role, calibration, and boundary/regime preservation separately. A failed bridge unlicenses that transported interpretation; it does not make the target negative or erase the source result.

Transport-bridge field	Response
Validated source-domain claim	
Target-domain realization	
Declared mapping T	
Construct preservation evidence	
Causal-role preservation evidence	
Calibration preservation evidence	
Boundary/regime preservation evidence	

Worksheet M2 - Transport result and scope limit

Complete the transport decision only after the four preservation records above. The transport license and the separate target-domain result answer different questions and must receive separate justifications.

- **Transport license:** ☐ licensed ☐ indeterminate ☐ unlicensed
- **Target-domain result:** ☐ positive ☐ negative ☐ indeterminate ☐ unlicensed

Transport-decision field	Response
Frozen tolerances	
Smallest observation that breaks the bridge	
Transport-license justification	
Target-domain-result justification	
Scope left intact if transport fails	

N. Phenomenal-character bridge, if applicable

Use this module only after freezing an independently supported positive presence gate for the tested contents. Keep causal-trajectory geometry and phenomenal-quality anchors in separate evidence streams, then test the bridge on held-out contents and a geometry-changing intervention. Failure here bears on the character extension, not automatically on minimal phenomenal presence.

Character-bridge foundation field	Response
Frozen presence-gate rule and result	
Contents admitted by the gate	
Causal-trajectory object G_c	
Independent quality-space object Q	
Frozen bridge F_ϕ	
Development, calibration, and held-out partitions	
Nuisance models and invariances	

Worksheet N2 - Held-out deformation and bridge result

Continue the character audit with the held-out prediction and geometry-changing intervention. Preserve the independently observed anchor change and the presence-gate retest before classifying the bridge.

- **Bridge outcome:** ☐ supported ☐ refuted ☐ indeterminate ☐ unlicensed

Character-bridge result field	Response
Held-out relational prediction	
Geometry-changing intervention and predicted deformation	
Anchor change observed independently	
Presence gate preserved after intervention?	
Bridge-outcome justification	
Symmetry or underdetermination that remains	
Scope left intact if the bridge fails	

O. Schematic-competence audit, if applicable

Use this module to test whether trained readers can reconstruct and fault-localize a public role-annotated model under blind conditions. Freeze the interpretation manual, trial-level unit, abstention rules, and scoring before reveal. Competence or legibility does not establish that the underlying consciousness identity is true.

Schematic-trial design field	Response
Public schematic and interpretation manual	
Declared reader population and training	
Blind trial-level unit	
Bearer-definition decision and evidence	
Operational/causal-readout decision and evidence	
Coverage/unity decision and evidence	
Fault-localization prediction	
Explanation/teaching criterion	
Permitted abstention and ambiguity rules	
Inter-rater reliability criterion	

Worksheet O2 - Blind outcomes, competence, and limits

Complete the outcome statuses only after the held-out fault is revealed. A licensed competence result does not confirm the underlying consciousness identity.

- **Trial license:** [] licensed [] unlicensed
- **Conjunctive competence:** [] supported [] not supported [] indeterminate

Schematic-trial outcome field	Response
Held-out fault outcome	
<i>D</i> decomposable-semantics status and evidence	
<i>C</i> compositional-readout status and evidence	
<i>F</i> fault-localization status and evidence	
<i>E</i> evidential-independence status and evidence	
Conjunctive competence justification	
Trial-license justification	
Identity-confirmation status or explicit limit	
Claim the trial cannot license	

P. Ethics, feasibility, and practical action

Use this worksheet to state burdens, oversight, and value/loss assumptions separately from the scientific result. Complete it for every project affecting people, animals, biological tissue, or systems with plausible welfare or safety stakes. Precaution may be justified under uncertainty, but it must not be counted as evidence for the target.

Ethics-and-feasibility field	Response
Participants or systems affected	
Burden and risk	
Potential benefit	
Reversibility	
Alternatives	
Consent or oversight requirements	
Data privacy and stewardship	
Stopping condition for welfare or safety	
Scientific result stated separately	
Value/loss assumptions	

Worksheet P2 - Outcome-to-action record

For each possible scientific outcome, state a proportional action and its value/loss rationale without changing the scientific label. Use the full response area to record reversibility, burden, uncertainty, and the condition that would change the action.

Scientific outcome	Proportional practical action and rationale
Favorable	
Disfavoring	
Indeterminate	
Unlicensed	

Q. Red-team disposition log

Use this worksheet to preserve every substantive criticism and its disposition. Accepted criticisms must produce a visible protocol change; rejected criticisms require a reason that another reviewer can audit. Prepare and retain at least five copies before review, use one concern per record, duplicate additional copies as needed, and never delete a record because the team reached consensus later.

Category: ☐ circularity ☐ validity ☐ selectivity ☐ boundary ☐ statistics ☐ transport ☐ ethics ☐ other

Red-team record field	Response
Concern number, reviewer, and date	
Reviewer concern	
Team response	
Change made	
Accepted by reviewer? Yes / No / Partly, with reason	
Residual limitation	

R. Final preregistration audit checklist

Use this worksheet only after every applicable common and branch module is complete. Mark **yes**, **no**, or **not applicable**; justify each N/A scientifically and identify replacement diagnostic leverage. Any unresolved **no** requires a written consequence for interpretation before the protocol is frozen.

Audit item	Yes	No	N/A
The target is narrower than “consciousness in general.”	☐	☐	☐
The candidate family was generated without the desired verdict.	☐	☐	☐
Grain, regime, and interval are explicit.	☐	☐	☐
Viability is defined independently.	☐	☐	☐
Full and rival models address a matched target.	☐	☐	☐
At least one diagnostic case makes their predictions differ.	☐	☐	☐
Diagnostic coverage has a minimum criterion.	☐	☐	☐
Component estimators have construct-validity evidence.	☐	☐	☐
Rule-in, rule-out, equality, and conflict rules are frozen.	☐	☐	☐
Uncertainty and specification fragility are handled prospectively.	☐	☐	☐
Retained components have equivalence margins.	☐	☐	☐
Intervention selectivity has a failure rule.	☐	☐	☐
Boundary, viability, content, and regime changes are monitored.	☐	☐	☐
At least one major negative control is present.	☐	☐	☐

Worksheet R1b - Final preregistration audit checklist continuation

Continue the checklist with the remaining outcome, branch, action, governance, and revision commitments. Mark every item; carry every **no** or **N/A** into R2 rather than squeezing a rationale into the checkbox table.

Audit item	Yes	No	N/A
A rescue is included or its infeasibility is explained.	☐	☐	☐
Component-construction measures and adjudicating anchors are separate; unavoidable overlap is declared, tested leave-channel-out, and reflected in conclusion strength.	☐	☐	☐
Development, calibration, and adjudication roles are separated.	☐	☐	☐
Missingness, exclusions, multiplicity, and stopping are specified.	☐	☐	☐
Simulated-data recovery or analysis rehearsal is planned.	☐	☐	☐
Favorable, disfavoring, indeterminate, and unlicensed outcomes are all defined.	☐	☐	☐
Every outcome maps to a next commitment.	☐	☐	☐
Transport claims have four-part bridge evidence.	☐	☐	☐
A character project freezes the presence gate, independent quality anchors, held-out bridge, invariances, and deformation intervention.	☐	☐	☐
A schematic project freezes the manual, trial-level unit, blind fault test, abstention rule, and competence outputs.	☐	☐	☐
Scientific status and practical action are separate.	☐	☐	☐
Rival representatives had symmetric input.	☐	☐	☐
Amendments and disagreements remain visible.	☐	☐	☐
At least one result would force a nontrivial revision.	☐	☐	☐

Worksheet R2 - Checklist exceptions and N/A rationales

Use one row for every **no** or **N/A** selected in the checklist. A no requires an inferential consequence and repair; an N/A requires a scientific rationale plus replacement diagnostic leverage. Duplicate this page when four rows do not suffice.

Checklist item	No consequence or N/A rationale	Replacement diagnostic leverage or required repair

Worksheet R3 - Final claims and unresolved items

Use this final page only after the checklist and exception log have been reviewed by a rival representative or red team. The final claims define the strongest conclusion the team will risk and the overreach it explicitly refuses.

Final-claim field	Response
Unresolved items and their consequence for inference	
Final claim the team is willing to risk	
Final claim the team refuses to make	

Appendix E - Capstone Assessment Guide

This guide makes the assessment standard visible without supplying a preferred theory verdict or result. Use it to plan work, request useful peer review, and check that every team member can defend the complete design. Unless the course specifies otherwise, the capstone grade is 70% team preregistration, 20% individual defense, and 10% collaboration record.

Five staged checkpoints

Preserve each submitted version and the reviewing team's response. A compressed course may retain Checkpoints 1, 3, and 5, but it should not skip the evidence-architecture or final-defense functions.

Checkpoint	Team deliverable	Review team must provide
1. Target and rivals	Exact target, exclusions, selected branch, and strongest rival	One target-drift risk and one stronger rival formulation
2. Diagnostic region	Disagreement set, expected coverage, candidate boundary, and branch routing	One ordinary agreement case and one possible boundary or scope reversal
3. Evidence architecture	Measures, anchors, partitions, intervention or decisive observation, and controls	One circularity route and one selectivity, validity, or bridge failure
4. Outcome freeze	Study dispositions, branch-specific outputs, and revision rules	One unclassified result and one commitment that remains movable
5. Mock defense	Complete versioned preregistration and disposition log	One fatal-unless-repaired judgment and one valuable-even-if-null judgment

Team-preregistration rubric

The following dimensions apply at the level required by the selected branch. A scientifically justified N/A receives no automatic penalty when its weight is reallocated prospectively to diagnostic leverage, construct validity, or robustness; an unjustified omission is a gap.

Dimension	Weight	What strong work makes visible
Target and scope	8	Exact explanandum or competence target, bearer or conceptual object, content, regime, interval, and excluded claims
Rival quality	10	Strong exact versions with genuinely different prospective predictions
Diagnostic identification	12	Correct disagreement set, coverage plan, uncertainty margin, and branch-appropriate leverage
Boundary discipline	8	Independent candidate proposal, stability, minimality, and separability, or a justified scope boundary for noncandidate work
Construct validity	10	Operational definitions, validation evidence, confounds, and validity envelopes for every operative measure

Dimension	Weight	What strong work makes visible
Anti-circularity	12	Enforceable evidence partitions, anchor independence, declared overlap, and no hidden dual use
Branch-specific diagnostic leverage	10	Selective intervention and rescue, boundary/transport reversal, bridge deformation, or blind fault/counterexample prediction
Outcome discipline	10	Prospective study disposition plus all special component, bridge, boundary, transport, or competence statuses
Robustness and transport	8	Reasonable alternative specifications, stability rules, and transfer limits; transport tested only when claimed
Feasibility, ethics, and data	6	Executable scope, governance, burden, sources, data structure, code environment, and reproduction materials as applicable
Revision commitments	6	Every declared result has a substantive consequence for the model, branch, or protocol

Four performance levels

Level	Anchor	Typical evidence in the submission
Complete and executable	100% of dimension points	All required choices are frozen, internally consistent, auditable, and ready to implement
Substantively sound with a repairable omission	85%	The central design is correct, but one bounded omission can be repaired without changing the claim
Developing with a consequential gap	65%	A missing license, control, bridge, or decision rule could change the interpretation
Insufficient or internally inconsistent	0-40%	The target, rival, evidence architecture, or outcome rules cannot jointly support the proposed inference

The assessor should name the decisive missing element rather than relying on a global impression. A polished document cannot compensate for a circular test, mismatched rival, or unclassifiable outcome.

Individual defense

Each student should be ready to answer one or two questions from five published domains after quiet preparation time, with a brief written supplement permitted:

1. reconstruct the target, selected branch, and full model without teammate prompting;
2. state the strongest rival fairly and identify the diagnostic disagreement;
3. explain one construct-validity, selectivity, boundary, transport, or independence risk;
4. classify an unanticipated result at both study and branch-specific levels; and
5. state a personal revision commitment if the favored interpretation loses.

Strong performance is accurate, fair to the rival, explicit about uncertainty, and able to connect a result to a nontrivial next commitment. The defense evaluates individual understanding, not confidence or speaking speed.

Collaboration record

The record should include the completed charter, role-rotation matrix, contribution log, accessibility and redistribution plan, preserved dissent, timely cross-team reviews, and the red-team disposition log. Accepted criticism should produce a visible protocol change; rejected criticism should receive a substantive reason. Collaboration quality is judged from auditable behavior and intellectual fairness, not from equal word counts or the absence of disagreement.

Collaboration dimension	Auditable evidence	Indicator requiring revision
Role rotation	Dated matrix showing actual chair and role changes	One member repeatedly controls the same high-authority role without a documented access reason
Contribution visibility	Behavior-specific log linked to versions and decisions	Credit rests on impressions, word counts, or unrecorded labor
Accessibility and redistribution	Agreed participation plan plus dated responses to missed work	Barriers or absences silently become lower intellectual authority
Preserved disagreement	Minority views remain in the versioned log with their evidential consequences	Consensus language erases a substantive objection
Cross-team review	Timely review artifacts identify a target, validity, or inference risk	Peer review supplies praise without a testable criticism
Red-team response	Every accepted concern changes the protocol; every rejection gives a reason	A criticism disappears after discussion without a disposition
Individual accountability	Each member can defend the target, rival, evidence architecture, and one revision rule	The submission depends on one spokesperson or hidden division of understanding

Glossary

Ablation: A planned removal used to test a claim. Criterion ablation removes a term from a rule; mechanism ablation intervenes on a proposed realization; measurement ablation removes an estimator or data route. These answer different questions.

Access consciousness: Availability of information for reasoning, report, and control in some philosophical and scientific traditions. It must not be assumed identical to minimal phenomenal presence.

Adversarial collaboration: A governed project in which advocates of competing views jointly specify predictions, controls, exclusions, and result language before critical data are inspected.

Anchor evidence: Held-out, convergent evidence used to evaluate a model prediction. An anchor is not an oracle and should not merely repeat the model's defining input.

Autonomy: The degree to which a candidate's organization is endogenously sustained rather than imposed from outside. Autonomy contributes to candidacy; it is not itself consciousness.

Bearer: The system to which a phenomenal attribution would apply. The bearer must be resolved rather than supplied by an unexamined pronoun such as "it."

Boundary audit: A prospective procedure for generating, freezing, and stress-testing candidate systems before measuring consciousness-relevant properties.

Boundary reversal: A change in classification caused by changing the candidate system. It reveals that individuation is part of the claim.

Calibration: The process of relating a measure to independently supported construct behavior, thresholds, uncertainty, and error rates in a declared domain.

Candidate system: A causally admissible possible bearer defined at a declared grain, regime, and horizon. Candidacy does not imply a positive consciousness result.

Causal availability (N_2): The content-specific capacity to modulate multiple functionally different recipient classes through supported routes within a bounded context. It does not require overt report.

Causal role: The difference a property makes in the running organization, not merely its anatomical location, name, or correlation.

Causally coupled: An audit outcome in which the available interventions cannot vary proposed components separately. It may motivate a composite factorization rather than an indispensability claim.

Constitutive identity hypothesis: The proposal that minimal phenomenal presence is identical with viable N^* organization in a resolved inclusion-minimal bearer. The biconditional states the hypothesis; it does not validate itself.

Construct: A theoretically defined property such as integration or availability that must be connected to observable estimators through validation evidence.

Construct preservation: Evidence that a measure estimates the same property after transport to a new domain.

Content selectivity: Evidence that an effect depends on the represented content rather than undifferentiated energy, arousal, or diffusion.

Counterfactual availability: A capacity revealed by appropriate interventions across nearby admissible contexts even when every recipient is not actually active at once.

Criterion ablation: Removal of a term from a decision rule to test incremental prediction.

Diagnostic case: A case on which frozen competing models make different predictions.

Diagnostic coverage: The amount and diversity of target-domain data that actually fall in a diagnostic region.

Equivalence margin: A justified interval within which a retained component counts as practically preserved. Nonsignificance alone is not equivalence.

Explanandum: The phenomenon a theory aims to explain. Here the core target is minimal phenomenal presence.

Four-way outcome: Positive, negative, indeterminate, or unlicensed. The last two prevent incomplete or invalid evidence from being forced into a binary verdict.

Functional recipient class: A group defined by a distinct causal response repertoire. It should not be created merely by splitting an anatomical parcel.

Grain: The spatial, functional, and temporal resolution at which a candidate and its dynamics are described.

Held-out evidence: Data not used to choose the final measures, thresholds, or models that it later evaluates.

Inclusion-minimality: The condition that no qualifying proper subsystem exists at the same specification. It does not mean smallest node count.

Indeterminate: A licensed outcome in which relevant evidence remains insufficient, threshold-straddling, or credibly conflicted.

Integration or synergy (N_1): Joint causal or informational contribution not reducible to independent parts. Activity and statistical correlation are not sufficient.

License condition: A prerequisite that determines whether an evidential interpretation is allowed, such as validity, boundary stability, selectivity, or transport.

Mechanism ablation: An intervention on a proposed physical or computational realization. It requires selectivity and re-evaluation of boundaries and retained components.

Measurement ablation: Removal of a sensor, estimator, or analysis route to test robustness of measurement.

Minimal phenomenal presence (C_0 or $C\emptyset$): The minimal fact that there is something it is like for a system in a specified interval. It is not equivalent by definition to report, wakefulness, intelligence, reflection, or richness.

Model indeterminacy: Disagreement among serious theories about how to interpret a case, even when evidence within each theory may be decisive.

N-star (N^*): The conjunction $N_1 \wedge N_2 \wedge N_3$: integration, system-wide causal availability, and recurrent stability.

Necessity threat: A well-supported C_0 case for which the recomputed frozen positive-bearer set excludes the candidate under the same target, regime, and interval.

Negative: A licensed rule-out for the exact target based on valid, sensitive, adequate evidence. It is not simply a failed positive test.

Negative control: A condition designed to reveal a specific confound because the target-specific effect should not appear there.

Operating regime: Relevant state conditions such as arousal, task, drug, developmental stage, temperature, or computational mode.

Phenomenal character: What an experience is like - its qualitative organization - rather than the bare fact that experience is present.

Positive: A licensed attribution of the exact target under declared model and evidence rules. It is not certainty.

Positive bearer: An inclusion-minimal all-pass candidate in the frozen positive-bearer set. Final attribution additionally requires a resolved bearer relation: either a unique retained bearer or a plurality that satisfies the declared pairwise-separability rule.

Precaution: A practical response to possible harm under uncertainty. It combines evidence with loss, value, and reversibility assumptions and is not itself evidence of consciousness.

Preregistration: A timestamped record of hypotheses, models, measures, validity rules, analyses, outcomes, and revisions before critical data inspection.

Recurrent stability (N_3): Bounded recurrent organization that sustains a content-relevant dynamic pattern. It is not equivalent to mere duration or late activity.

Report-independent: Not requiring speech, button press, or another overt report as a constitutive criterion. Reports may still contribute as independently calibrated evidence.

Rescue: Restoration of a proposed causal route or component used to test whether an observed effect is pathway-specific.

Rule-in region: A preregistered evidence region sufficient for a positive model-relative classification under passed licenses.

Rule-out region: A validated evidence region sufficient for a negative classification under fair detection conditions.

Schematic legibility: A publicly teachable ability to reconstruct bearer, operation, unity, system-level presence, temporal presence, and fault consequences from a role-annotated model. It is intelligibility, not proof.

Separability: Preservation of each candidate's organization under reciprocal channel interruption. It helps assess whether plural candidates are distinct.

Specification neighborhood: A prospectively justified set of nearby analytic choices used to assess conclusion stability.

Substrate independence: The claim that a form of causal organization is not restricted by definition to one material. In this program it is constrained by transport evidence.

Sufficiency threat: A resolved inclusion-minimal all-pass bearer with strong, licensed independent negative evidence for C_0 at the same target, candidate, regime, and interval.

Surrogate conflict: Credible measures of a target that disagree beyond a preregistered tolerance.

Target: The precise property or capacity being assessed for a declared bearer, content, regime, and interval.

Transport: Application of a source-domain interpretation to a new domain after construct, causal-role, calibration, and boundary/regime preservation are supported.

Unlicensed: A meta-level result indicating that a prerequisite for interpretation failed. It is not evidence of absence.

Validity envelope: The domain in which the relationship between evidence and target interpretation has been established.

Version card: A source-backed record of an exact theory formulation, including target, bearer, commitments, report role, temporal assumptions, and prediction.

Viability (V): Independently assessed capacity of the candidate to realize the relevant dynamics in the operating regime. It is not a post hoc exception.

References and Source Trail

This textbook follows a seven-paper project by Phil Stilwell. The student-edition baseline is the core Version 2.0, revised July 8, 2026, plus the companion manuscript files present in repository commit 400f44a745e5425232a62f7a71487552fdfa9a0d on July 18, 2026. The project-paper links below are pinned to that commit, and the machine-readable build manifest records each corpus file’s SHA-256 digest. Check the current project before quoting a later revision. The taxonomy link is a dated retrieval rather than an immutable snapshot. Chapter source notes identify the principal paper or synthesis used locally.

Corpus item	Baseline date or version	Verification source
Core model	Version 2.0, July 8, 2026	Core PDF
Boundary criterion	July 10, 2026	Source manuscript
Availability criterion	July 11, 2026	Source manuscript
Four-outcome framework	Repository baseline, July 18, 2026	Source manuscript
Component-necessity audit	July 12, 2026	Source manuscript
Character extension	July 11, 2026	Source manuscript
Schematic legibility	July 16, 2026	Source manuscript
Taxonomy placement source	Retrieved July 9, 2026	Closer To Truth consciousness taxonomy

Project site: <https://philstilwell.github.io/C0/>

1. Cø as N*: A Minimal Network-Dynamics Model of Phenomenal Consciousness

Role in the project: the core hypothesis. It narrows the explanandum to minimal phenomenal presence and proposes that viability plus integration, system-wide availability, and recurrent stability are jointly necessary and sufficient for that target when realized by a resolved inclusion-minimal bearer for the same content and interval. It exposes both necessity and sufficiency failure directions and separates the presence gate from richer reflection and phenomenal character.

Use this source when: defining Cø, V, N₁, N₂, N₃, the conjunction, the principal applications, or the overall research program.

2. Where Is the Conscious Subject? A Dynamical Criterion for System Boundaries

Role in the project: the scope and bearer condition. It argues that candidate systems should be generated from independent signs of causal autonomy, frozen prospectively, assessed for role completeness, and filtered through inclusion-minimality and separability before final subject attribution. It distinguishes constituents, interfaces, and enabling supports and permits plural or unresolved boundary outcomes.

Use this source when: a claim depends on where the system begins and ends, especially for split brains, brain-body loops, coupled agents, organoid-controller hybrids, or distributed artificial systems.

3. Consciousness Without Report: What System-Wide Availability Actually Requires

Role in the project: the detailed operationalization of N_2 . It distinguishes actual activation from availability, report from causal role, broadcast from diffusion, and functional diversity from node count. It proposes content-specific interventions, independently defined recipient classes, route tests, bounded latency, and report/executive removal audits.

Use this source when: analyzing no-report paradigms, locked-in or command-following cases, dream states, posterior or subcortical organization, species differences, or distributed artificial architectures.

4. Indeterminacy as a Scientific Result: A Four-Outcome Framework for Consciousness Attribution

Role in the project: the evidence and reporting discipline. It distinguishes positive, negative, indeterminate, and unlicensed results and decomposes uncertainty into evidential scarcity, surrogate discordance, model disagreement, boundary/grain uncertainty, and transport uncertainty. It also separates scientific status from practical action.

Use this source when: a test is invalid, conflicting, underpowered, out of domain, or ethically urgent; when interpreting abstention rates; or when defining outcome precedence.

5. Ablating N^* : Does Every Conjunct Earn Its Place?

Role in the project: the component-necessity audit. It explains why a conjunct does not earn necessity merely by covarying with the others. It compares the full model with leave-one-out and grouped rivals on diagnostic cases, separates criterion, mechanism, and measurement ablations, protects held-out anchor independence, and allows indispensable, redundant, causally coupled, indeterminate, and unlicensed outcomes.

Use this source when: designing selective interventions, testing minimality, interpreting nulls, measuring diagnostic coverage, or deciding whether failure calls for component removal, combination, or better operationalization.

6. From Phenomenal Presence to Phenomenal Character: A Dynamical-Geometry Extension of the N^* Model

Role in the project: the second-layer account of what experience is like. After a positive presence gate, it proposes mapping content-specific causal-trajectory geometry to independently anchored relations in an empirical quality space. It requires held-out tests, geometry-changing interventions, invariance analysis, and explicit transport licenses.

Use this source when: moving from whether experience is present to how color, sound, pain, valence, temporal flow, or cross-modal unity might be structured.

7. Reading Consciousness from the Schematic: Structural Legibility in the N* Model

Role in the project: the intelligibility bridge. Using the limited analogy of expert schematic reading, it asks whether a trained reader can reconstruct the bearer, operation, unity, system-level presence, temporal presence, and localized fault consequences from a role-annotated model. It distinguishes conceptual legibility from empirical confirmation.

Use this source when: asking whether the proposed identity is explanatorily readable, whether experts can abstain appropriately, or whether fault localization transfers across independently tested cases.

How the sources fit together

The core paper states the risky identity hypothesis. The boundary paper fixes the candidate system to which it applies. The availability paper makes one conjunct experimentally sharper without defining it by report. The indeterminacy paper governs what evidence licenses. The ablation paper asks whether every conjunct is truly needed. The character paper extends the project beyond bare presence. The schematic paper asks what would make the whole architecture structurally intelligible. No single companion paper independently establishes the core identity, and success in one work package does not automatically validate the rest.

Primary manuscript references

Stilwell, P. (2026). *CØ as N*: A minimal network-dynamics model of phenomenal consciousness* (Version 2.0) [Manuscript].

Stilwell, P. (2026). *Where is the conscious subject? A dynamical criterion for system boundaries* [Manuscript].

Stilwell, P. (2026). *Consciousness without report: What system-wide availability actually requires* [Manuscript].

Stilwell, P. (2026). *Indeterminacy as a scientific result: A four-outcome framework for consciousness attribution* [Manuscript].

Stilwell, P. (2026). *Ablating N*: Does every conjunct earn its place? A necessity audit without a consciousness oracle* [Manuscript].

Stilwell, P. (2026). *From phenomenal presence to phenomenal character: A dynamical-geometry extension of the N* model* [Manuscript].

Stilwell, P. (2026). *Reading consciousness from the schematic: Structural legibility in the N* model* [Manuscript].

Selected contextual readings

These readings place empirical examples and rival families in context. Their inclusion does not imply that they endorse Cø / N*.

Albantakis, L., Barbosa, L., Findlay, G., et al. (2023). Integrated information theory (IIT) 4.0: Formulating the properties of phenomenal existence in physical terms. *PLOS Computational Biology*, 19(10), e1011465. <https://doi.org/10.1371/journal.pcbi.1011465>

Birch, J., Schnell, A. K., & Clayton, N. S. (2020). Dimensions of animal consciousness. *Trends in Cognitive Sciences*, 24(10), 789-801. <https://doi.org/10.1016/j.tics.2020.07.007>

Bodien, Y. G., Allanson, J., Cardone, P., et al. (2024). Cognitive motor dissociation in disorders of consciousness. *New England Journal of Medicine*, 391(7), 598-608. <https://doi.org/10.1056/NEJMoa2400645>

Butlin, P., Long, R., Elmoznino, E., et al. (2023). *Consciousness in artificial intelligence: Insights from the science of consciousness*. arXiv. <https://doi.org/10.48550/arXiv.2308.08708>

Casali, A. G., Gosseries, O., Rosanova, M., et al. (2013). A theoretically based index of consciousness independent of sensory processing and behavior. *Science Translational Medicine*, 5(198), 198ra105. <https://doi.org/10.1126/scitranslmed.3006294>

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Reading-record template

Use this worksheet for every primary or contextual source that materially affects an argument or capstone decision. Record the exact location, target, evidence, remaining burden, and inferential limit before importing the claim. The guardrail is traceability: a citation should let another reader reconstruct what the source actually supports.

Reading-record field	Response
Paper title, author, year, and exact version	
Stable link, DOI, or repository location	
Section, page, figure, or passage	
Claim in your own words	
Exact target, population, regime, and scope	
Role in the seven-paper program or rival comparison	
Evidence actually offered	
Evidence still required	
Strongest objection or limitation acknowledged	
One prohibited inference	
How it changes the capstone or argument	